INSTRUCTION MANUAL
FOR
MOTORISED CABLE REEL TYPE
660DM.NCC(2X2kW).8 M1956-51GI
K460/4517/6FO/FC/R M9P/OT/W/LCBT

ZERTIFIKAT ◆ CERTIFICATE ◆ 認証書 ◆ СЕРТИФИКАТ ◆ CERTIFICADO ◆ CERTIFICAT

TÜV SÜD
Italia

CERTIFICATO

Nr 50 100 1352 - Rev. 04

Si attesta che / *This is to certify that*

IL SISTEMA QUALITÀ DI
THE QUALITY SYSTEM OF

CAVOTEC
Connecting mobile equipment

CAVOTEC SPECIMAS S.p.A.

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Questo certificato è valido per il seguente campo di applicazione
This certificate is valid for the following product or service range

Progettazione fabbricazione ed assistenza di avvolgicavi e avvolgitubi, panzerbelt, collettori rotanti, giunti idraulici. Fabbricazione di sistemi automatici di ancoraggio navi. Commercializzazione di accessori per impianti di movimentazione (EA 18, 14, 29a)

Design, manufacture and service of cable reels and hose reels, panzer belt, electrical sliprings and hydraulic swivels. Manufacturing of automated mooring systems. Trade of accessories for moving systems (EA 18, 14, 29a)

ACCREDIA
L'ENTE ITALIANO DI ACCREDITAMENTO

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SGA N° 018D ITX N° 001L ISP N° 057E
SCR N° 009F

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Per l'Organismo di Certificazione
For the Certification Body
TÜV Italia S.r.l.

Andrea Vivi
Amministratore Delegato - *CEO*

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"La validità del presente certificato è subordinata a sorveglianza periodica a 12 mesi e al riesame completo del sistema di gestione aziendale con periodicità triennale"

"The validity of the present certificate depends on the annual surveillance every 12 months and on the complete review of company's management system after three-years"

DISCLAIMER

This Instruction Manual is intended as a guide to the installation, usage and maintenance of the specified Cavotec equipment.

Cavotec Specimas SpA does not accept liability for any loss or damage suffered as a result of using this manual.

Prior to installing, operating or maintaining the equipment, read the GENERAL WARNINGS AND SAFETY GUIDELINES located on the following page.

If in doubt, always refer to the original equipment manufacturer.

The equipment warranty will be voided if the technical guidelines for the installation, usage and maintenance have not been correctly followed.

Only suitably qualified maintenance personnel who have read and understood every part of this document are authorised to maintain the specified equipment.

The information contained within this document is subject to change at any time without notice.

VERSION HISTORY

Rev.	Revised by:	Approved by:	Revision Date	Comments
A	E. Rossetto	P. Rosenwald	26.10.12	Primary issue

GENERAL WARNINGS AND SAFETY GUIDELINES





Reel operation is remotely controlled by the machine on which it is installed. Drum rotation can therefore occur at any time, so it is important to keep well clear during machine operation.

If it is necessary to be near the reel during operation (eg. while commissioning), be aware of the location of the nearest emergency stop button and exercise caution.

Prior to performing maintenance on the reel, ensure that the machine long travel and reel motor are isolated by an authorised person.

If work is to be done inside the collector enclosure, the drive centre or the motor terminal blocks, ensure that all relevant circuits are isolated by an authorised person prior to removing any covers.

Warning labels on the collector enclosure are there to remind personnel of the danger.

Prior to touching any electrical connection, confirm with an appropriate meter that there is no power. Failure to do so may lead to serious injury or death.

Note that any heaters or travel limit switches in the collector enclosure may still have tension on the terminals even if the main power is switched off.

Only suitably qualified maintenance personnel who have read and understood every part of this document are authorised to maintain the specified equipment.

STORAGE INSTRUCTIONS

For correct storage of Cavotec cable reels, maintain the goods in the original packing in a closed warehouse, protected from the rain and in an ambient that prevents the formation of condensation.

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SECTION 1 – SPECIFICATIONS

Client	ZPMC	
Project	Sri Lanka	
Works Order	121216	
Reel Type	660DM.NCC(2X2kW).8 M1956-51GI K460/4517/6FO/FC/R M9P/OT/W/LCBT	
Delivery	Drive module	Type 660DM.NCC(2X2kW).8 ..... K460/4517/6FO/FC/R Fixing with 8 off M16 high tensile bolts Weight 1205kg
	Drum	Type M1956-51GI Monospiral composite Weight 1622kg
	Guide	Type M9P/OT/W/LCBT Weight 95kg
	Drive + Panel	Type: Electrical panel for Vacon drive Weight 65kg
	Brake resistor	Type 80 Ohm, 1kW + Rain cover in AISI 316 Weight 20kg
Drawings	Installation reel	SC-121216-IS1
	Installation J.box	SC-120121-JB1
	Installation guides	STD-AD-034
	Typical guide installation	STD-AD-083
	Box + Drive	STD-BE-004
	Braking resistor	STD-BE-001
	Schematics	SC-121216-SE1 STD-SE-002
Cable	Manufacturer	Tratos
	Type	Tratosflex ESDB FO
	Voltage rating	6/10kV
	Composition	3x35+2x25/2+6FO(50/125)
	Dimensions	ϕ47.2mm, 3.05kg/m
	Allowable tension	3000N
Duty	Reeling length	400m
	Dead turns	3
	Speed	46m/min
	Acceleration	0.10m/s^2
	Duty cycle	60% ED
	Ambient temp.	0°C to + 50°C
Drum	Manufacturer	Cavotec Specimas S.p.A.
	Type	Monospiral
	Model	M1956-51GI
	Inner diameter	1900mm
	Outer diameter	5600mm
	Width	51mm
	Construction	Composite
	Main spokes	36 off 120x40x3 per side
	Material	Mild steel
	Treatment	Hot-dip galvanised
	Fasteners	AISI 304
	Weight	1622kg

Torque Unit	Manufacturer	Cavotec Specimas SpA	
	Type	3-stage bevel spur gearbox	
	Model	660DM for two motors flanged	
	Ratio	1:67.3	
	Main shaft ID	92mm	
	Overhung load	3000kg	
	Lubricant	Shell Spirax ASX 75W 90	
	Quantity	36 litres	
	Finish	Painted RAL 1004 (Yellow)	
	Weight	405kg	
Motor 1	Manufacturer	Cavotec Specimas SpA	
	Type	Totally enclosed, fan cooled (TEFC) 8 pole motor with Klixon, heater brake & encoder	
	Model	V112M8/1-B5	
	Power	2kW, 380V, 50Hz	
	Insulation	Class F, tropicalised	
	Nominal speed	706 rpm	
	Nominal current	5.95A @ 380V	
	Protection rating	IP65	
	Encoder	Type	Heidenhain ERN 430
		Resolution	1024 ppr
		Output	HTL half-wave
		Protection	IP67
		Supply	10-30VDC, 150mA
	Brake	Type	Intorq type BFK.458.12
		Coil voltage	205VDC
		Rectifier	4-pole full-wave type BEG-142-270
		Supply voltage	230VAC
		Torque range	14-46Nm
		Setting torque	23Nm
		Options	Release lever
	Heater	220V, 25W	
	Over-temperature	Klixon 140°C trip, Thermik S01	
	Finish	Painted RAL 5009 (Azure blue)	
	Weight	38kg	
Motor 2	Manufacturer	Cavotec Specimas SpA	
	Type	Totally enclosed, fan cooled (TEFC) 8 pole motor with heater and Klixon	
	Model	V112M8/1-B5	
	Power	2kW, 380V, 50Hz	
	Insulation	Class F, tropicalised	
	Nominal speed	706 rpm	
	Nominal current	5.95A @ 380V	
	Protection rating	IP65	
	Heater	220V, 25W	
	Over-temperature	Klixon 140°C trip, Thermik S01	
	Finish	Painted RAL 5009 (Azure blue)	
	Weight	32kg	
Drive	Manufacturer	Vacon	
	Type	NXP 5.5 kW + filter 380-500VAC, IP21	
	Model	NXP0012 5 A2 H1SSV A1A2	
	Accessories	Encoder board for type HTL encoder code NCOPTA5 + Profibus DP board with DSUB connector code NXOPTC5	
	Note	Refer to document in SECTION 15	

Drive Panel	Manufacturer	Cavotec Specimas SpA
	Type	600x800x300, AISI 304, door RH
	Model	AS-518
	Protection rating	IP54
	Finish	Painted RAL 7035 (Light grey)
	Weight	50kg
Braking resistor	Manufacturer	Cavotec Specimas SpA
	Model	AS-521
	Type:	80 Ohm, 1kW, + Rain cover in AISI 316
	Weight	20kg
Collector	Manufacturer	Cavotec Specimas SpA
	Type	15kV 3 phase + earth, multi-contact copper-graphite brushes on brass rings
	Model	K460/4517/R
	Earth	1 off 600A nominal Ø450 standard ring with 2 off BG300/450 brush per ring, 300A current rating @ 100% ED
	Power	3 off 600A nominal Ø450 standard rings with 2 off BG300/450 brush per ring, 300A current rating @ 100% ED
	Fixtures	Ring side - 6 x M12, tighten 30Nm Brush side - 1 x M12, tighten 30Nm
	Insulator type	Epoxy resin finned post insulator
	Insulation distance	170mm
	Highest voltage U_m	17.5kV nominal
	Test voltage	35kV
	Impulse withstand	95kV
	Short circuit rating	9.2kA for 1 second
	Housing material	AISI 316
	Protection	IP55
	Heater	Raychem 29W/m 220V, 60W, self-regulating
	Accessories	Hinged doors, portal kit
	Other	Junction box mounted on collector for pre-wired: motors, limit switch, heaters
	Weight	195kg
Junction box	Manufacturer	Cavotec Specimas SpA
	Type	Junction box mounted on collector housing With Pilot light red for heater on Pre-wired: motors, limit switch, heaters
FO Rotary Accumulator	Manufacturer	Cavotec Specimas SpA
	Type	Spool to spool
	Model	40406-5S0
	Useful turns	50
	No. of ways	6
	Fibre type	50/125µm
	Connector type	ST
	Sheath diameter	4mm
	Losses	<3dB including connectors
	Operating temp.	-30°C to +50°C
	Heater	Raychem 29W/m, 220V, 40W, self-regulating
	Weight	24kg
Reeling Cable Terminations	Slipring core lengths	Earth 245mm Phase 1 465mm Phase 2 685mm Phase 3 905mm
	Fibre bundle length	FO 1145mm

Limit Switch	Manufacturer	Stromag AG	
	Type	48 turn geared rotary limit switch with 1+4 switches + transducer	
	Model	48BM 199/499 WGMU36S	
	Contacts	5 x silver, changeover	
	AC-15 contact rating	1.5A, 230VAC in accordance with IEC 60947-5-1	
	DC-13 contact rating	0.5A, 60VDC in accordance with IEC 60947-5-1	
	Thermal current	10A	
	Insulation voltage	250V	
	Switch life	$1x10^7$ operations	
	Connections	Screw-type terminals with self-lifting pads	
	Gear ratio	1:30.35	
	Cams	1-2	15º (48 revs off, 1.5 revs on)
		3-4	15º (48 revs off, 1.5 revs on)
	Transducer	Type	WGMU36S
		Output	4-20mA
		Supply	18-30VDC, 60mA
		Electrical angle	0-345º
		Linearity	0.15%
		Mech. angle	360º without stop
	Protection rating	IP20	
	Driven by	Gear from FO connector drive	
Guide	Manufacturer	Cavotec Specimas SpA	
	Type	One cable, Bi-directional roller guide	
	Model	M9P/OT/W/LCBT	
	Material	Mild steel	
	Treatment	Hot-dip galvanised	
	Fasteners	AISI 304	
	Options	2 x overtension switches	
		1 x undertension switch	
		1 x direction sensing switch	
		Prewired switches to terminal box with control buttons	
	Switch type	Telemacanique type XC2	
	Rollers	Makralon	
	Weight	95kg	

SECTION 2 – CABLE REELING SYSTEM OVERVIEW

Large mobile machines such as cranes, stackers and ship loaders rely on an external electrical power source to operate. In many cases, these machines also require analogue and/or digital data conduits for control and feedback. This power and data is typically transmitted through a composite flexible cable comprising of copper conductors for power and hard wire control circuits, and optical fibres for digital data. Since these machines can travel over relatively long distances, this cable needs to be quite long, and a means of handling the cable in a controlled manner is required. A Cable Reeling System typically carries out this function.

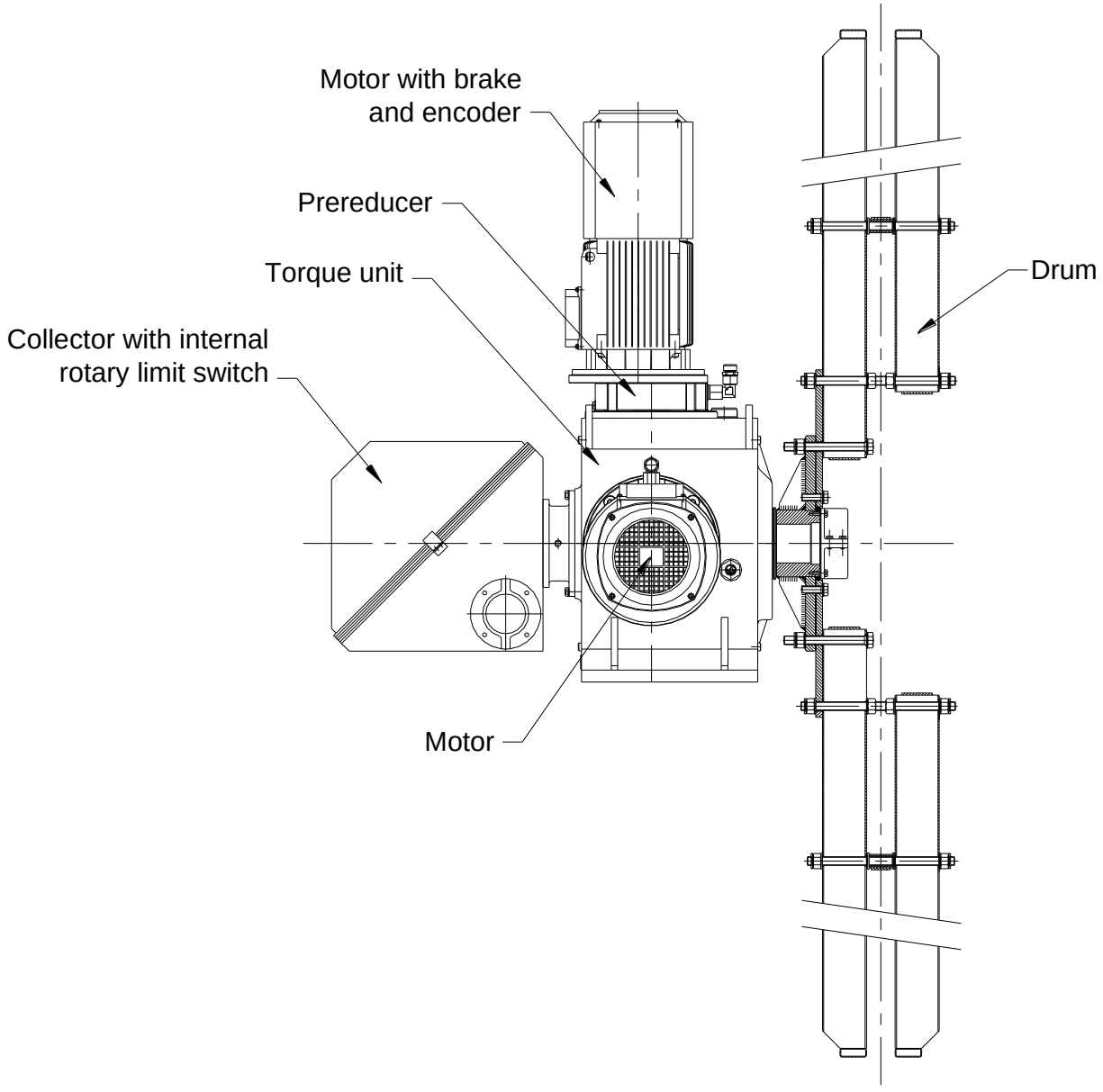


Figure 1 – Typical cable reeling system

2.1 DRUM

2.1.1 A drum stores cable by rotating in the appropriate direction as the machine travels towards the feedpoint. The internal diameter of the drum is selected to meet the minimum bend radius requirements of the cable and to ensure an optimal relationship between reeling torque and cable tension. The external diameter of the drum is selected to ensure that the full cable length can be accommodated. The compartment width of the drum

depends on the cable diameter and the type of lay selected (eg. monospiral, 3-2-3, random).

2.1.2 Drums are typically fabricated from mild steel RHS, and can be painted or hot-dip galvanised. The drum sides can be either fully welded (for smaller drums) or composite designs. All fasteners are stainless steel.

2.2 TORQUE UNIT

2.2.1 A torque unit is typically a specially designed reduction gearbox with a large bore main shaft through which the cable can be fed, and a high overhung load capacity. The torque output of this unit is regulated by mechanical or electrical means. An example of a mechanical torque regulator is the Hydrodynamic oil immersed friction drive. Electrical torque regulators include torque motors, permanent magnet clutches and flux vector controlled AC motors.

2.2.2 The optimum torque output of a torque unit is just sufficient to recover and pay out cable in all modes of operation without placing undesirable stress into the typically fragile conductors. The speed output of the torque unit can vary greatly in relation to the machine speed because of the changing winding diameter, and the differences are generally taken up by mechanical or electrical slippage.

2.3 MOTOR

2.3.1 Motors are typically Totally Enclosed Fan Cooled (TEFC) type, are fitted with sealed for life bearings, and have IP65 protection ratings. When used with flux vector drives, the motors are oversized to preclude the need for forced cooling.

2.3.2 Cavotec motors are supplied with class F insulation. Optional Klixon switches or thermistors are embedded in the windings to provide a means of protecting the motor from over-temperature conditions.

2.4 BRAKE

2.4.1 The brake prevents the reeler from unwinding cable when the power supply to the reeler motor is turned off.

2.4.2 Cavotec brake motors are equipped with failsafe electromagnetic brakes utilising asbestos free brake pads. These brake pads have a high resistance to wear and operate reliably at high temperatures.

2.4.3 The electromagnet is driven by direct current, and is characterised by a very simply design, low hum level and no vibrations. Due to direct current use, the current intensity is very low.

2.4.4 The electromagnetic coil is designed to be continuously activated. A special compound surrounds the coil in the electromagnetic housing and makes the coil humidity and vibration proof.

2.4.5 The rectifier is housed in the motor terminal box, and it is specifically designed for electromagnetic brakes. It is compact, high temperature resistant and unaffected by voltage variations.

2.5 ENCODER

2.5.1 Flux vector controlled AC drives require speed and rotation direction feedback from the motor in order to accurately control the motor’s torque output. A dual phase incremental encoder attached to the end of the motor shaft carries out this function.

2.6 COLLECTOR

2.6.1 The collector facilitates the transmission of power and data between the rotating reeling cable and the fixed machine cables.

2.6.2 The collector consists of a number of rotating brass sliprings (typically 4 for power applications) against which a number of fixed copper-graphite brush tips rub. Insulators and air separate each phase.

2.6.3 The collector may also include a fibre optic rotary joint used to transmit optical data across the rotating interface.

2.6.4 Collectors are typically housed in stainless steel enclosures with anti-condensation heaters.

2.7 ROTARY LIMIT SWITCH

2.7.1 A geared rotary limit switch for reel full and reel empty detection is typically supplied with the collector, mounted either internally or externally of the collector enclosure.

2.7.2 The rotary limit switch can be fitted with a potentiometer or rotary transducer to give a precise indication of the amount of cable on the drum at any time.

2.8 CABLE GUIDE

2.8.1 Cable guide is essential in both modes of operation, i.e. reeling and unreeling to ensure that regular laying of the cable is achieved.

2.8.2 Normally all cable all cable reels are fitted with a cable guide.

2.8.3 The cable guide is normally fitted with devices to sense crane location in respect to centre feed point (left/right) and slack cable.

2.9 DRIVE NCC

2.9.1 The reeler control centre houses the flux vector controlled AC drive and the various other electrical components required to control the reeler motors.

2.9.2 During normal operation, the flux vector drive operates in torque control mode. Torque output is adjustable via potentiometers.

2.9.3 It is possible to operate the reeler manually in speed control mode in order to perform maintenance tasks. A selector switch and two push buttons mounted on the enclosure door facilitate this.

SECTION 3 - DRIVE MODULE

3.1 DELIVERY

3.1.1 The torque unit, base frame (if supplied), collector, prereducer/s (if supplied) and motor/s are supplied pre-assembled as a completed drive module. Check that this module has arrived in good condition.

3.1.2 Note any special instructions in the *Delivery* subsection of SECTION 1.

3.2 INSTALLATION

3.2.1 Check that the structure onto which the drive module is to be mounted is flat and free from distortion.

3.2.2 Lift the drive module using the lifting points provided. The total weight of the module is indicated in the *Delivery* subsection of SECTION 1.

3.2.3 Bolt the drive module to the mounting structure using high tensile bolts of appropriate length with the dimensions indicated in the *Delivery* subsection of SECTION 1, ensuring that it is level by using a spirit level. Ensure that the face of the drum mounting flange is parallel to the direction of machine travel. Use shims or packing pieces as necessary and bolt down securely.

3.3 ELECTRICAL INSTALLATION

3.3.1 Referring to the electrical schematics; run, gland and terminate appropriate cables between the reeler control centre and the various components supplied.

3.3.2 Run, gland and terminate an appropriate cable/s between the collector brushgear and the machine's power/control inlet.

SECTION 4 – DRUM

4.1 DELIVERY

4.1.1 Check that the following separate items have arrived and are in good condition:

Pos.	Codice/Part No.	Description	Qty
1	SC-101745-001	External cpoke for cable entry M1956-59G	1
2	SC-101745-002	Internal spoke for cable entry M1956-59G	1
3	SC-101745-003	External spoke M1956-59G	35
4	SC-101745-004	Internal spoke M1956-59G	35
5	SC-101745-005	Outer ring D=5600 36 spokes	12
6	SR4-05330-005	Inner ring 36 spokes galv.	1
7	S42-93108-020	Socket head cap screw M8x 20 AISI 304	72
8	S49-7473X-008	Self-locking nut M8	72
9	S42-73912-050	Hexagon setscrew M12x50 AISI 304	12
10	S47-6592X-012	Plain washer M12 AISI 304	12
11	S42-73920-080	Hexagon setscrew M20x80 AISI 304	24
12	S49-5588B-020	Hexagonal full-nut M20, zinc-plated brass	312
13	S47-6592X-020	Plain washer M20 AISI 304	408
14	S49-8835X-020	Safety locknut M20 AISI 304 (Palmutter)	240
15	S45-2000X-210	Tie rod M20x210 AISI 304	36
16	S45-2000X-380	Tie rod M20x380 AISI 304	36
18	SR4-03597-000	Clamp for spoke 40 - L=80, AISI 304	108
19	SR4-05792-000	Clamp for spoke 40 - L=40 , AISI 304	70
20	S45-2000X-480	Tie rod M20x480 AISI 304	1
21	S92-2125X-165	S.s. tube L = 165	1
22	S42-73910-065	Hexagon setscrew M10x65 AISI 304	1
23	S49-5588X-010	Hexagonal full-nut M10, AISI 304	1
24	S47-6592X-010	Plain washer M10 AISI 304	3
25	S48-00137-010	Elastic washer M10	1
26	SR4-05756-000	Flange intermediate type K dia. 1010, 20t	1
27	SR4-05330-005	Tie rod M20x360 AISI 304	35

4.1.2 Note any special instructions in the *Delivery* subsection of SECTION 1.

4.2 REQUIRED TOOLS & ACCESSORIES

4.2.1 The following tools and accessories will be required to carry out the drum assembly:

- 4 off blocks of wood 200x200x2800
- 1 off block of wood 350x350x200
- 6mm hex (Allen) key
- Small socket wrench with 13mm socket
- 2 off 30mm open ended spanners
- 5 tonne lifting sling, 10m long

4.3 DRUM ASSEMBLY

4.3.1 Position the blocks of wood as shown in Figure 2 on a flat piece of ground 10m x 10m near the machine onto which the drum will be mounted.

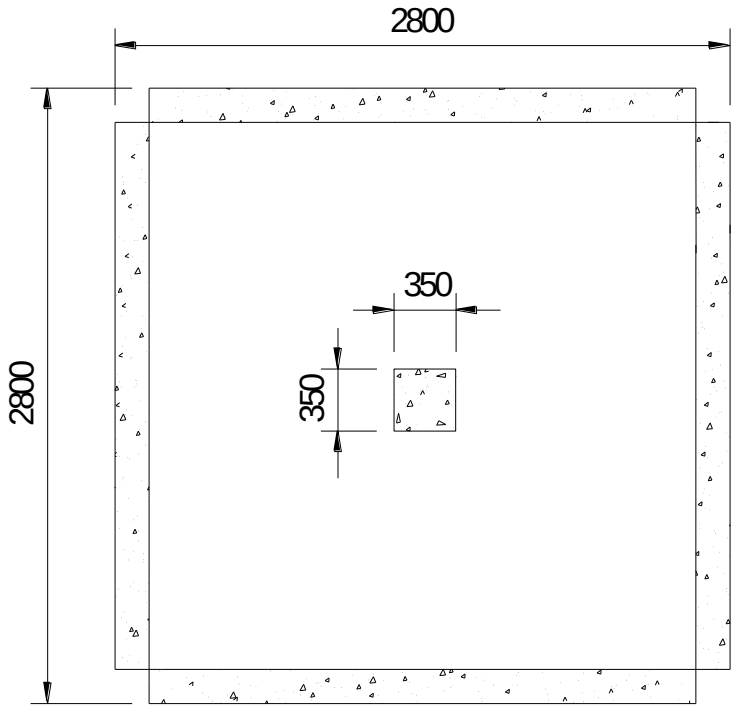


Figure 2 - Drum support blocks

4.3.2 Carefully lift the intermediate flange (26) onto the centre block as shown in Figure 3.

4.3.3 Mount the 36 internal spokes (2+4) to the flange according to the sequence shown in Figure 3 fixing them with the M20 tie-rods, nuts & washers (12+13+15) on Ø560, and with tie-rods, clamps, nuts & washers (12+13+16+18) on Ø960, as shown in the Figure 4. Tighten to 155Nm. Fit the supplied Pal-nuts (14) as shown in Figure 5.

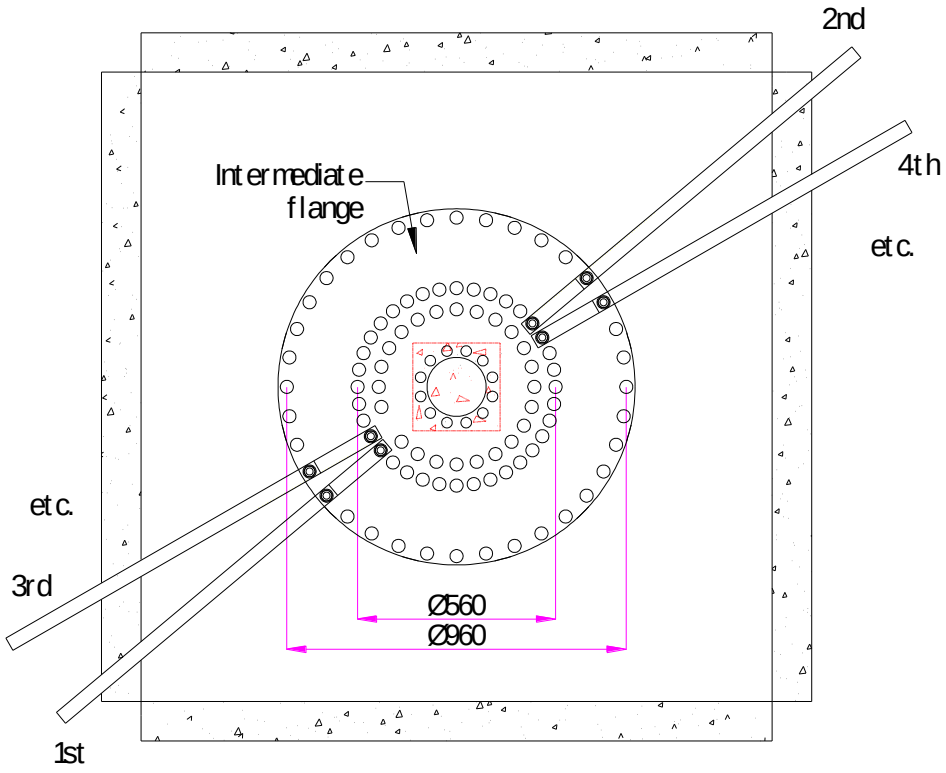


Figure 3 - Internal spokes

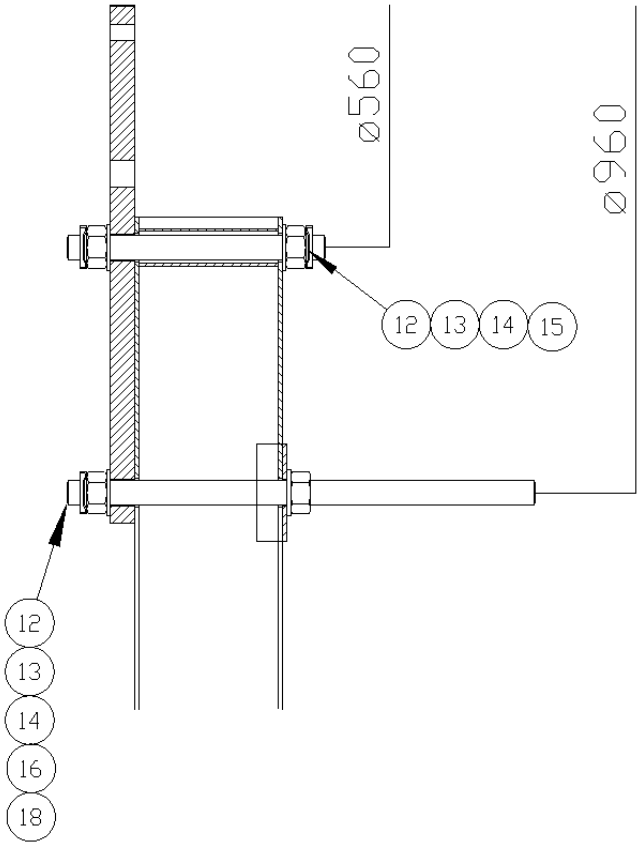


Figure 4 - Drum fixing

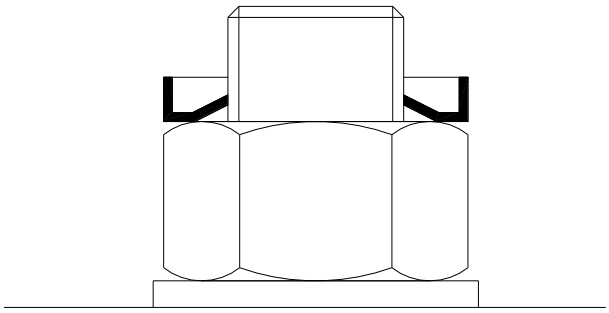


Figure 5 - Pal nut installation

4.3.4 Fix the outer ring sections (5) on the internal spokes using the screws and nuts M8 (7+8). and an appropriate locking fluid (eg. Loctite, Stalock...), as shown in Figure 6. The outer rim should be flush with the spoke inner surface or protruding a maximum 0.5mm from the spoke surface. Otherwise the drum can in a later stage not be adjusted to the correct width to accommodate the cable correctly. The outer rim should rest on the retaining finger when the inner surface is flush. If it is needed the retaining finger should be bent to correct position on mounting.

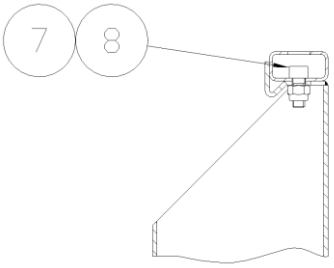


Figure 6 - Outer rim sections

4.3.5 Fit the inner ring (6) as shown in Figure 7 preparing the 35 M20 tie-rods (27) + clamps + nuts & washers (12+13+14+18+19) on Ø1900 (see Figure 8), making sure to use the correct number of spacer washers (13) to adjust the drum width; see SPACING ADJUSTMENT section 4.5.

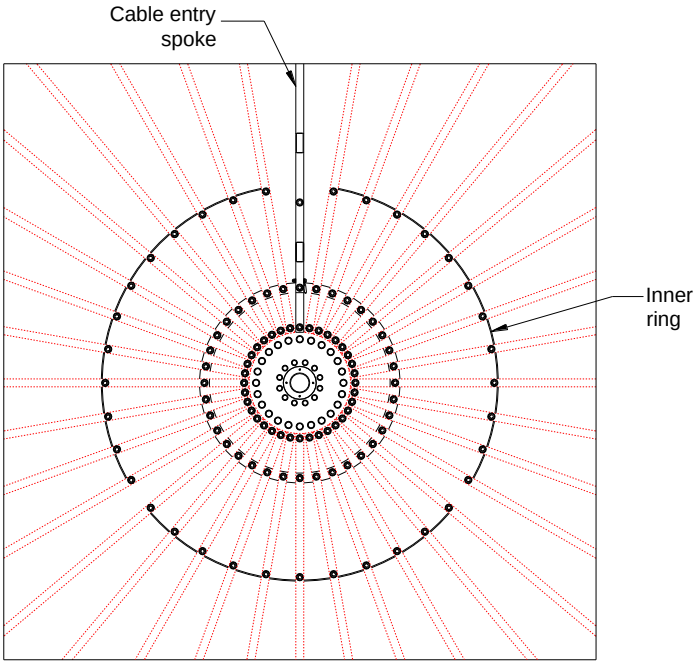


Figure 7 - Inner ring assembly

4.3.6 Mount the 35 outer spokes (3), fixing them with the M20 nuts & washers (12+13) on Ø960, and with the clamps + M20 nuts & washers (12+13+18+19) on Ø1900 following the sequence as described in the Figure 3. Tighten to 155Nm. Fit the supplied Pal-nuts (14) as shown in Figure 5.

4.3.7 Mount the external cable entry spoke (1) in correspondence with the internal cable entry spoke using longer tie-rod (20) + clamps + nuts & washers (12+13+18) and the spacer tube (21) as shown in Figure 8. Tighten to 155Nm. Fit the supplied Pal-nuts (14) as shown in Figure 5.

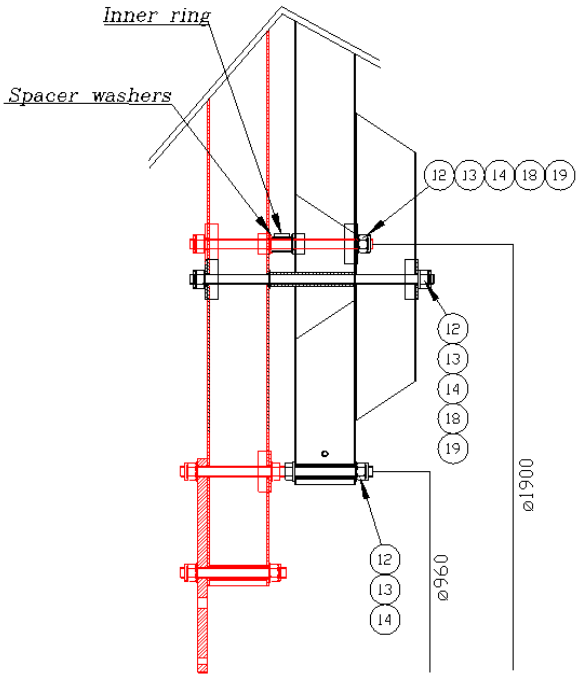


Figure 8 - Cable entry spokes

4.3.8 Fit the section (5) of the outer ring on the outer spokes, as in point 3, with the M8 cap-screws and nuts (7+8) , fixing screws with locking fluid (eg. Loctite, Stalock..). See Figure 6.

4.4 DRUM INSTALLATION

4.4.1 Carefully lift the drum using a crane and sling as shown in Figure 9.

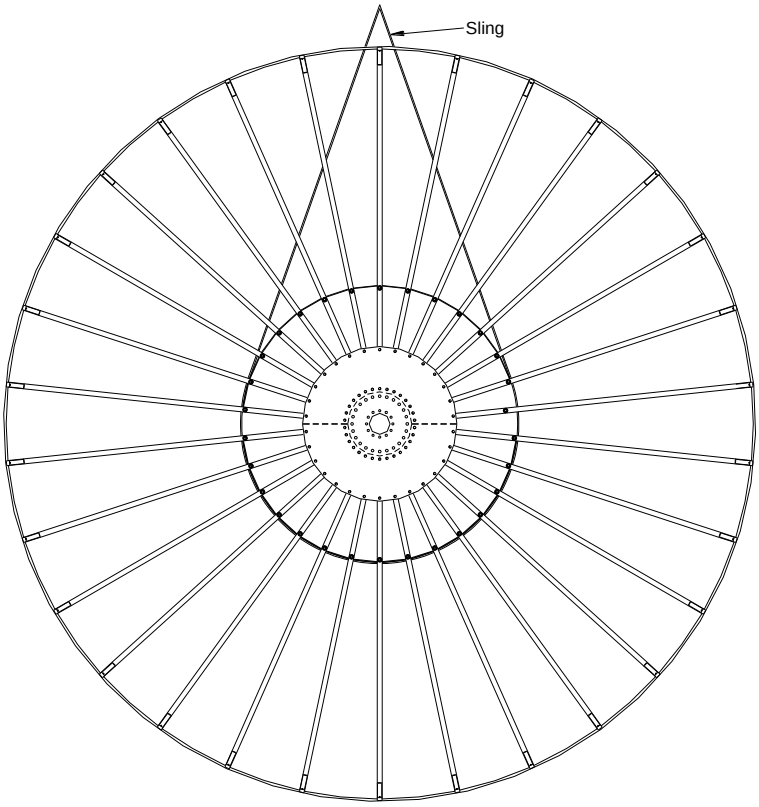


Figure 9 – Lifting drum

4.4.2 Position the drum so that the flange is centred onto the drive flange.

4.4.3 Fit the 12 off M16x60 setscrews and washers (10,11), and torque them to 70Nm.

4.4.4 Fit the 24 off M20x80 setscrews, washers and nuts (12,13,14), and torque the nuts to 155Nm. See Figure 10.

4.4.5 Fit the supplied Pal nuts, noting the orientation shown in Figure 5.

4.4.6 Verify tightening of all nuts prior to winding on the cable.

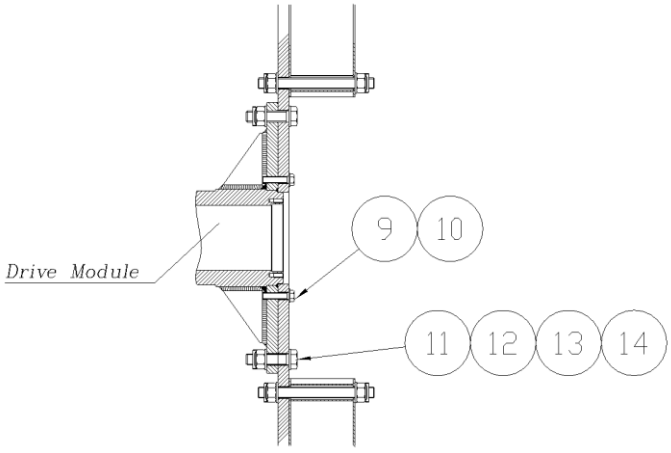


Figure 10 - Fixing drum

4.5 SPACING ADJUSTMENT

4.5.1 In order to ensure reliable performance of the reel, it is very important that the spacing between the spokes is accurately set.

4.5.2 Measure the exact diameter of the cable at a minimum of 3 points along its length and calculate the average value D.

4.5.3 Referring to Figure 11, remove all Pal-nuts from (b) then loosen all nuts (a) and (b) on the external main spokes.

4.5.4 If the drum width S1 is greater than D+4mm, or less than D+1mm, partially remove each tie-rod (c) one at a time, and remove or insert washers (w) to bring the width to within the correct range.

4.5.5 Re-insert each tie-rod, refit each washer and loosely refit each full-nut.

4.5.6 Once the adjustment has been completed, tighten all the full-nuts to 155Nm. Refit the Pal-nuts and tighten.

4.5.7 Adjust the parallelism of every main spoke pair (internal and external) so that S2 is between D+2mm and D+3 mm.

4.5.8 To increase S2, loosen nut (a) and tighten nut (b). To decrease S2, loosen nut (b) and tighten nut (a).

4.5.9 Tighten all the full-nuts to 155Nm, and check the spacing again. If no changes need to be made, refit the half-nuts and tighten.

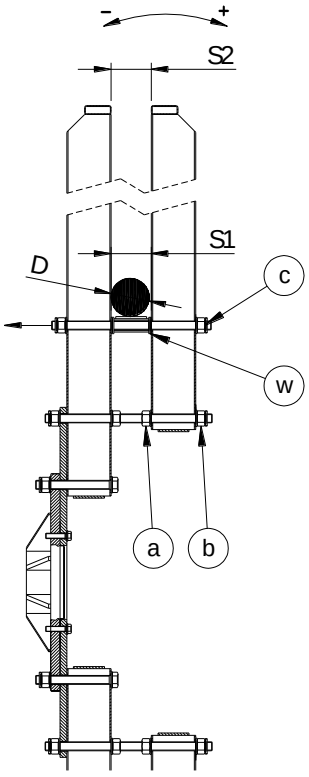


Figure 11 - Spacing adjustment

4.6 PERIODIC MAINTENANCE

4.6.1 Verify the tightness of all fasteners.

4.6.2 Check the drum width, and adjust as necessary following the procedures described in section 4.5.

4.6.3 Verify the fixing of the first turn of cable on the drum.

SECTION 5 - CABLE

5.1 CABLE PREPARATION

5.1.1 Check that the cable diameter and weight are as originally specified (see *Cable* subsection of SECTION 1 for cable details) and ensure that the cable length does not exceed that for which the reel was designed.

5.1.2 Generally, reputable manufactures will supply their cables free of induced twist on storage drums. If, however, the cable on the storage drum is second hand, or the supplier cannot guarantee the cable to be free of induced twist, the cable should be prepared in accordance with the following.

5.1.3 The transport drum should be supported on stands or rollers to allow free rotation. The cable must always be pulled from the top of the drum. Refer to Figure 12.

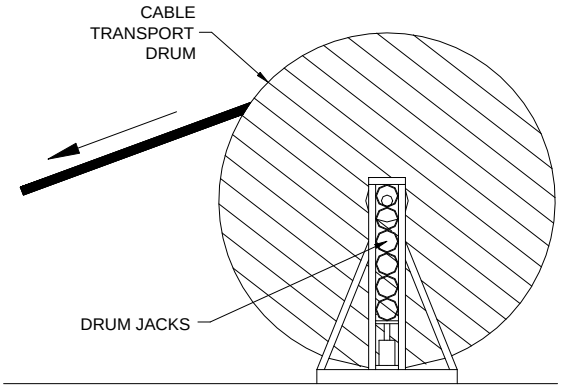


Figure 12 – Cable Transport Drum

5.1.4 Lay out the full length of the cable as it is pulled off the drum, either by manhandling with one person every 5m approximately or by pulling over cable rollers. ***On no account should the full length of the cable be dragged over the ground, nor should a dozer, tractor or any other powered machine be used to pull or drag the cable***.

5.1.5 There may be some twist evident in the cable, which is due to the manufacturing process, and this should now be removed. Leave one end of the cable fixed (eg. still attached to the transport drum) and working from this end, run the twist out of the cable by either (a) placing a suitable bar under the cable and walking towards the free end while gently moving the bar up and down (Figure 13), or (b) forming a coil of cable as near as possible to the fixed end and rolling it along to the free end (Figure 14). If there is a lot of twist in the cable, the process described above may need to be repeated several times.

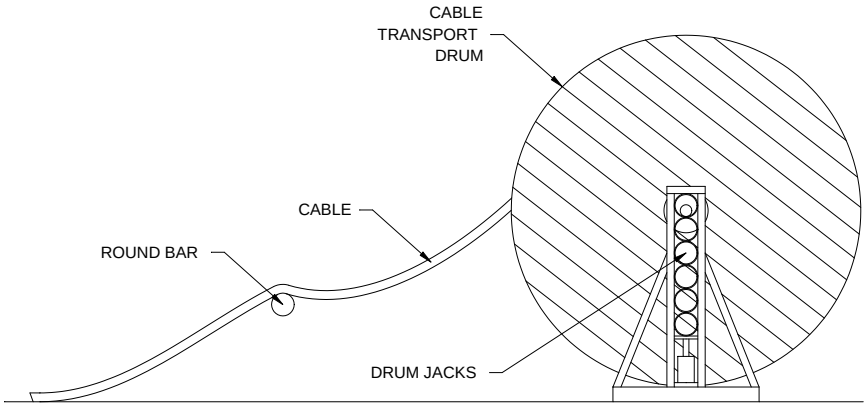


Figure 13 – Removal of Cable Twist Using Round Bar

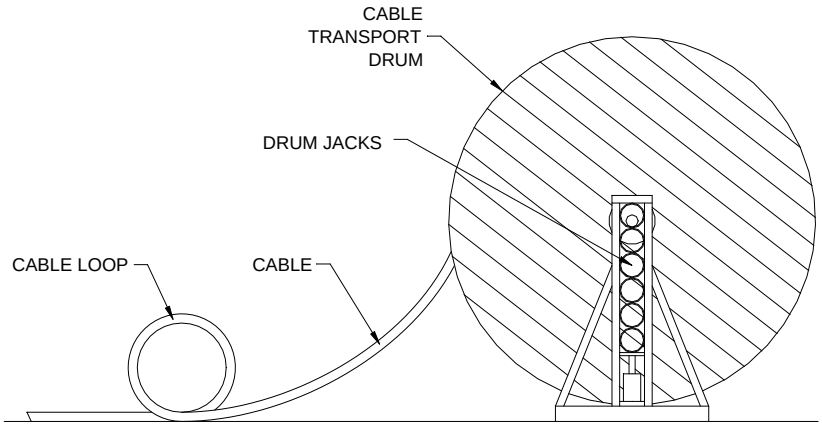


Figure 14 – Removal of Cable Twist Using Cable Coil Method

5.2 CABLE INSTALLATION

5.2.1 In order to ensure satisfactory performance of a reeling cable, the following installation procedure should be adopted. If the cable has been layed out according to the instructions in Section 5.1, then it should be rewound onto the storage drum before being transferred to the reeler using the method described below.

5.2.2 Manually rotate the reeler drum so that the opening in the inner ring is pointing vertically upwards.

5.2.3 With the machine located as close to the feed point as possible, feed the cable from the **top** of the cable transport drums directly onto the **top** of the reeler drum (refer to Figure 16).

5.2.4 Continue feeding it through the opening of the inner ring of the drum, out through the space between the spokes and then through the cable gland as shown in Figure 15

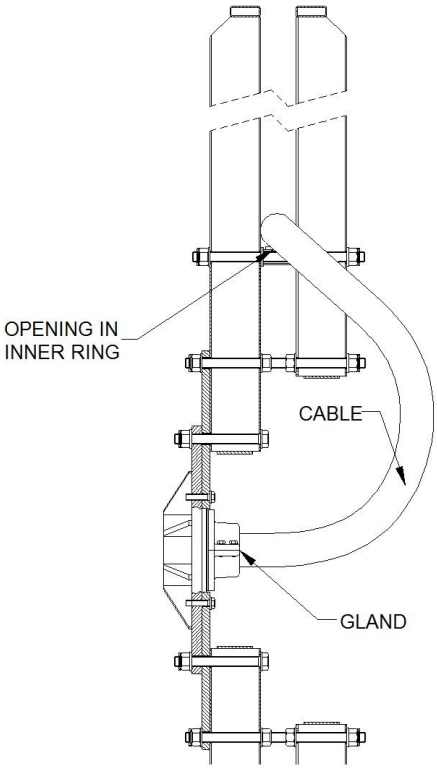


Figure 15 – Cable entry

5.2.5 Once the cable has been terminated to the sliprings in the collector, tighten the cable gland to exclude moisture and dirt.

5.2.6 Secure the cable onto the drum inner band with a number of heavy duty cable ties.

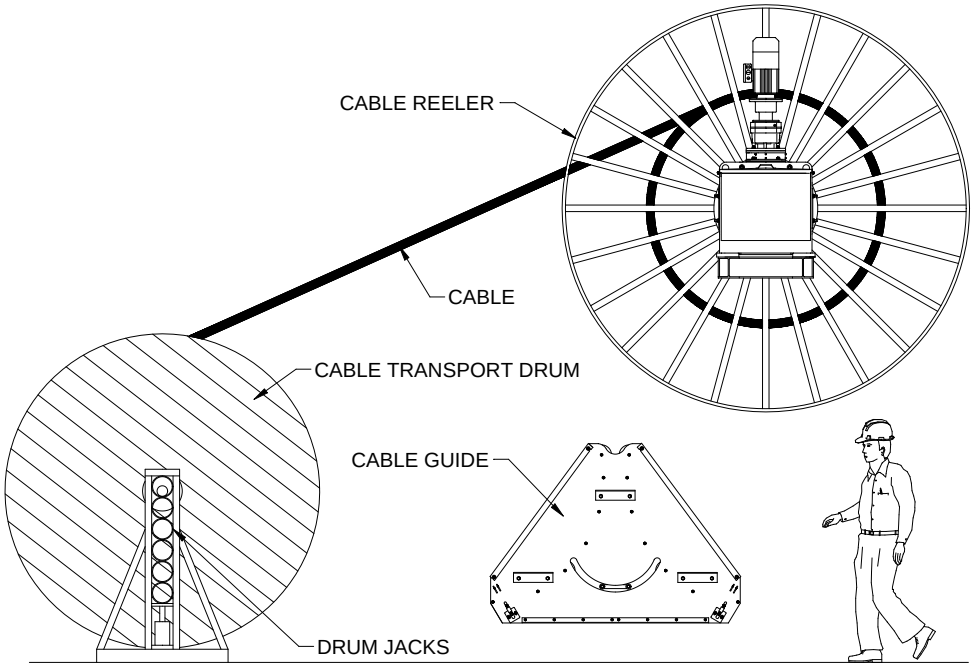


Figure 16 – Correct Method of Cable Installation

5.2.7 The drive unit needs to be energised to rotate the reel for cable fitting.

Ensure that an operator is available to immediately switch off the motor in the event of a problem.

5.2.8 The cable can now be wound onto the reel, and it is essential that this be done with the cable under some tension to ensure uniform tight layering on the reel. It is most important that the cable be wound onto the reel drum without any induced twist. Figure 16 shows the correct method of cable transfer from the storage drum. The use of any other method will induce twist in the cable.

5.2.9 The ends of the cable can now be passed down through the cable guide (if fitted) and manually pulled along the ground or cable tray to the nearby feedpoint.

5.2.10 Lay out the cable as it is pulled off the reel, either by manhandling with one person approximately every 5 m or by pulling over cable rollers. ***On no account should the cable be dragged over the ground, nor should a dozer, tractor or any other powered machine be used to pull or drag the cable***.

5.3 PERIODIC MAINTENANCE

5.3.1 The cables should be inspected after a short time in operation. Twisting may occur as the cable runs in due to the lay of the cable relaxing slightly under operation. This twisting will tend to move towards the feed point over time, and can eventually cause damage to the cable. If twisting is apparent at any time, the cable should be freed at the feed point with the reel at the furthest point of travel (ie. with all the cable paid out), and the twist should be removed by following the instructions in Section 5.1. Due to variations in operating conditions, it is impossible to state an exact number of cycles which need to be carried out, or period of time after which twisting will cease, and in some cases it may be several weeks. Periodic inspections are advisable after the machine has begun operation.

SECTION 6 – COLLECTOR

Ensure that all circuits are isolated by an authorised person prior to removing the cover of the collector housing. Electrical connections should only be carried out by qualified persons.

6.1 REELING CABLE CONNECTION

6.1.1 Remove the outer and inner covers of the collector housing.

6.1.2 If the reeling cable diameter, plus an allowance for a termination boot if fitted, is smaller than the main shaft ID specified in the *Torque Unit* subsection of SECTION 1, cable termination is easiest done before feeding the cable through the torque unit shaft into the collector.

6.1.3 Strip the outer sheath in order to expose sufficient core length to respect the core dimensions specified in the *Reeling Cable Terminations* subsection of SECTION 1.

6.1.4 Terminate each core with an appropriately sized ring terminal to suit the ring side fixtures specified in the *Collector* subsection of SECTION 1. Apply an appropriate medium voltage termination kit to each core, carefully following the manufacturer's instructions.

6.1.5 Remove the E90 cable gland that is mounted on the drum side of the torque unit. Feed the cable through the torque unit shaft and the internal cable support clamp (removing the appropriate number of concentric sealing rings to suit the cable diameter) and into the collector.

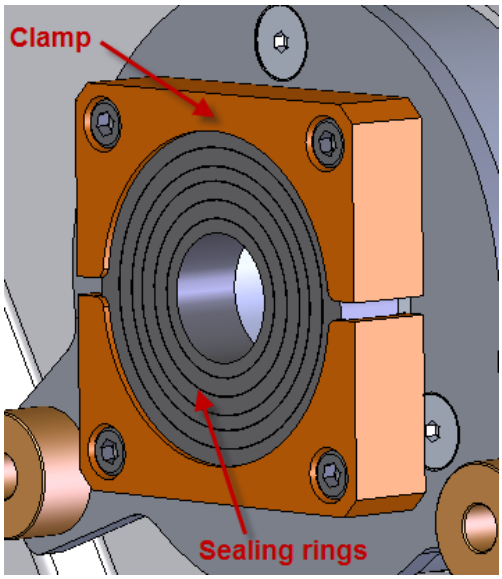


Figure 17 – Internal cable support clamp

6.1.6 Connect each core to its corresponding slipring, ensuring that there is sufficient clearance between individual cores, and also between each core and live metal parts.

6.1.7 Re-fit the E90 cable gland in accordance with the following instructions:

- Using electrical tape, fix one end of the sealing tape to the cable in the appropriate position.
- Tightly wrap the sealing tape around the cable until the overall diameter is approximately 95-96mm.
- Cut off the remaining tape, and fix the loose end with 3-4 wraps of electrical tape.

- Fit the 2 clamp halves and tighten the 4 bolts. Ensure that the cable is rigidly clamped. Apply additional electrical tape as required to increase the diameter.
- Fit the gland assembly to the torque unit shaft, ensuring that the cable enters the gland perpendicularly. Any bending of the cable at the gland will reduce the effectiveness of the sealing.

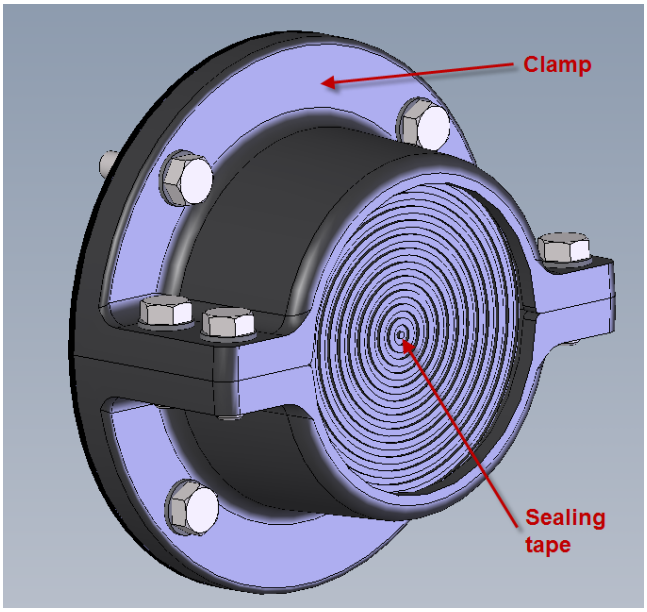


Figure 18 – E90 gland

6.1.8 In cases where there are either fibre optic or signal cores in the cable, these should be tie-wrapped to the support rod once they have been fed into the second compartment. Refer to Figure 19 below. Do not feed the fibre optic cables through until the FORA start up procedure has been carried out.

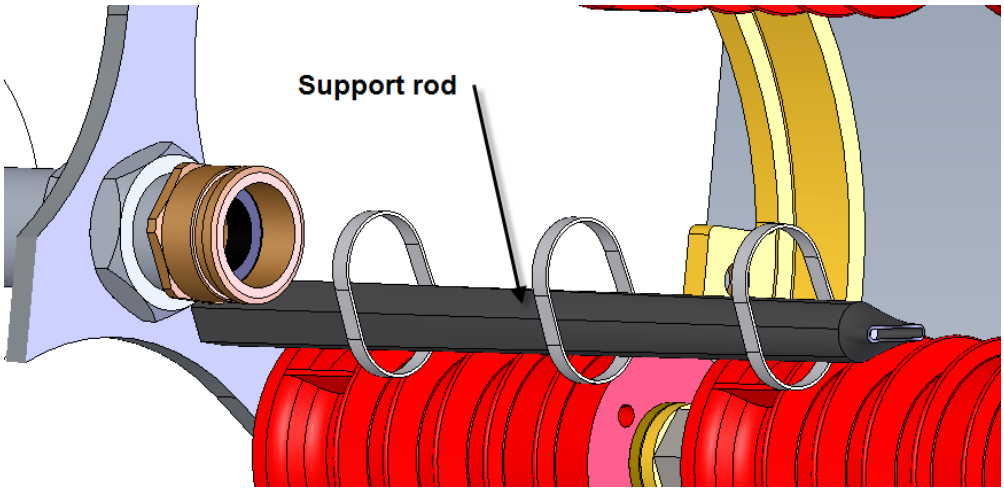


Figure 19 – Support rod for FO and signal cables

6.2 FIXED CABLE CONNECTION

6.2.1 Feed the fixed cables through the glands mounted on the bottom of the collector housing, ensuring that they are well supported so as not to put any bending force on the gland.

6.2.2 Pull through sufficient core length to neatly arrive at the brushgear connection points.

6.2.3 Terminate each core with an appropriately sized ring terminal to suit the brush side fixtures specified in the *Collector* subsection of SECTION 1. Apply an appropriate MV termination kit to each core.

6.2.4 Ensure that the cores are neatly arranged and properly supported inside the collector housing, and ensure that the cores are sufficiently clear of rotating and live metal parts.

6.2.5 Tighten the cable glands.

6.3 ANTI-CONDENSATION HEATER

6.3.1 Already connected to Junction box.

6.3.2 The heater should remain energised at all times in order to prevent the formation of condensation.

6.4 FINAL CHECKS

6.4.1 Before closing the housing and applying current to the collector for the first time, ensure that:

- all the carbon brushes are correctly aligned in the slipring grooves.
- all the brush arms are correctly aligned with the sliprings.
- all the electrical connections are correctly tightened.
- all cores are sufficiently clear of each other and rotating and live metal parts.

6.5 PERIODIC MAINTENANCE

6.5.1 The sliprings and brushgear should be inspected to ensure good contact and alignment is maintained between the sliprings and brushes. Worn brushes should be replaced.

6.5.2 Verify tightness of all electrical connections.

6.5.3 If necessary, the sliprings can be lightly cleaned with glass paper, and then wiped clean with a cloth dampened with methylated spirit.

6.5.4 Vacuum the carbon dust from the insulators, cable terminations and floor of the collector housing. Alternatively, use a brush or a lint free cloth. ***Never use water or solvents for cleaning***.

6.5.5 The housing cover seal should be checked and replaced if necessary.

SECTION 7 - FIBRE OPTIC ROTARY ACCUMULATOR

Ensure that all circuits are isolated by an authorised person prior to removing the cover of the collector housing. Electrical connections should only be carried out by qualified persons.

7.1 GENERAL

7.1.1 The Fibre Optic Rotary Accumulator (FORA) is a rugged and simple device that facilitates the uninterrupted transmission of multi-mode or single-mode light signals across a rotating interface.

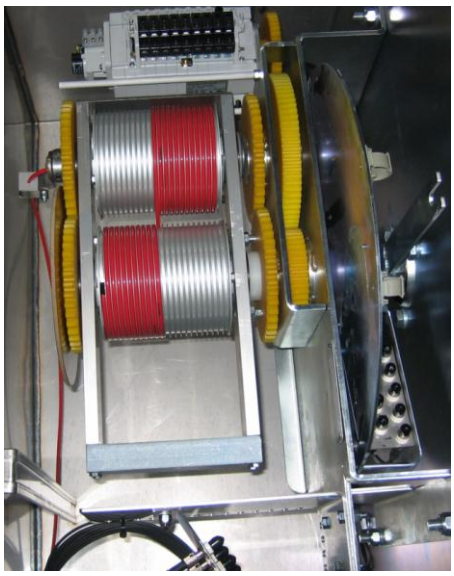


Figure 20 - FORA

7.1.2 The FORA works on a twin drum principal where a special fibre optic cable is reeled in a highly controlled manner between a coaxial drum and a planetary drum. The coaxial drum rotates at the same speed as the reel drum, and the planetary drum remains fixed relative to its centre axis while it circumnavigates the coaxial drum at half a revolution per revolution of the coaxial drum. Both drums are grooved to ensure the cable layers perfectly, and small rotary coil springs inside the drums maintain a constant cable tension between the two drums.

7.1.3 The technical specification of the supplied FORA can be found in the *FO Rotary Accumulator* subsection of SECTION 1.

The FORA is a limited turn device. Turning the FORA beyond its limits will result in irreparable cable damage, and cable replacement will be required.

7.2 START-UP PROCEDURE

7.2.1 In order to prevent damage to the fibre optic cable during transport, and also during installation of the reeling cable, the FORA is mechanically disconnected from the main drive shaft. The reeling drum and the FORA can therefore be rotated independently of each other until the drive pin has been fitted. It is important that the drive pin is not fitted until all the steps of this start-up procedure have been carried out.

7.2.2 Ensure that the reeling cable has been correctly installed, that the fibre optic cores have not yet been fed through the FORA drive shaft, and that these cores are adequately supported inside the main collector assembly.

7.2.3 Ensure that the machine is capable of movement.

7.2.4 Travel the machine towards the feedpoint and note the rotation direction of the reeling drum when viewed from the collector end.

7.2.5 Position the machine so that the roller guide is directly above the feedpoint (ie. in the drum full position), and set the FORA as follows:

- If the reeling drum was turning clockwise during recovery, manually rotate the FORA in the clockwise direction until the planetary drum is nearly full, leaving one turn on the coaxial drum as shown in Figure 21.

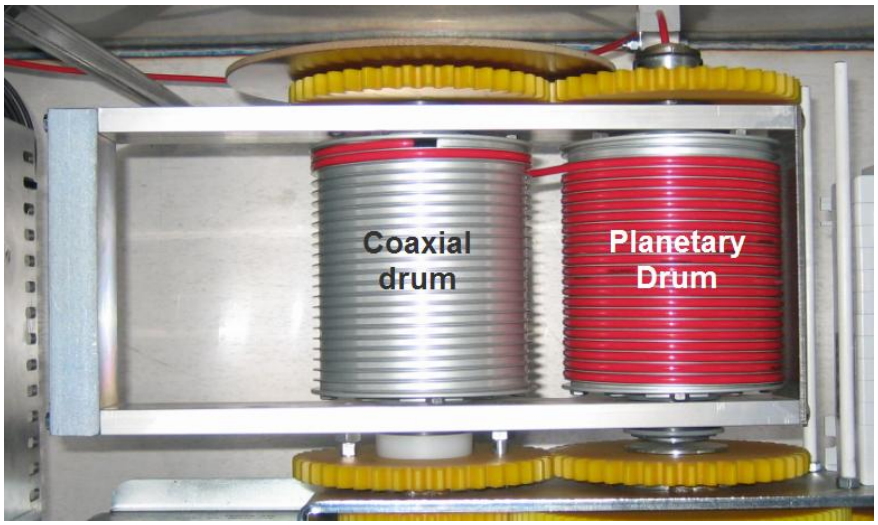


Figure 21 – FORA setting at reel full position for clockwise recovery

- If the reeling drum was turning anti-clockwise during recovery, manually rotate the FORA in the anti-clockwise direction until the coaxial drum is nearly full, leaving one turn on the planetary drum as shown in Figure 22.



Figure 22 – FORA setting at reel full position for anti-clockwise recovery

7.2.6 Manually unwind a small amount of cable from the reeling drum until the FORA torque arm points vertically downwards as shown in Figure 23.

7.2.7 Manually rotate the FORA in the same direction that the reeling drum went until the reeling cable FO connection points are facing upwards as shown in Figure 23.

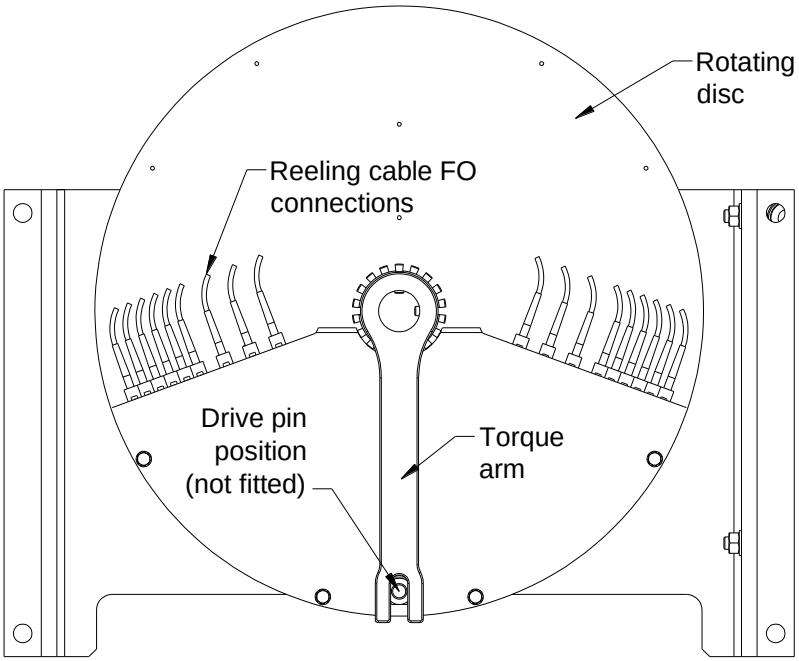


Figure 23 – Disc and torque arm positions

7.2.8 Isolate and lock-out the main power to the machine so that the collector assembly can be accessed safely.

7.2.9 Remove the covers to the collector assembly, and feed about 60cm of the fibre optic bundle through the cable gland into the FORA drive shaft and on to the rotating disc of the FORA. Tighten the gland.

7.2.10 Split the FO bundle and terminate the fibres with the appropriate connectors. Connect the terminated fibres to the appropriate bulkhead adaptors mounted on the rotating disc.

7.2.11 Neatly coil and fix excess fibre cable as shown in Figure 24.

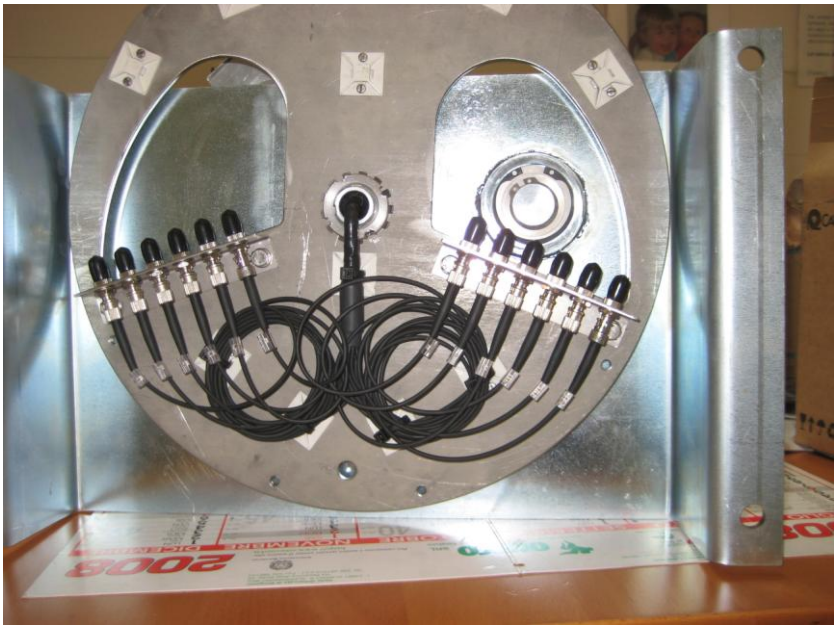


Figure 24 – FO cable storage

7.2.12 Manually wind or unwind a small amount of cable from the reeling drum until the FORA torque arm points vertically upwards as shown in Figure 25. ***At exactly the same time***, manually rotate the FORA in the same direction in order to prevent twisting of the fibre bundle.

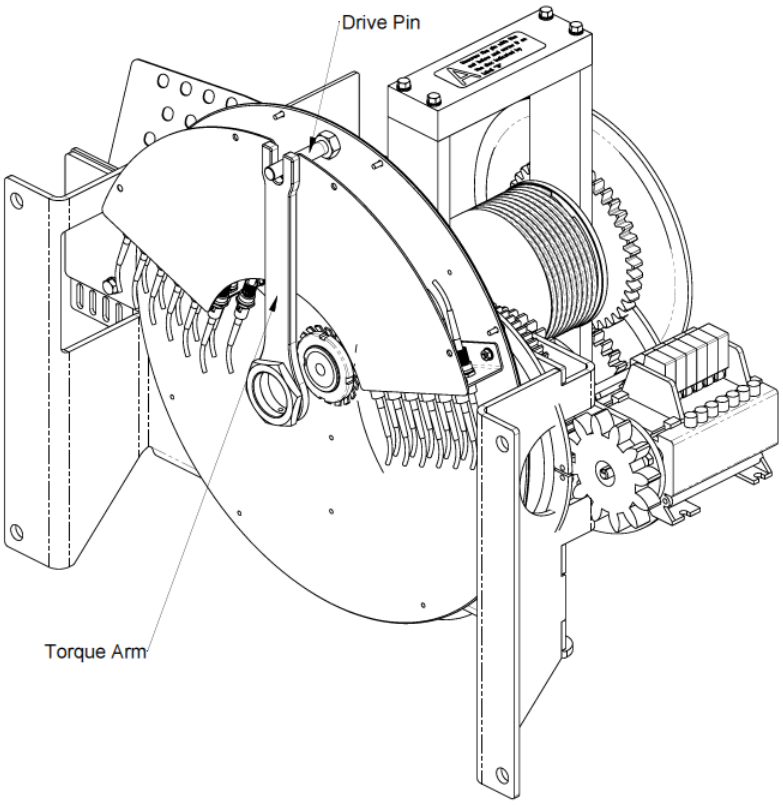


Figure 25 – Drive pin mounting

7.2.13 Remove the drive pin from the storage bag mounted on the side wall of the housing, and fit same to the rotating disc as shown in Figure 25. Ensure that there is sufficient clearance between the torque arm slot and the pin to give adequate misalignment allowance. Also ensure that both the drive pin and torque arm are well fixed.

7.2.14 Refit the internal cover on the collector assembly, and apply power to the machine.

7.2.15 Travel the machine away from the feedpoint, and ensure that the FORA cable is being transferred from the full FORA drum to the empty FORA drum. ***Immediately stop the machine*** if this is not the case in order to avoid damaging the FORA cable. The reeling cable fibres and drive pin will need to be removed, and the procedure from paragraph 7.2.4 will need to be repeated.

7.2.16 Travel the machine towards the furthest position from the feedpoint, observing the FORA in the process. Ensure that there is at least one turn remaining on the empty FORA drum when the endpoint is reached. Contact the manufacturer if this is not the case.

7.2.17 Replace all covers.

7.2.18 Note that before making any modifications to the cable reeling system that may result in the FORA going out of sync, such as:

- changing the reeling cable length;
- repositioning the cable feedpoint;
- replacing the reeling cable;
- disconnecting the collector assembly or FORA from the main drive;

the reeling cable fibres and drive pin will need to be removed, and the procedure from paragraph 7.2.4 will need to be repeated once the work has been completed.

7.2.19 If it is not possible to position the machine over the feedpoint when setting up the FORA, it will be necessary to calculate the number of turns required on the relevant FORA drum to

allow for the number of active cable turns actually on the reeling drum. The following procedure needs to be followed:

- Count the total number of turns of cable on the reeling drum, and subtract the number of dead turns as indicated in the *Duty* subsection of SECTION 1 to give the number of active turns N on the drum.
- If the reeling drum rotates clockwise when the machine moves towards the feedpoint, manually rotate the FORA until the number of turns on the planetary drum corresponds to N/2+1 turns.
- If the reeling drum rotates anti-clockwise when the machine moves towards the feedpoint, manually rotate the FORA until the number of turns on the coaxial drum corresponds to N/2+1 turns.
- Continue following the instructions from paragraph 7.2.6.

7.3 PERIODIC MAINTENANCE

7.3.1 The FORA is maintenance free, however, it is prudent to periodically check that all fasteners and fittings are properly tightened, particularly those holding the torque arm and the drive pin.

7.3.2 Ensure that the FORA cable is not damaged in any way. Contact the manufacturer for replacement instructions if necessary.

SECTION 8 - STROMAG ROTARY LIMIT SWITCH

The limit switch is located inside the collector enclosure. Ensure that all circuits are isolated by an authorised person prior to removing any covers. Electrical connections should only be carried out by qualified persons.

8.1 GENERAL

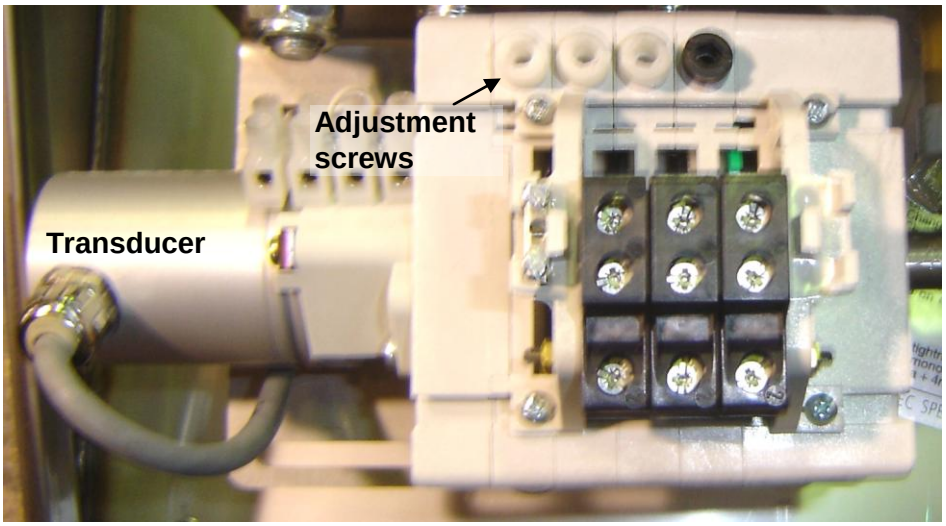


Figure 26 – Stromag rotary limit switch (Example)

8.1.1 The geared rotary limit switch (GRLS) consists of a number of cam activated switches used for drum full/drum empty detection. These cams rotate via a gear reduction connected to the input shaft, with one turn of the input shaft resulting in a partial rotation of the cams that is proportional to this gear reduction.

8.1.2 Drum full detection is carried out using two switches in order to minimise the effects of hysteresis. The resulting closed contacts are used to bypass the undertension switch as the machine moves over the feedpoint.

8.1.3 Additional cam operated switches are available for other functions if required. The transducer mounted on the end of the switch is used to provide a 4-20mA position signal to the reeler control system.

8.1.4 The technical specification of this limit switch can be found in the *Limit Switch* subsection of SECTION 1.

8.2 ELECTRICAL CONNECTION

8.2.1 Already connected to the Junction box.

8.3 SETTING

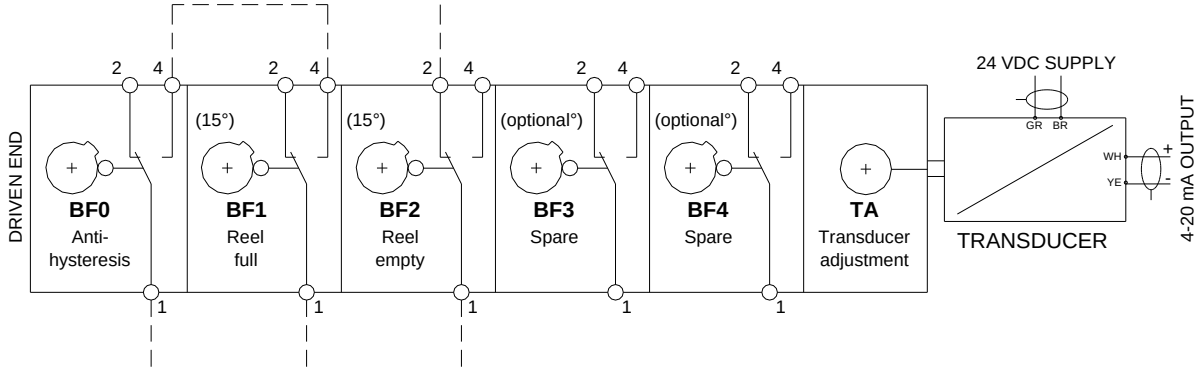


Figure 27 – Rotary limit switch contacts

8.3.1 Travel the machine towards the feedpoint and note the rotation direction of the limit switch shaft when viewed from the rear.

8.3.2 Position the machine so that the roller guide is about 2 m from the feedpoint, and set the anti-hysteresis BF0 and drum full BF1 switches as follows:

- If the GRLS shaft was turning clockwise during recovery, turn the black adjustment screw (BF0) in the clockwise direction until the first cam just activates the switch (there will be an audible click). Turn the first white adjustment screw (BF1) in the same direction until the first cam just activates the switch.
- If the GRLS shaft was turning anti-clockwise during recovery, turn the black adjustment screw (BF0) in the anti-clockwise direction until the first cam just activates the switch (there will be an audible click). Turn the first white adjustment screw (BF1) in the same direction until the first cam just activates the switch.
- Ensure that the incoming wires on switches 1 and 2 are connected to the normally open terminals marked 1 and 4.

8.3.3 Set the transducer as follows:

- If the GRLS shaft was turning clockwise during recovery, turn the last adjustment screw in the clockwise direction until the transducer output is 4.2mA.
- If the GRLS shaft was turning anti-clockwise during recovery, turn the last adjustment screw in the anti-clockwise direction until the transducer output is 19.8mA.

8.3.4 Travel the machine over the feedpoint. Note that as the machine passes the feed point, the rotation direction of the cams will change. Ensure that the cams do not over-run the switch before this direction change occurs. Ensure that no fault occurs as the cable/hose passes through the undertension sensor.

8.3.5 Repeat step 8.3.4 from the other side of the feedpoint.

8.3.6 Move the machine to the longest extremity of travel from the feedpoint. Note that if the feedpoint is not in the centre of travel, or if the reeler is not positioned in the centre of the machine, the amount of cable/hose payed out will differ in each travel direction. Set the drum empty BF2 switch to trip at this point by following the instructions below:

- If the GRLS shaft was turning clockwise during recovery, turn the second white adjustment screw (BF2) in the anti-clockwise direction until the third cam just activates the switch (there will be an audible click).
- If the GRLS shaft was turning anti-clockwise during recovery, turn the second white adjustment screw (BF2) in the clockwise direction until the third cam just activates the switch (there will be an audible click).

- Ensure that the incoming wires on switch 3 are connected to the normally closed terminals marked 1 and 2.

8.3.7 Travel the machine back towards the feedpoint, then back to the end of travel. Ensure that the drum empty switch does not trip within the normal travel limits. Travel the machine to the other end of travel to prove the operation.

8.3.8 Note that if the anti-hysteresis adjustment screw is turned after the switches and potentiometer have been set, it will be necessary to re-set the switches and potentiometer.

8.3.9 The spare switches BF3 and BF4, if used, should be set using the procedures described above.

8.4 PERIODIC MAINTENANCE



Ensure that all live electrical circuits have been isolated before performing any maintenance inside the collector housing.

8.4.1 Check that the wires are properly connected to the terminals, and that the terminal screws are correctly tightened.

SECTION 9 - TORQUE UNIT

9.1 GENERAL

9.1.1 The torque unit is the primary support for all the components of the cable-reel. It reduces the motor speed to the value that is required for the cable reel, at the same time multiplying the torque output of the motor.

9.1.2 The 660DM torque unit has one primary conical gear driving second and third stage spur gears.

9.1.3 An additional pre-reducer can be fitted to provide additional speed reduction/torque multiplication.

9.1.4 Up to two motors can be installed.

9.1.5 All gears are immersed in oil and the gearboxes require very little maintenance.

9.2 PERIODIC MAINTENANCE

9.2.1 After the first 500 hours of operation, and every 10,000 hours thereafter, drain the oil from the torque unit and replace it with the type specified (or approved equivalent) in the *Torque Unit* subsection of SECTION 1.

9.2.2 Between oil change intervals, perform occasional checks on the oil level using the level gauge and top up as necessary with the oil specified.

9.2.3 After every 10,000 hours of operation, replace the grease in the pre-reducer/s with the type specified (or approved equivalent having Brookfield viscosity of 4000) in the *Pre-reducer* subsection of SECTION 1.

9.2.4 Check for oil leakage or unusual noises, and rectify as necessary.

9.2.5 Check for water ingress and replace oil if necessary.

9.2.6 Ensure that the breather is clean and not blocked.

SECTION 10 - MOTOR



Ensure that all circuits are isolated by an authorised person prior to removing any covers. Electrical connections and motor maintenance should only be carried out by qualified persons.

10.1 ELECTRICAL CONNECTION

10.1.1 Ensure nameplate data matches the available power supply.

10.1.2 Remove the cover of the motor terminal box, ensuring beforehand that all live electrical circuits have been isolated.

10.1.3 Measure the winding resistances between each phase and earth, and between individual phases, and ensure that they are greater than 5 MΩ.

10.1.4 Motor is supplied pre-cabled to junction box.

10.1.5 Prior to closing the terminal boxes, ensure that all connections have been properly tightened.

10.2 PERIODIC MAINTENANCE

10.2.1 Clean cooling air passages using low-pressure water spray.

10.2.2 Ensure that all electrical connections are properly tightened.

10.2.3 Ensure that all glands are correctly tightened and sealing correctly.

10.2.4 Ensure that the encoder (if fitted) is properly fixed to the motor shaft.

10.2.5 Ensure that the brake (if fitted) is functioning correctly.

10.2.6 The ball bearings are sealed for life, and therefore maintenance free.

SECTION 11 - BRAKE

Ensure that all circuits are isolated by an authorised person prior to removing any covers. Electrical connections and brake maintenance should only be carried out by qualified persons.

Operation of the manual brake release (if supplied) can result in uncontrolled unwinding of the cable/hose on the reel. Ensure that this does not put personnel at risk.

11.1 GENERAL

11.1.1 The fail-safe brake is mounted on the motor shaft and prevents unwinding of the reel when the motor is not energized. Its DC coil is powered from the AC supply via a rectifier mounted in the motor terminal box. The torque setting of the brake is set such that if the cable were to be pulled off the reel with the motors de-energised, the resulting tension would not damage the cable.

11.2 MANUAL RELEASE

11.2.1 The optional manual brake release is activated by a lever which protrudes from the fan cowling. This lever can be removed if required by unscrewing it.

11.3 BRAKE ADJUSTMENT

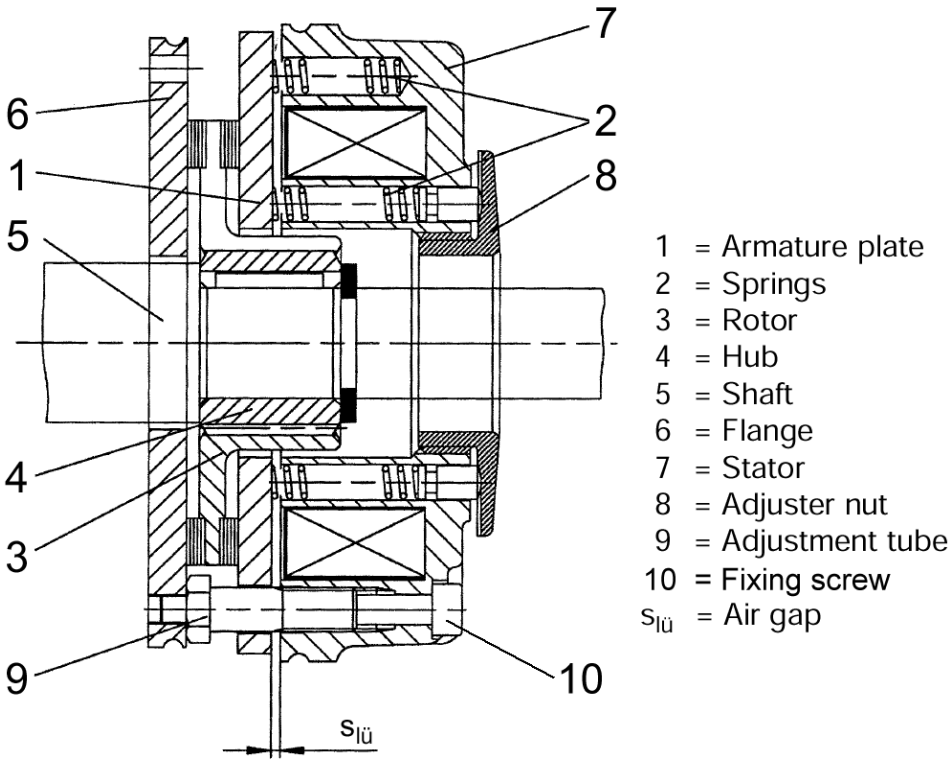


Figure 28 – Brake

11.3.1 In order to ensure that the correct braking torque is maintained, it is important to occasionally adjust the air gap $s_{lü}$ so that it remains in the range 0.2 to 0.5 mm. This can be done as follows:

- Isolate the motor and brake from the supply.
- Clean the motor with a low pressure water spray.

- Remove the cowlings, encoder (if fitted) and fan.
- Loosen the 3 fixing screws (10).
- Turn the adjustment tubes (9) in the anti-clockwise direction until the air gap $s_{lü}$ is 0.2 mm all around.
- Re-tighten the fixing screws.
- Refit the fan, encoder (if fitted) and cowlings.

11.3.2 Measure the thickness of the rotor, and replace it if the dimension is less than 7.5 mm.

SECTION 12 - DRIVE

12.1 DELIVERY

12.1.1 The NCC cable reel drive can either be delivered as a complete mounted panel with all drive and auxiliary logic components mounted and pre-wired, IP22 to be mounted in the clients cabin or optionally delivery can be made in a housing IP54 for indoor mounting or in special housings for outdoor mounting. The braking resistor is delivered separately and to be mounted and cabled by the client.

See the *Drive* subsection of SECTION 1.

12.2 INSTALLATION

12.2.1 The drive and it's components are to be installed in a ventilated ambient suitable for electric equipment. The braking resistor can be mounted in the same area next to the drive housing or elsewhere. If drive cannot be mounted in a controlled ambient a housing with appropriate heating and ventilation is needed.

12.2.2 Standard drive is specified for 0 deg C to 40.

12.3 INTERFACE AND SIGNALS

12.3.1 Refer to documents in SECTION 15.

SECTION 13 - RECOMMENDED SPARES

Drive System Spares

Part Number	Part Description	Qty	
		Inst'd	Spares
S01-1286F-663	Torque unit 660 for 2 motors flanged	1	1
SC-121216-MC1	8 pole, 2kW 112 frame flange mount motor complete with Klixon brake, encoder and heater	1	1
SC-121216-MC2	8 pole, 2kW 112 frame flange mount motor complete with Klixon and heater	1	1
S64-20075-001	Oil synthetic Shell Spirax ASX 75W-90	36ℓ	80ℓ

Collector Assembly Spares

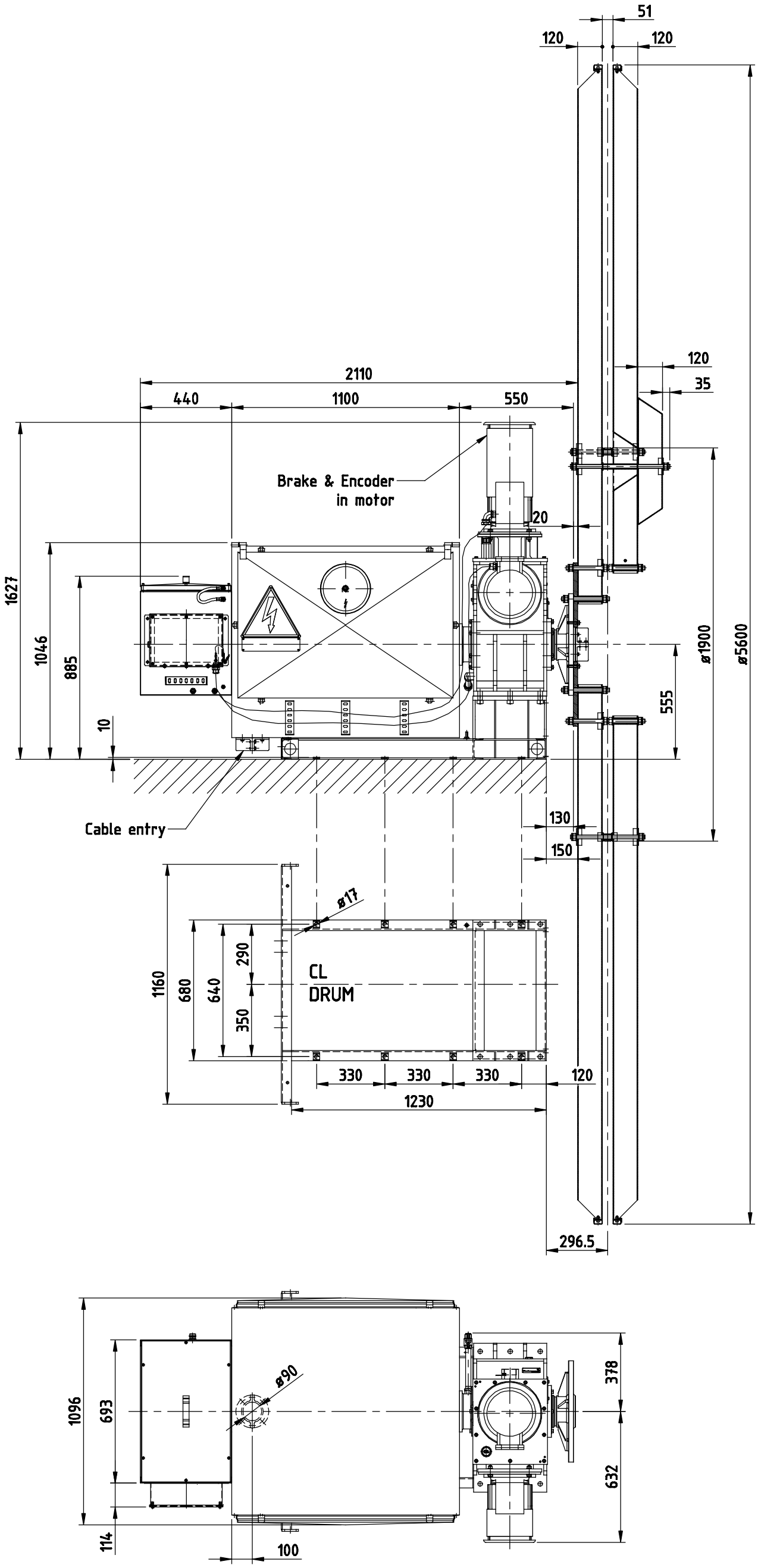
Part Number	Part Description	Qty	
		Inst'd	Spares
S22-H0015-F01	F460/4517 ring assembly 15KV prep new GFO	1	1
SR2-05417-000	Slipring 450 microgrooved for BG brushes	4	4
SR2-S0077-003	Brushgear BG300/450	4	4
SR2-00995-000	Insulator IGn 15-700 H= 175 mm	18	10
S22-40406-5S0	FO joint 40 turns, 6 ways, ST 50/125 um	1	1
S99-SG050-402	Limit switch Stromag 1:48/4 IP20+ transducer	1	1

Guide Assembly Spares

Part Number	Part Description	Qty	
		Inst'd	Spares
SR6-S0146-000	Roller DB 120	18	9
SR6-S0325-002	Overtension kit	1	1

SECTION 14 - MECHANICAL DOCUMENTATION

Drawing No.	Description
SC-121216-IS1	Reel assembly
SC-120121-JB1	Installation J.box
STD-AD-034	Cable guide
STD-AD-083	Typical guide installation
STD-BE-004	Box + Drive
STD-BE-001	Braking resistor + Rain cover



TITLE	CTC Cable Reel 660.DM.NCC.(2x2kW).8 M1956-51GI K460/4517/6FO/FC/R		
CLIENT	ZPMC	PROJECT	Srilanka

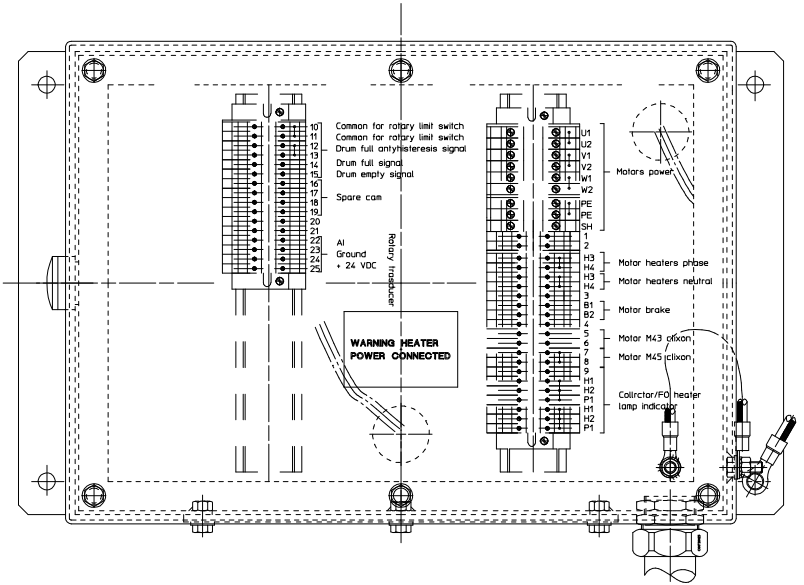
No.	REVISION	BY	DATE
B	ADDED DIMENSION E POSITION CABLE ENTRY	AR	05.07.12
A	ISSUED FOR APPROVAL	AR	02.07.12

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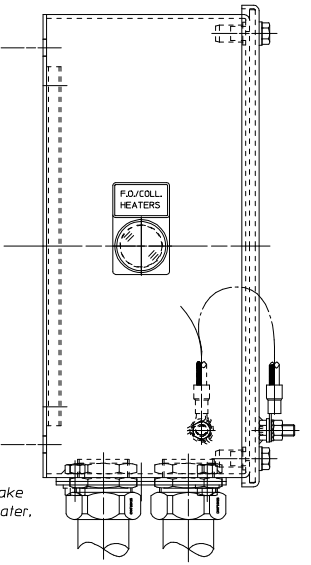
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DATE	02.07.12		
CHKD		APPD	
SCALE	1:22 ON A3		
MAT'L			
WGHT	2820 kg		
DWG No	SC-121216-IS1		B

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ALL DIMENSIONS IN MM UNLESS OTHERWISE NOTED



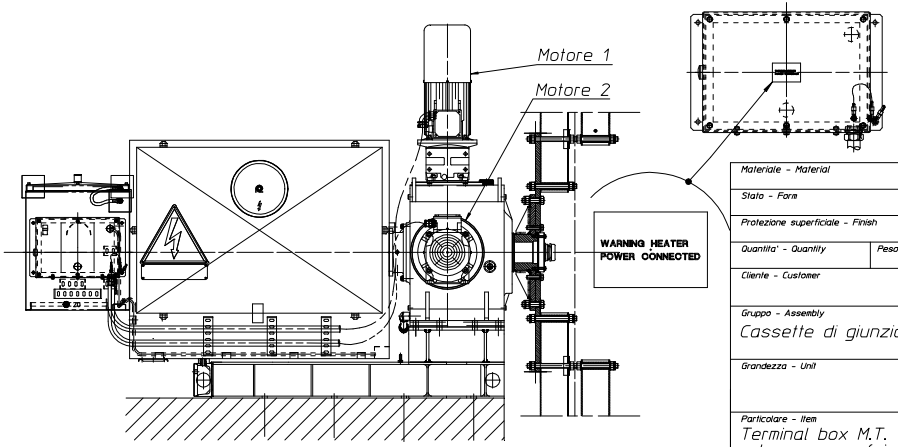
FISSARE CON "LOCTITE"



Power,brake
Clixon,Heater,
Motor 1

Power,
Clixon,Heater,
Motor 2

FISSARE CON "LOCTITE"



Materiale - Material				
Stato - Form				
Protezione superficiale - Finish				
Quantita' - Quantity	Peso - Weight Kg 6,6 ~			
Cliente - Customer				
Gruppo - Assembly: Cassette di giunzione				
Grandezza - Unit				
Particolare - Item: Terminal box M.T. + lamp. rossa fzionam scald				

Ind.	Modifiche	Data	Sigla
A	Emessa per fabbricazione	16-02-12	SP

CAVOTEC

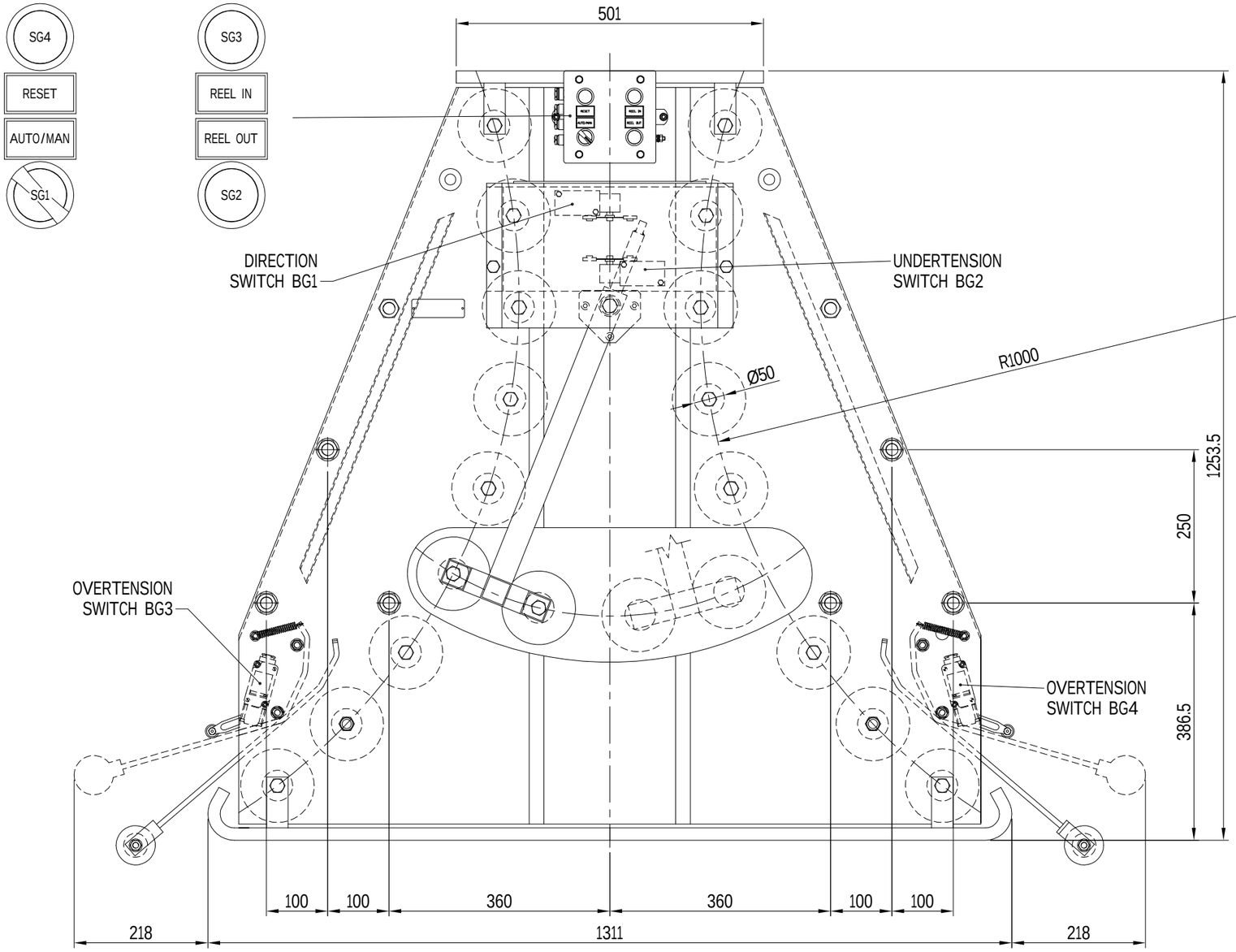
CAVOTEC SPECIMAS S.p.A.
A MANUFACTURING COMPANY OF THE CAVOTEC GROUP
I - 20054 NOVA MILANESE (MI) - VIA GALVANI, 1
TEL. (0362) 451111 - FAX. (0362) 41233

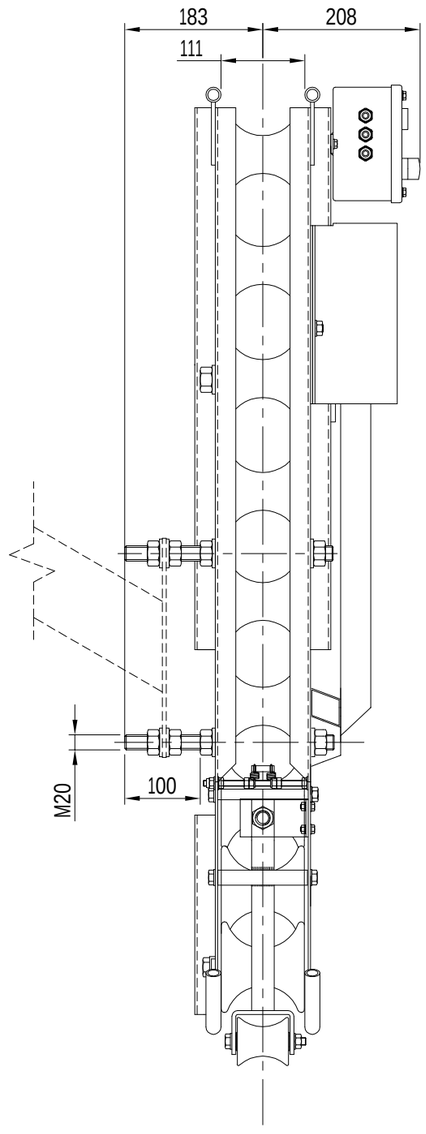
Disegnato: S.P.	Data - Date: 16-02-2012
Controllato	Scala - Scale: 1 : 2
Approvata	Codice - Code

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Sostituito dal N° | Sostituisce il N°

N° Disegno - Drawing Nr: SC-120121-JB1





MATERIAL - HOT-DIP GALVANISED MILD STEEL (ALSO AVAILABLE IN AISI 304 & AISI 316)

MAXIMUM CABLE DIAMETER (OD) - 95 mm

ROLLER TYPE - DB120
ROLLER PART NO. - SR6-S0146-000

LIMIT SWITCHES - TELEMECANIQUE TYPE XC2-J, IP65 OR OPTIONAL TELEMECANIQUE TYPE XCK-J, IP66

REFER TO STD-SE-002 FOR ELECTRICAL SCHEMATIC

No.	REVISION	BY	DATE
A	ISSUED FOR INFORMATION	PR	18/7/05

TITLE: M9P/OT/W CABLE GUIDE WITH CONTROLS - 1000 RADIUS
MAXIMUM CABLE DIAMETER 95 mm

CLIENT

PROJECT

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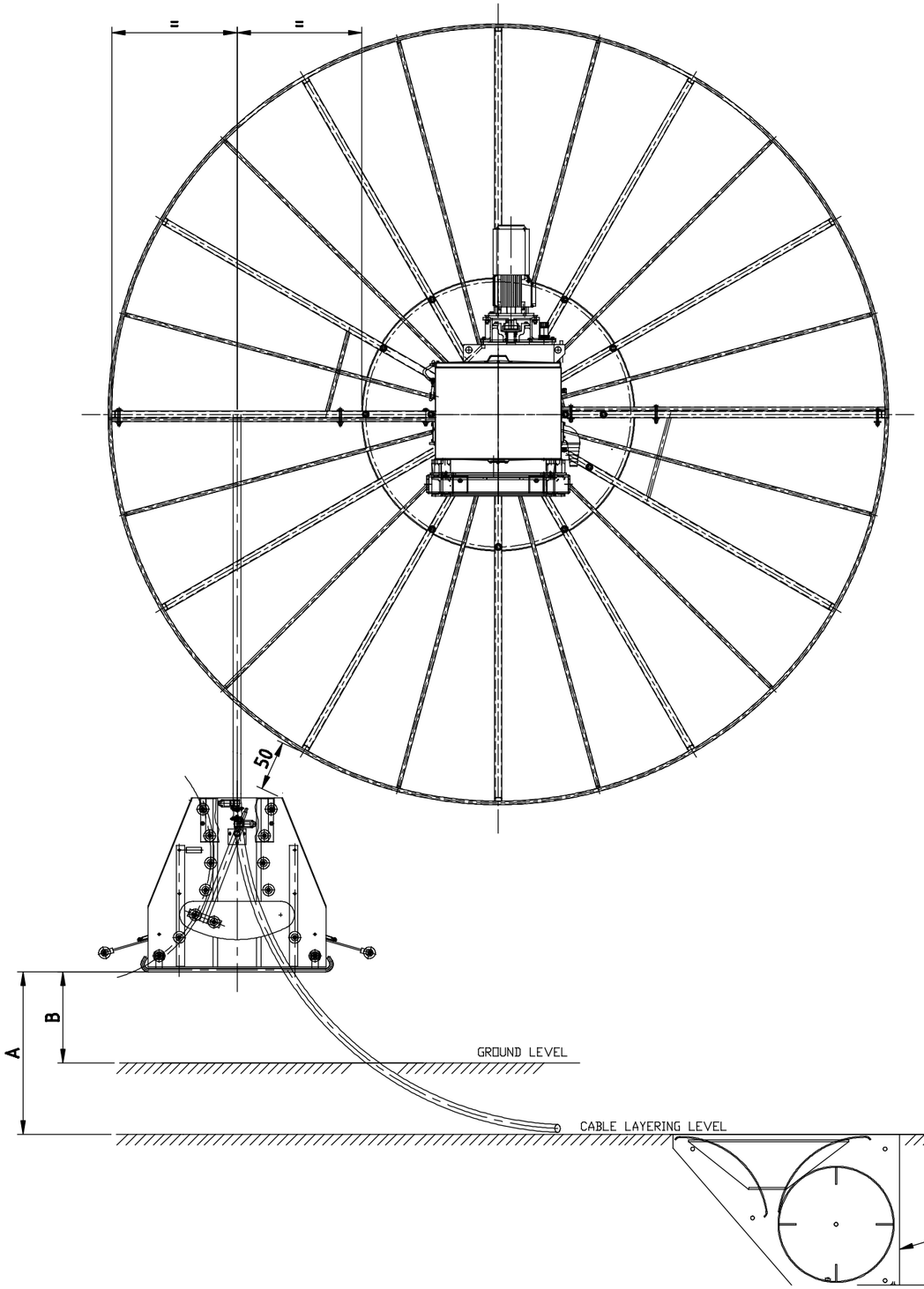
ALL DIMENSIONS IN MM UNLESS OTHERWISE NOTED

DRWN PR
DATE 18/7/05
MAT'L
WGHT 95 kg
SCALE 1:8 ON A3

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DWG No STD-AD-034 A

ALL DIMENSIONS IN MM UNLESS OTHERWISE NOTED



A=1m or 15 x cable OD
whichever is greater, contact Cavotec.
if lower mounting is required.

B=270±10 mm when PANZERBELT is used

No.	REVISION	BY	DATE
B	CHANGED NOTES	GJM	13.03.12
A	ISSUED FOR INFORMATION	GJM	02.02.12

TITLE: Typical cable guide installation

CLIENT

PROJECT

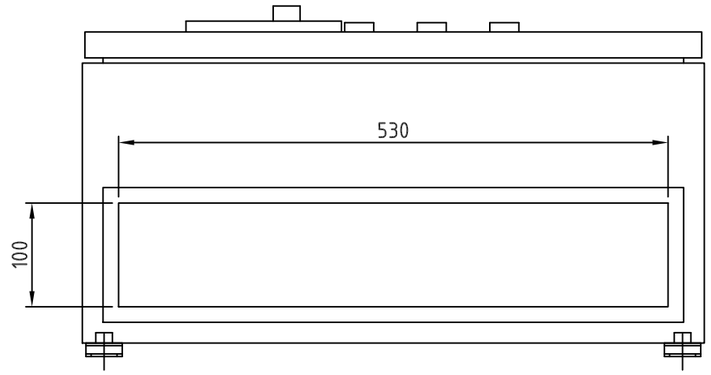
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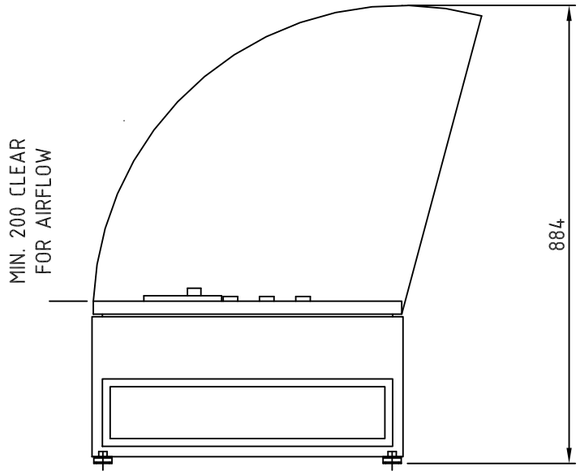
DRWN GJM
DATE 02.02.12
CHKD
APPD

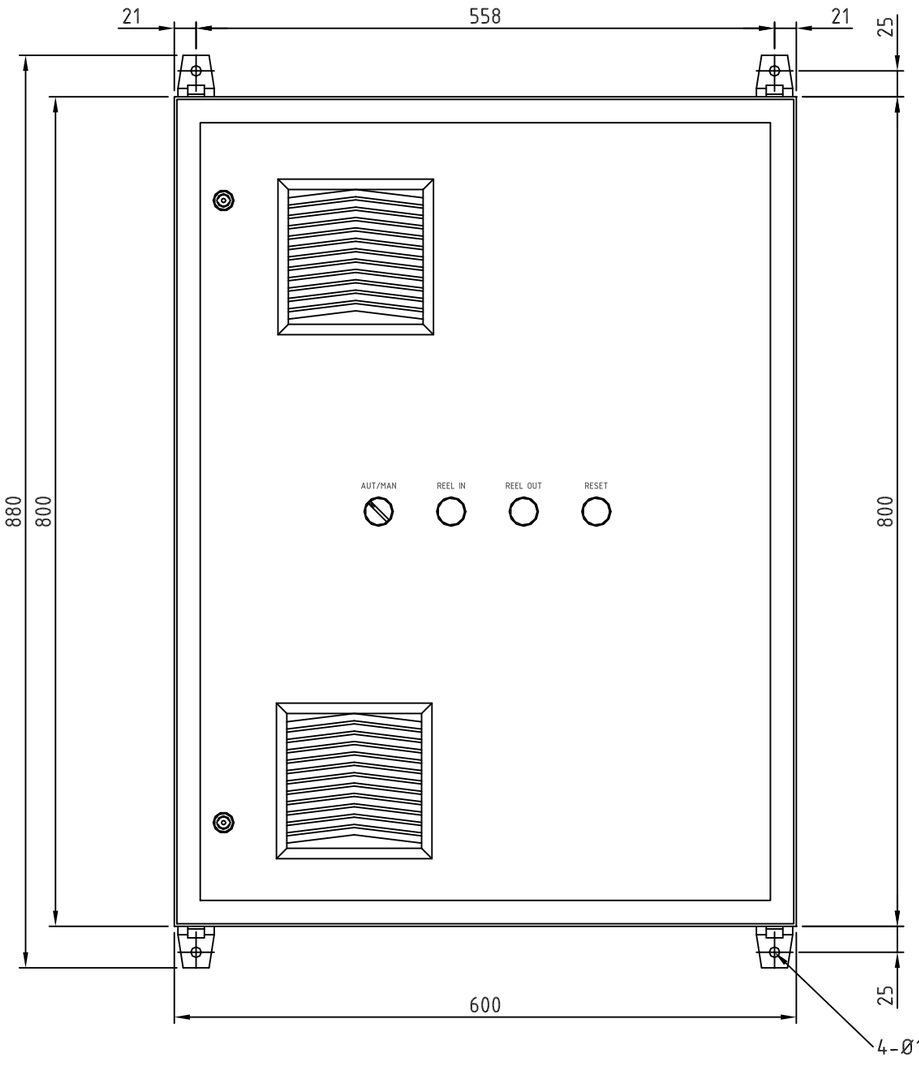
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MAT'L
WGHT

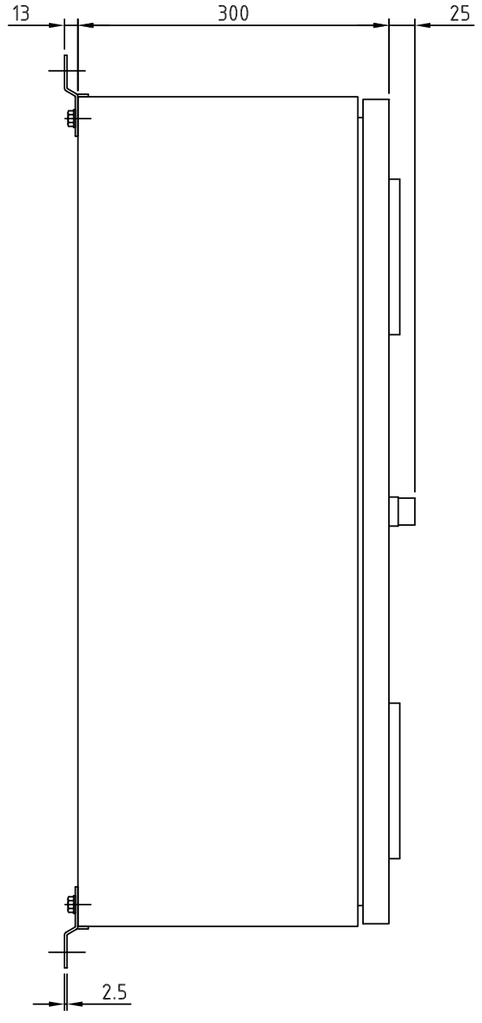
CAVOTEC
Connecting mobile equipment

DWG No STD-AD-083 B









Notes

Protection rating - IP54

To be installed in a climate controlled environment eg. drive cabin or electrical house.

Colour - RAL 7035

Box part nos.

S01-S0518-000 (door hinged on right as drawn)
S01-S0523-000 (door hinged on left)

No.	REVISION	BY	DATE
D	COLOUR WAS 7032	DR	24/09/10
C	VENTS MOVED TO FRONT	PR	1/11/04
B	REVISED CLEARANCE NOTE - WAS 100 mm	PR	16/6/04
A	ISSUED FOR INFORMATION	PR	26/4/04

TITLE: CTC REEL CONTROL CENTRE
BOX PAINTED 600x800x300

CLIENT

PROJECT

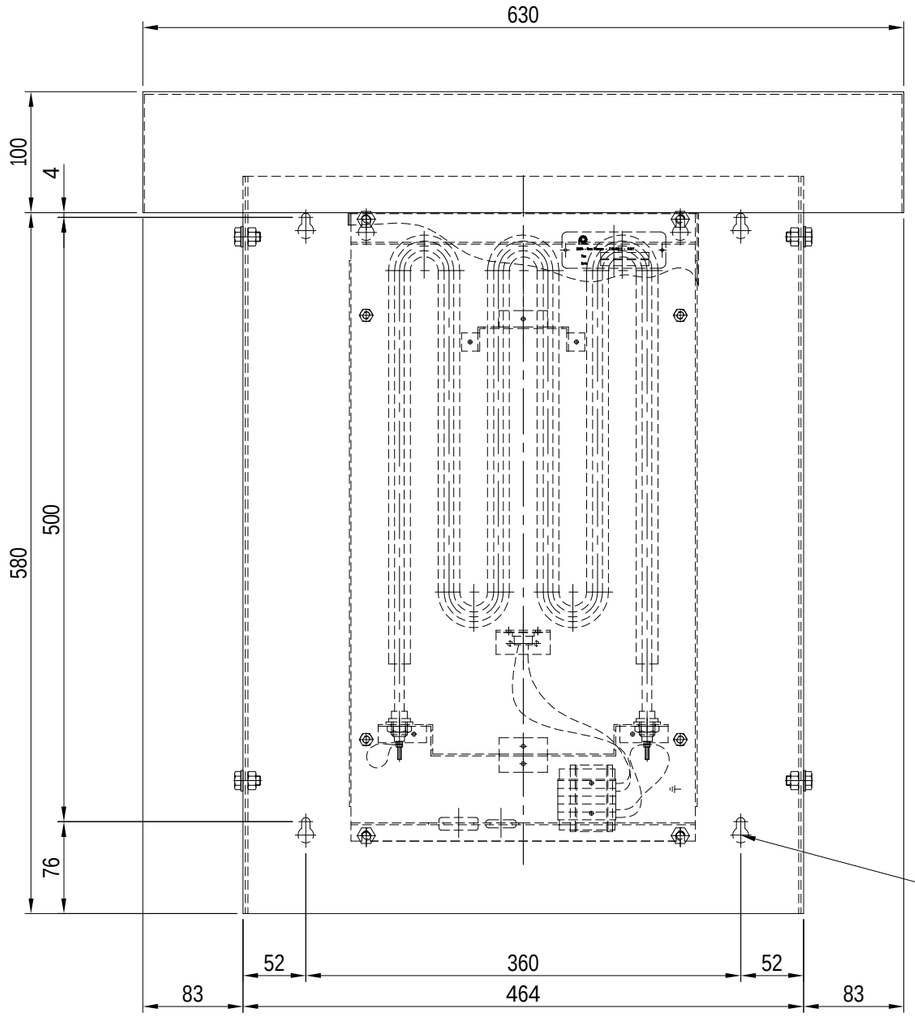
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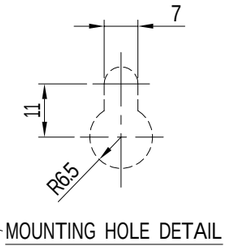
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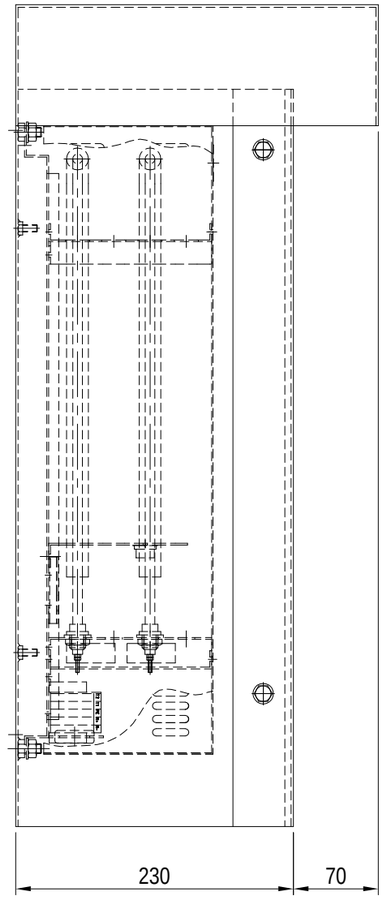
DRWN PR
DATE 26/3/04
MAT'L MS
WGHT 50 kg
SCALE 1:6 ON A3

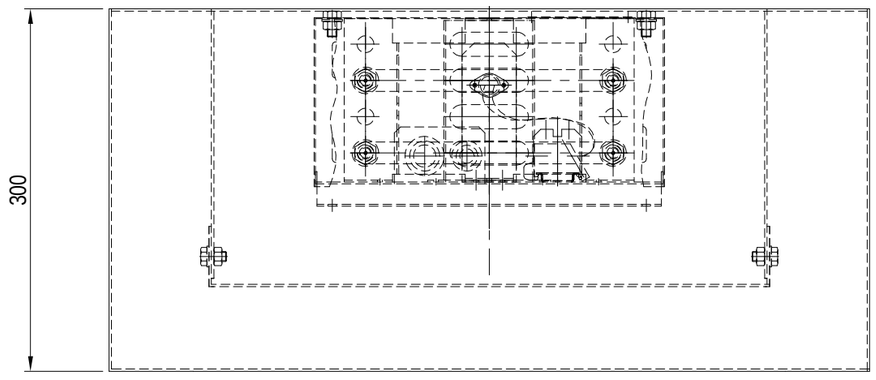
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PART NUMBERS

AISI 304 CONSTRUCTION - S01-S0603-000
AISI 316 CONSTRUCTION - S01-S0603-T00

NOTE

COVER SUITS BOTH CAVOTEC STANDARD RESISTOR AND ABB SACE RESISTOR.
CAVOTEC RESISTOR SHOWN.

No.	REVISION	BY	DATE
B	MODIFIED NOTES, CHANGED TO FIRST ANGLE PROJECTION	PR	25/1/05
A	CERTIFIED ISSUE	PR	22/3/03

TITLE: CTC REELING SYSTEM
RAIN COVER FOR BRAKE RESISTOR

CLIENT:

PROJECT:

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DRWN PR
DATE 22/3/03
MAT'L AISI 304/316
WGHT 15 kg
SCALE 1:5 ON A3

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DWG No STD-BE-001 B

SECTION 15 - ELECTRICAL DOCUMENTATION

Drawing No.	Description
SC-121216-SE1	Power & control schematic
SC-121216-FD1	Functional description
STD-SE-002	Schematic for cable guide J-box with controls
STD-TD-003	Roller Guide Functional Description
SC-121216-PAR1	Parameters
Commissioning procedure	Gantry cable reel linear torque levels Vacon NXP Drive
Commissioning procedure	Control Keypad
ASITQ224_v320	Vacon NXP frequency converters, application Cavotec

CAVOTEC SPECIMAS S.P.A.

PROJECT: SC-121216-SE1 CAVOTEC NCC SYSTEM - 2 MOTORS, LINEAR CONTROL

INDEX REVISION:

Rev	A	B	C	D	E	F			
Pag.1	X								
Pag.2		X					POWER DISTRIBUTION		
Pag.3		X					VACON NXP DRIVE 5.5 kW		
Pag.4		X					REEL MOTOR - BRAKE RESISTOR		
Pag.5		X					CONTROL 230 VAC		
Pag.6		X					CONTROL 230 VAC		
Pag.7	X						CLIENT FEEDBACK		
Pag.8	X						NC0PTA1 OPTION BOARD		
Pag.9	X						DRIVE BOARD, ENCODER, ROTARY LIMIT SWITCH, PROFIBUS CARD		
Pag.10	X						PANEL LAYOUT		
Pag.11		X					POWER TERMINAL		
Pag.12	X						CONTROL TERMINAL		
Pag.13	X						MOTOR TERMINAL		
Pag.14	X						ENCODER TERMINAL		
Pag.15	X						MOTOR TERMINAL		
Pag.16	X						JUNCTION BOX LEFT SIDE		
Pag.17	X						JUNCTION BOX RIGHT SIDE		
Pag.18	X						CABLE GUIDE TERMINAL		
Pag.19									
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Pag.37									

NOTES

- Standard wire numbers have the form xxyy where xx is the sheet number on which it first appears and yy is a progressive number for that sheet, eg. 0514 = sheet number 05, wire number 14.
- Component identifiers have numbers xy where x is page number and y column number, eg. M34 - motor page 3 column 4.
- Position references are indicated by sheet number followed by coordinate, eg. /4.E2 = sheet number 4, position E2.
- All dashed interconnecting wiring to be done by others.
- Wire colours are indicate in parentheses (BK=Black, BU=Blue, BN=Brown, RD=Red, WH=White, YE=Yellow, GN=Green, GY=Grey, PK=Pink).
- Standard cable are indicate with -Wxx.

NOTES

Colour Code for internal wiring CEI EN 60204-1 (13.2.3-13.2.4) :

AC and DC power circuits - BLACK (BK)
Neutral - LIGHT BLUE (BU)
AC control circuits - RED (RD)
DC control circuits - DARK BLUE (BU)

Screened and component specialized cable can be coloured differently.

No.	REVISION	BY	DATE
A	MODIFIED TITLE OF PAGES 5 & 6	DD	05/07/2012

TITLE: CAVOTEC NCC SYSTEM - 2 MOTORS, LINEAR CONTROL

CLIENT

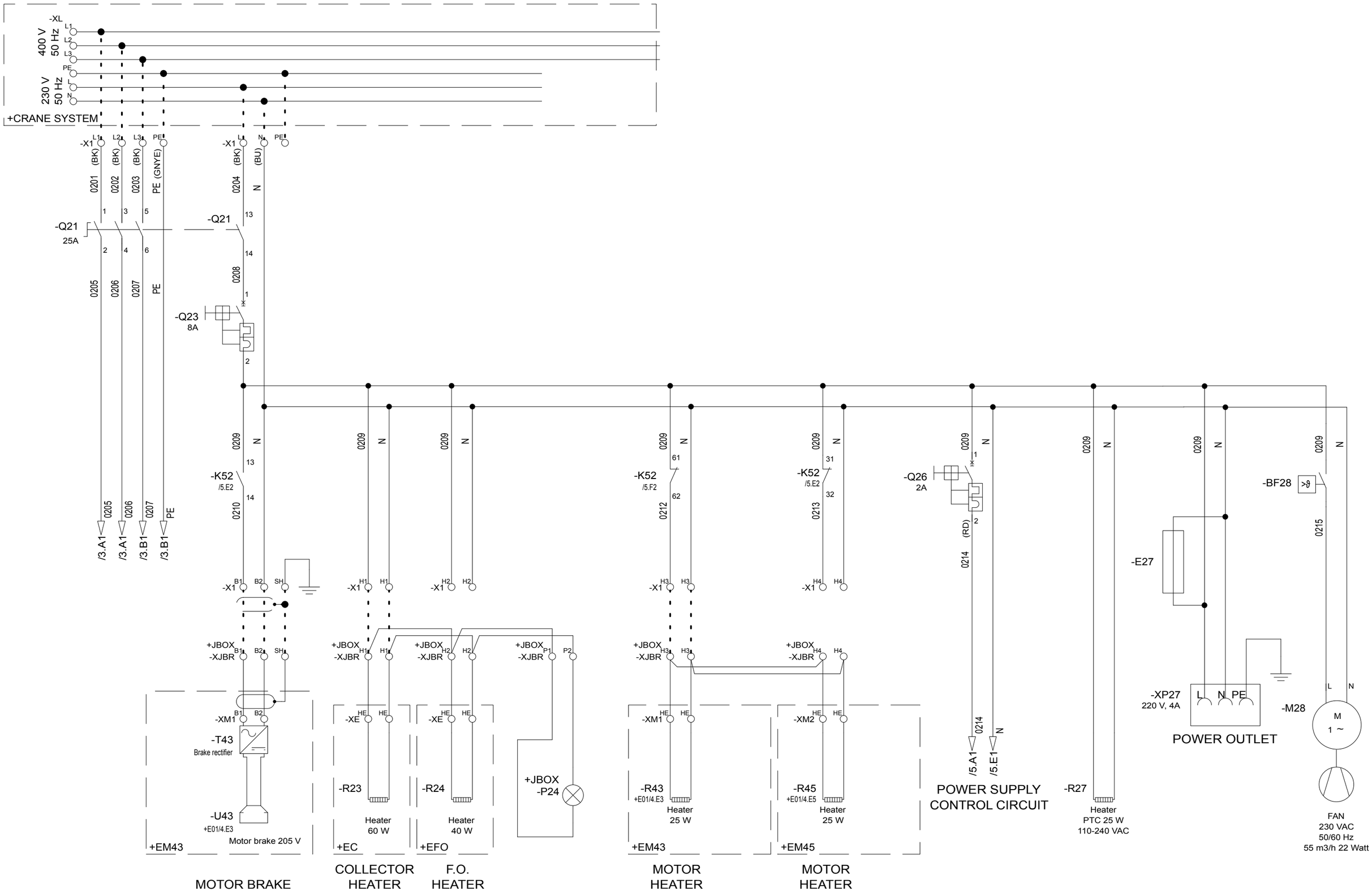
PROJECT: ZPMC SRILANKA

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DRWN DD
DATE 28/06/2012
CHKD
APPD

CAVOTEC
Connecting mobile equipment

SHEET No 1 OF 18
DWG No SC-121216-SE1
A



Refer to sheet 1 for notes

No.	REVISION	BY	DATE
B	MODIFIED TENSION LEVELS	DD	05/07/2012
A	ISSUED FOR APPROVAL	DD	28/06/2012

TITLE
CAVOTEC NCC SYSTEM - 2 MOTORS, LINEAR CONTROL POWER DISTRIBUTION

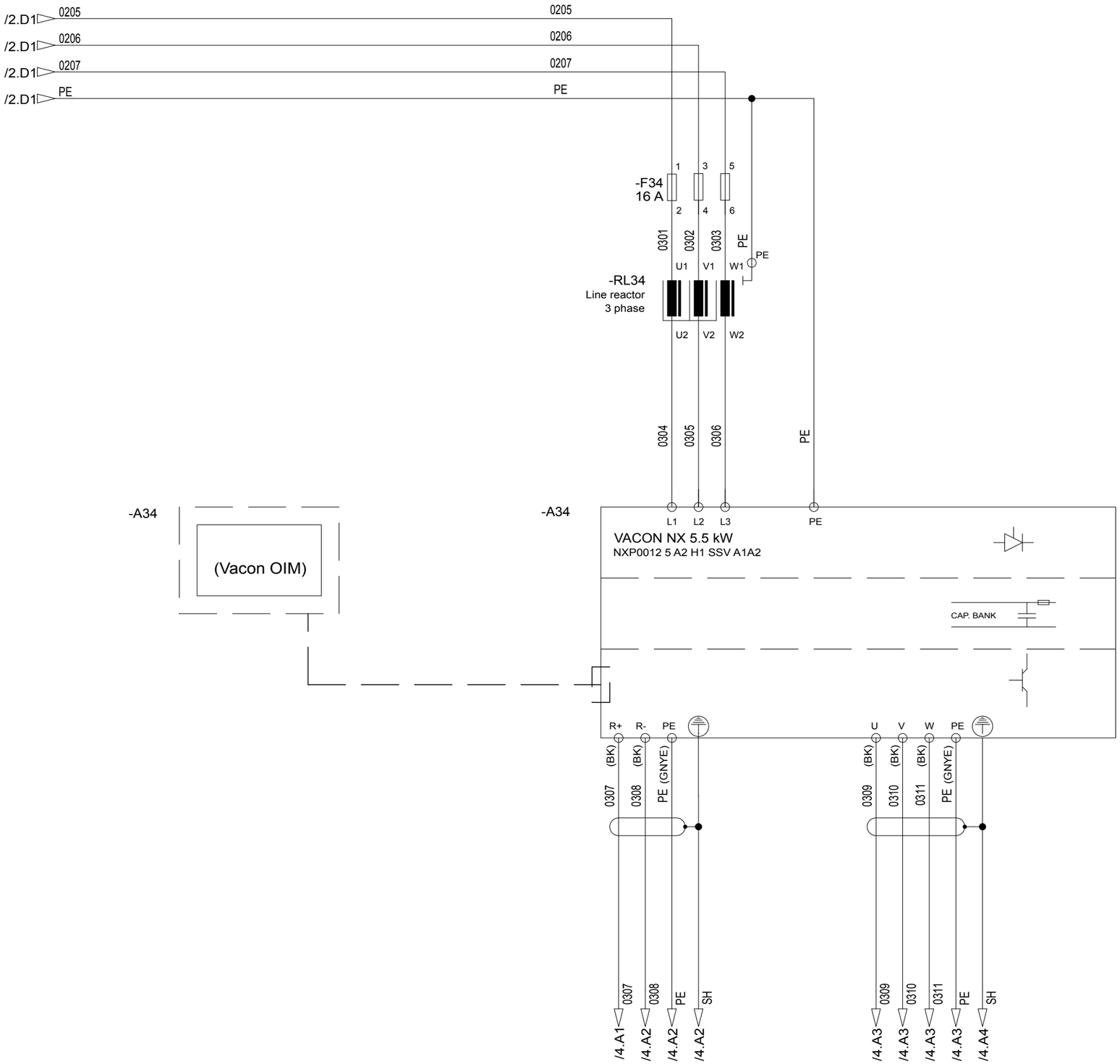
CLIENT

PROJECT
ZPMC SRILANKA

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DRWN	DD
DATE	28/06/2012
CHKD	
APPD	
SHEET No	2 OF 18
DWG No	SC-121216-SE1
	B

CAVOTEC
Connecting mobile equipment
+E01



MAXIMUM ALLOWABLE CABLE LENGHT BETWEEN MOTORS AND PANEL 300 m

WARNING: IF THE POWER SUPPLY TO THE DRIVE IS DISCONNECTED, WAIT AT LEAST 1 MINUTE PRIOR TO RECONNECTING. NOT DOING SO MAY RESULT IN DAMAGE TO THE DRIVE.

Refer to sheet 1 for notes

No.	REVISION	BY	DATE
B	MODIFIED DESCRIPTION INSIDE DRIVE	DD	23/10/2012
A	ISSUED FOR APPROVAL	DD	28/06/2012

TITLE
CAVOTEC NCC SYSTEM - 2 MOTORS, LINEAR CONTROL
VACON NXP DRIVE 3 kW

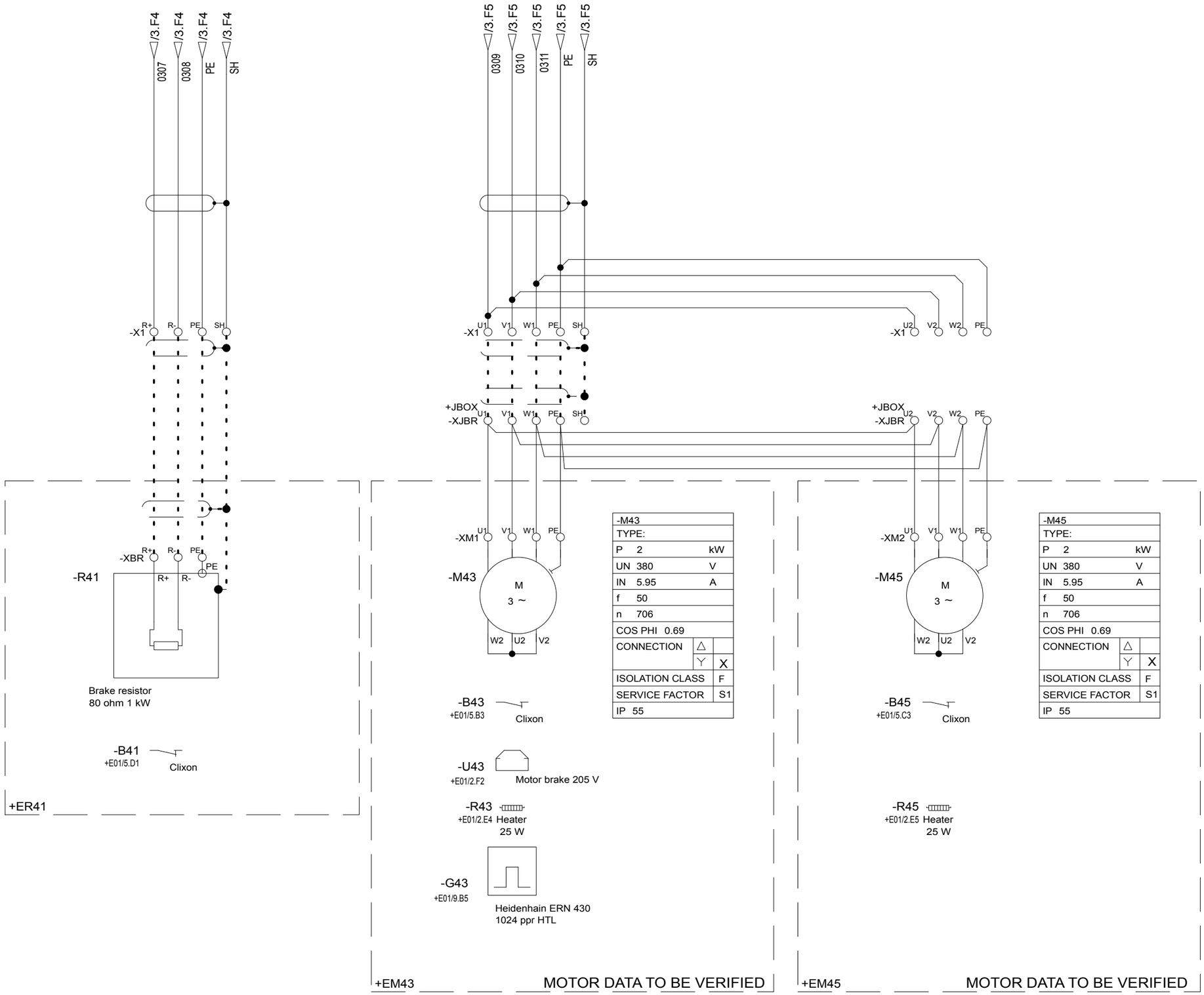
CLIENT

PROJECT
ZPMC SRILANKA

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DRWN	DD
DATE	28/06/2012
CHKD	
APPD	
SHEET No	3 OF 18
DWG No	SC-121216-SE1
	B
	+E01

CAVOTEC
Connecting mobile equipment



Refer to sheet 1 for notes

No.	REVISION	BY	DATE
B	MODIFIED MOTOR DATA	DD	25/10/2012
A	ISSUED FOR APPROVAL	DD	28/06/2012

TITLE: CAVOTEC NCC SYSTEM - 2 MOTORS, LINEAR CONTROL
REEL MOTOR - BRAKE RESISTOR

CLIENT

PROJECT: ZPMC SRILANKA

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DRWN	DD
DATE	28/06/2012
CHKD	
APPD	
SHEET No	4 OF 18
DWG No	SC-121216-SE1
	B

CAVOTEC
Connecting mobile equipment
+E01

DRIVE READY | DRIVE RUNNING | OVERHEATING MOTOR | DRUM FULL | DRUM EMPTY

230 V /2.E6 50 Hz — 0214 (RD) — 0214 /6.A1

-A34 /9.E1 (23/22) — 0501 — -K53 /5.E3 (11/14) — 0502 — -X2 3 — -XBR T1 — 0502 — -B41 +E01/4.E1 Clixon (T1/T2) — 0503 — -XBR T2 — +ER41 — -X2 4 — 0503 — A1 -K51 /5.E1 A2

-A34 /9.E2 (26/25) — 0504 — A1 -K52 /5.E2 A2

-X2 5 — +JBOX -XJBR 4 — -XM1 C T1 — -B43 +E01/4.D3 Clixon — +EM43 -XM1 C T2 — Motor M43 clixon — +JBOX -XJBR 5 — 0505 — +JBOX -XJBR 6 — -XM2 C T1 — -B45 +E01/4.D5 Clixon — +EM45 -XM2 C T2 — Motor M45 clixon — +JBOX -XJBR 7 — -X2 6 — 0506 — A1 -K53 /5.E3 A2

-X2 9 — +JBOX -XJBL 10 — 0214 — +EFO -BF91 /9.B1 Cam 0 (14/11) — +JBOX -XJBL 11 — 0507 — +JBOX -XJBL 12 — +EFO -BF91 /9.B2 Cam 1 (24/21) — +JBOX -XJBL 13 — -X2 10 — 0508 — A1 -K54 /5.E4 A2

-X2 11 — +JBOX -XJBL 14 — 0214 — +EFO -BF91 /9.B2 Cam 2 (32/31) — +JBOX -XJBL 15 — -X2 12 — 0509 — A1 -K55 /5.E5 A2

Brake resistor clixon

NEUTRAL /2.E6 N (BU) — N /6.E1

-K51 /5.E1: /6.D5 (12/14/11), /6.C2 (22/24/21), /7.C1 (32/34/31), /6.C3 (42/44/41)

-K52 /5.E2: /2.D2 (14/13), /2.D5 (22/21), /7.C2 (32/31), /2.D4 (44/43), /6.C1 (54/53), (62/61), (72/71), (84/83)

-K53 /5.E3: /5.B1 (12/14/11), /7.C6 (22/24/21), (32/34/31), (42/44/41)

-K54 /5.E4: /6.B2 (12/14/11), /7.C8 (22/24/21), (32/34/31), (42/44/41)

-K55 /5.E5: /6.B3 (12/14/11), /7.C7 (22/24/21), (32/34/31), (42/44/41)

Refer to sheet 1 for notes

No.	REVISION	BY	DATE
B	MODIFIED PAGE TITLE	DD	05/07/2012
A	ISSUED FOR APPROVAL	DD	28/06/2012

TITLE: CAVOTEC NCC SYSTEM - 2 MOTORS, LINEAR CONTROL
CONTROL 230 VAC

CLIENT

PROJECT: ZPMC SRILANKA

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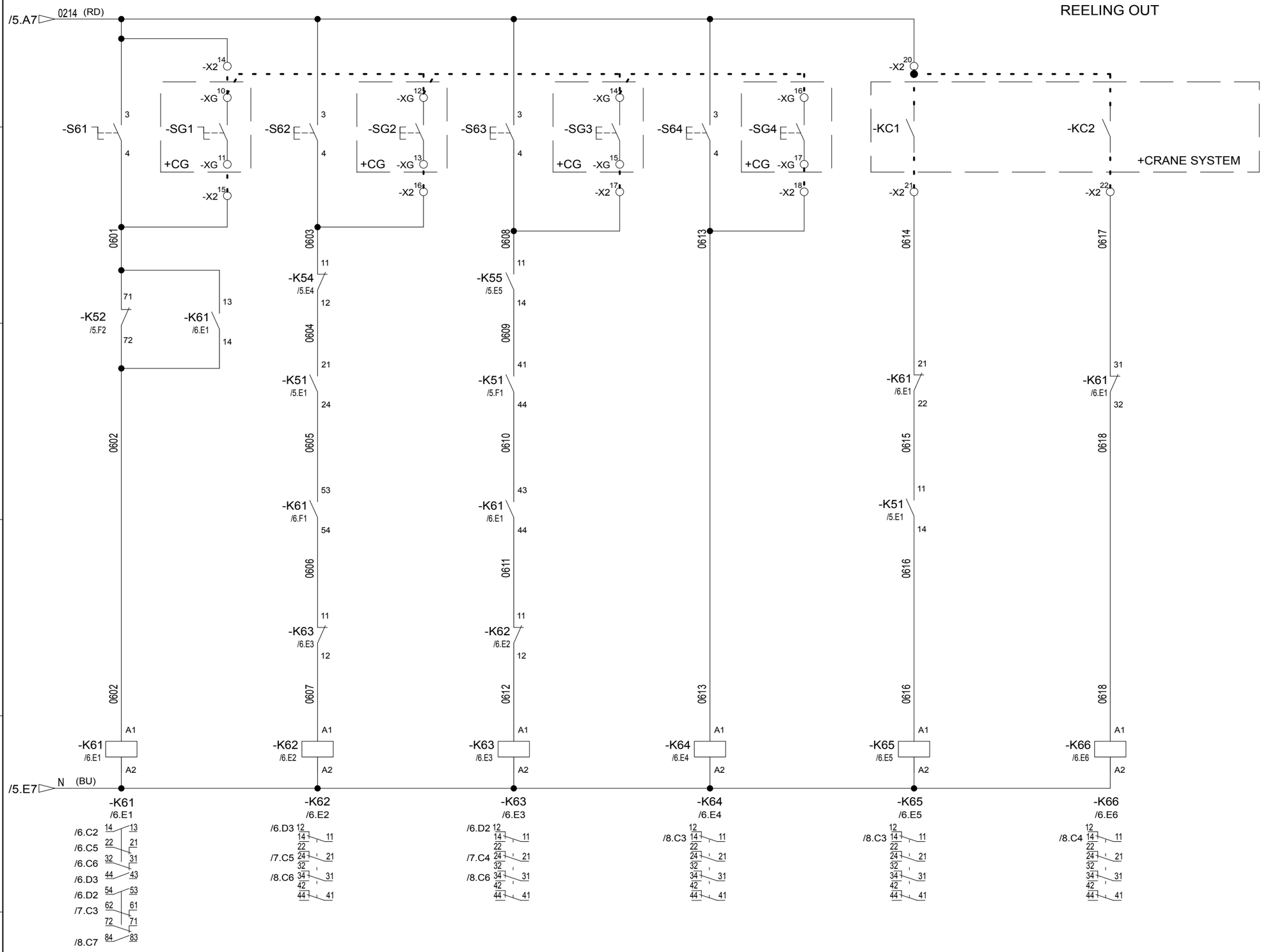
DRWN DD | DATE 28/06/2012 | CHKD | APPD

CAVOTEC Connecting mobile equipment

+E01

SHEET No 5 OF 18 | DWG No SC-121216-SE1 | B

AUTO/MANUAL | MANUAL REEL IN | MANUAL REEL OUT | RESET | REELER RUN | REELING OUT



NOTES

The following control signals are to be provided by others:

- KC1 machine moving, to be active whenever the machine is moving. 6 second delay off except in case of emergency.
- KC2 reeling out signal. Note that this signal must be deactivated as soon as possible

-S61 is auto/manual selector on control panel
-S62 is manual reel in push button on control panel
-S63 is manual reel out push button on control panel
-S64 is fault reset push button on control panel

-SG1 is remote manual/auto selector on cable guide
-SG2 is remote manual reel in push button on cable guide
-SG3 is remote manual reel out push button on cable guide
-SG4 is remote fault reset push button on cable guide

Refer to sheet 1 for additional notes.

No.	REVISION	BY	DATE
B	MODIFIED PAGE TITLE	DD	05/07/2012
A	ISSUED FOR APPROVAL	DD	28/06/2012

TITLE: CAVOTEC NCC SYSTEM - 2 MOTORS, LINEAR CONTROL
CONTROL 230 VAC

CLIENT

PROJECT: ZPMC SRILANKA

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DRWN DD
DATE 28/06/2012
CHKD | APPD

CAVOTEC Connecting mobile equipment

+E01

SHEET No 6 OF 18 | DWG No SC-121216-SE1 | B

NOTES

24VDC/3A, 250VAC/3A maximum allowable switching load for each contact.
Total maximum allowable load at the common terminal is 10A.

-X2 25 (RD) 0701

-K51 /5.E1 31 34
-K52 /5.E2 43 44
-K61 /6.F1 61 62
-K63 /6.E3 21 24
-K62 /6.E2 21 24
-K53 /5.E3 21 24
-K55 /5.E5 21 24
-K54 /5.E4 21 24

-X2 26 (RD) 0702 DRIVE READY
-X2 27 (RD) 0703 DRIVE RUNNING
-X2 28 (RD) 0704 AUTOMATIC MODE SELECTED
-X2 29 (RD) 0705 MANUAL REELING OUT
-X2 30 (RD) 0706 MANUAL REELING IN
-X2 31 (RD) 0707 MOTORS OVER-TEMPERATURE FAULT
-X2 32 (RD) 0708 DRUM EMPTY FAULT
-X2 33 (RD) 0709 DRUM FULL

Refer to sheet 1 for notes

No.	REVISION	BY	DATE
A	ISSUED FOR APPROVAL	DD	28/06/2012

TITLE: CAVOTEC NCC SYSTEM - 2 MOTORS, LINEAR CONTROL
CLIENT FEEDBACK

CLIENT:
PROJECT: ZPMC SRILANKA

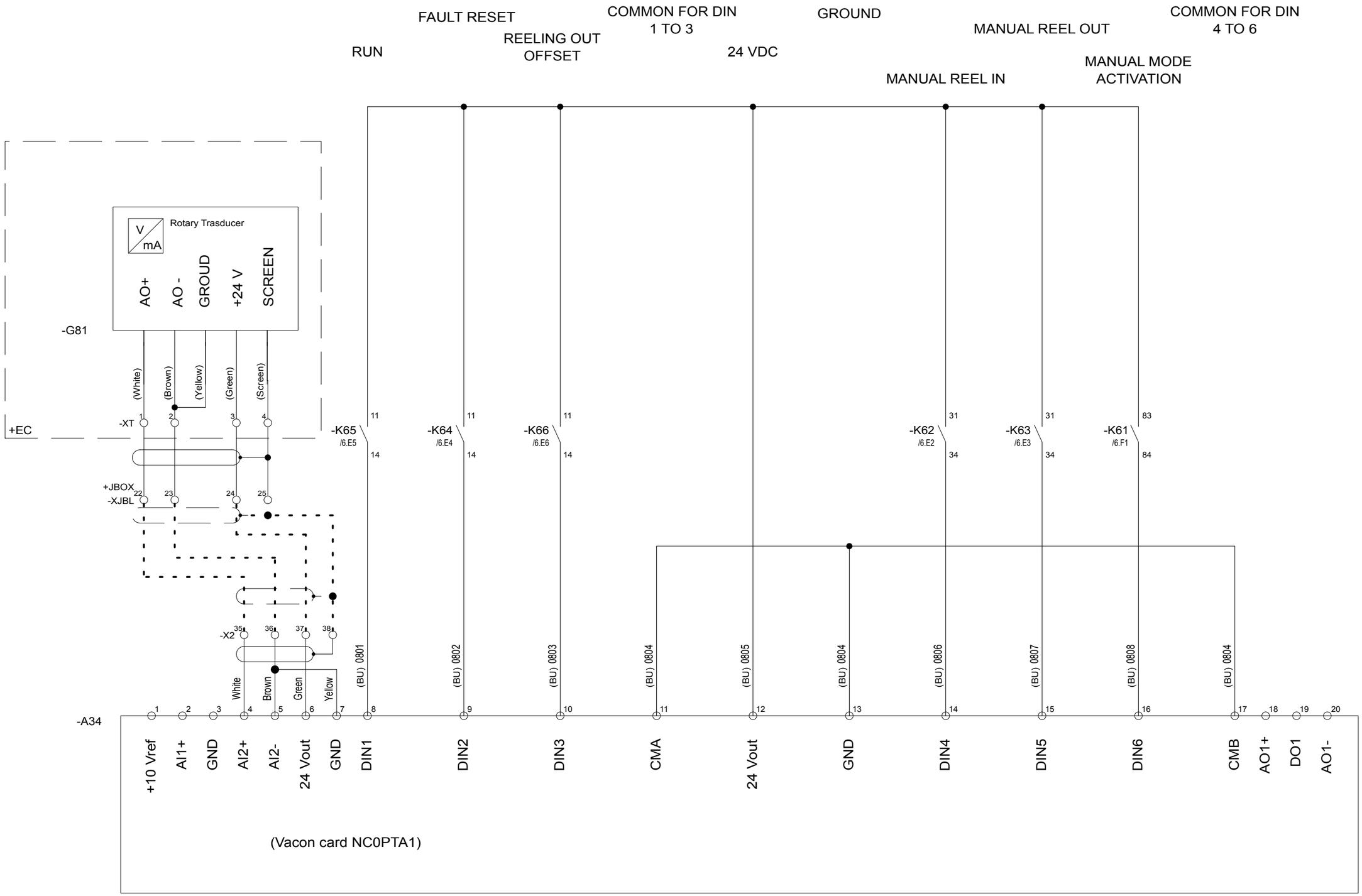
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DRWN DD
DATE 28/06/2012
CHKD
APPD

CAVOTEC Connecting mobile equipment

+E01

SHEET No 7 OF 18
DWG No SC-121216-SE1
A



Refer to sheet 1 for notes

A	ISSUED FOR APPROVAL	DD	28/06/2012
No.	REVISION	BY	DATE

TITLE: CAVOTEC NCC SYSTEM - 2 MOTORS, LINEAR CONTROL
NC0PTA1 OPTION BOARD

CLIENT

PROJECT: ZPMC SRILANKA

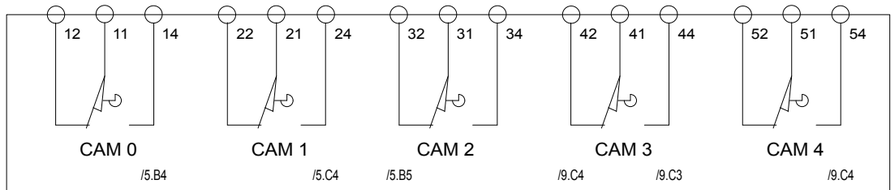
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DRWN	DD
DATE	28/06/2012
CHKD	APPD
SHEET No	8 OF 18
DWG No	SC-121216-SE1
	A

CAVOTEC
Connecting mobile equipment
+E01

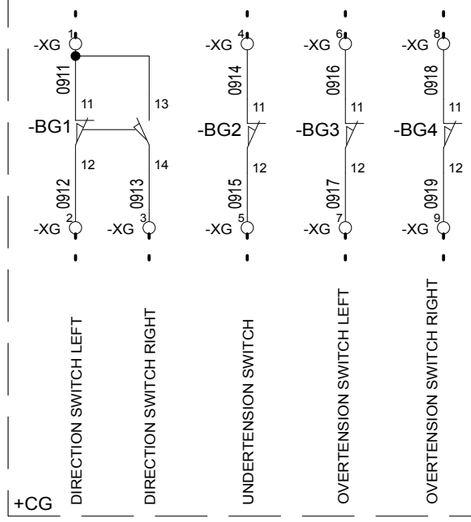
DRUM FULL DRUM EMPTY

+EFO
-BF91

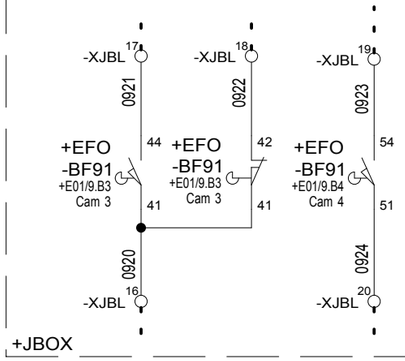


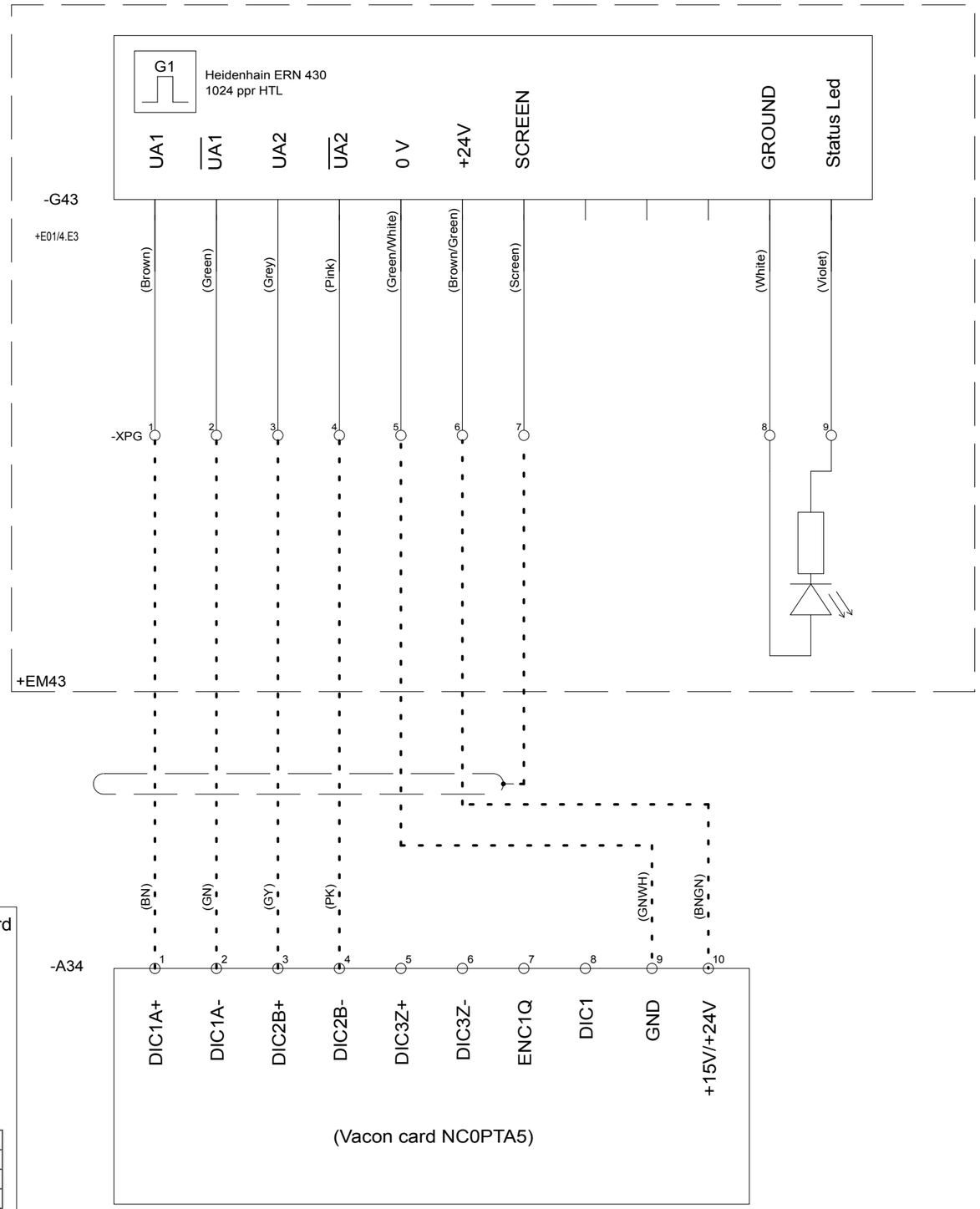
LS Stromag 1:48/4
2x15° Cam
2x45° Cam

CABLE GUIDE SWITCH
PUSH BUTTON (TO BE INCLUDED IN CRANE LOGIC)
ACCORDING TO DIAGRAM STD-SE-006

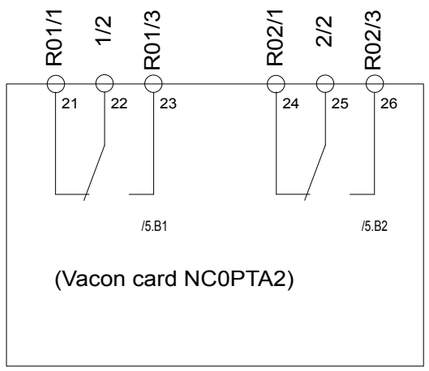


SPARE CAM PRE WIRED
ON JBOX



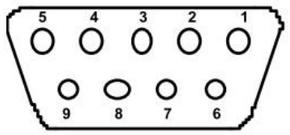


DRIVE READY DRIVE RUNNING



Profibus card

Connector on
Profibus card



Signal	Connector	Description
Shield	1	Cable shield
RxD/TxD-P	3	Receive/Transmit data - plus (B), RED
DGND	5	Data Ground (reference potential for VP)
VP	6	Supply voltage – plus (5V)
RxD/TxD-N	8	Receive/ Transmit data - minus (A), GREEN

Refer to sheet 1 for notes

No.	REVISION	BY	DATE
A	ISSUED FOR APPROVAL	DD	28/06/2012

TITLE
CAVOTEC NCC SYSTEM - 2 MOTORS, LINEAR CONTROL
DRIVE BOARD, ENCODER, ROTARY LIMIT SWITCH

CLIENT

PROJECT
ZPMC SRILANKA

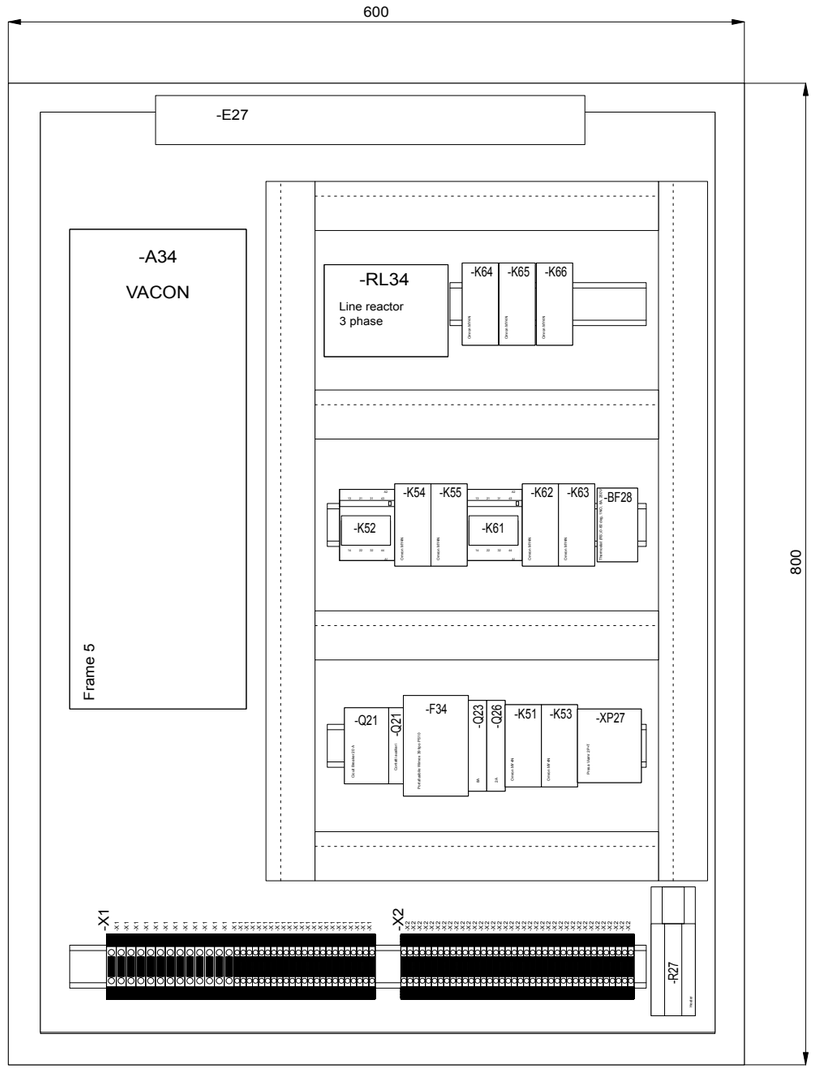
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DRWN DD
DATE 28/06/2012
CHKD
APPD

CAVOTEC
Connecting mobile equipment

+E01

SHEET No 9 OF 18
DWG No SC-121216-SE1
A



INSTALLATION DRAWING
STD-BE-004 - Box CTC painted 600x800x300

Refer to sheet 1 for notes

No.	REVISION	BY	DATE
A	ISSUED FOR APPROVAL	DD	28/06/2012

TITLE: CAVOTEC NCC SYSTEM - 2 MOTORS, LINEAR CONTROL PANEL LAYOUT

CLIENT:

PROJECT: ZPMC SRILANKA

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DRWN DD
DATE 28/06/2012
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APPD

CAVOTEC
Connecting mobile equipment

+E01

SHEET No 10 OF 18
DWG No SC-121216-SE1
A

terminal strip

-X1

| | | |

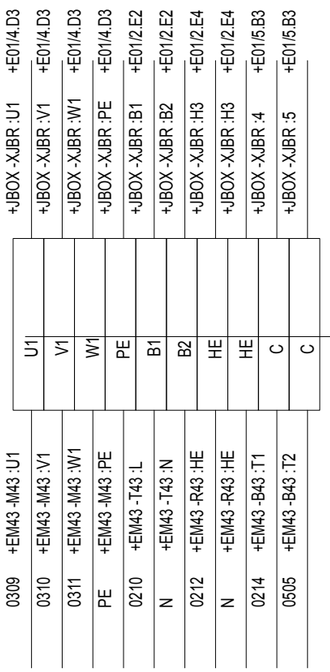
terminal strip

-X2

| | | |

terminal strip

-XM1



terminal _____

terminal _____

terminal _____

terminal _____

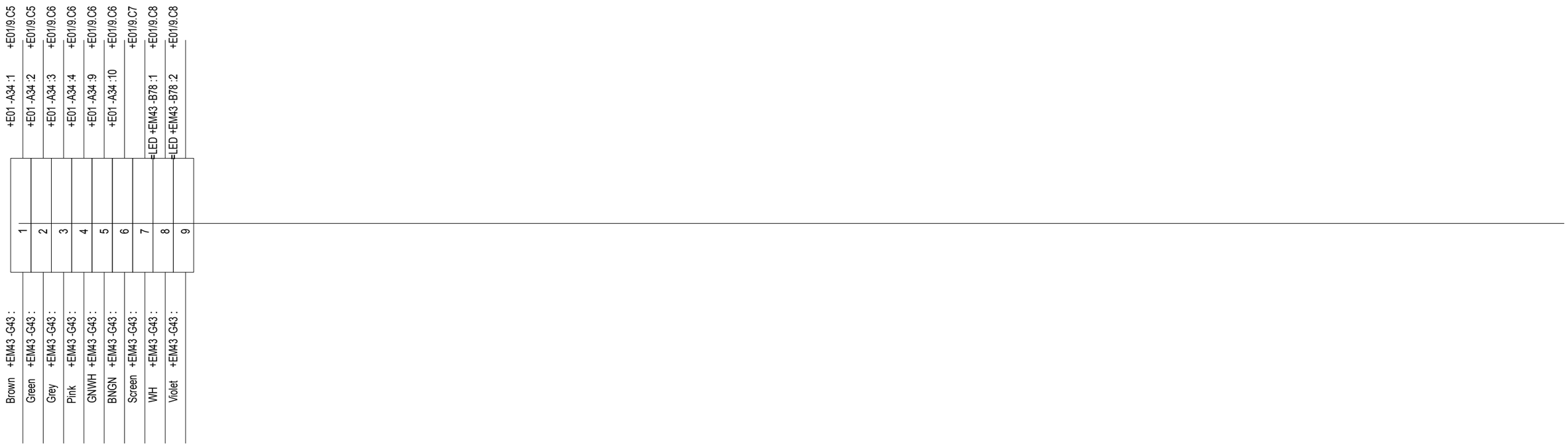
A	ISSUED FOR APPROVAL	DD	28/06/2012	TITLE	CAVOTEC NCC SYSTEM - 2 MOTORS, LINEAR CONTROL MOTOR TERMINAL	DRWN DD DATE 28/06/2012 CHKD APPD	CAVOTEC Connecting mobile equipment
No.	REVISION	BY	DATE	CLIENT	PROJECT ZPMC SRILANKA	SHEET No 13 OF 18	DWG No SC-121216-SE1 A

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+EM43

terminal strip

-XPG



terminal _____

terminal _____

terminal _____

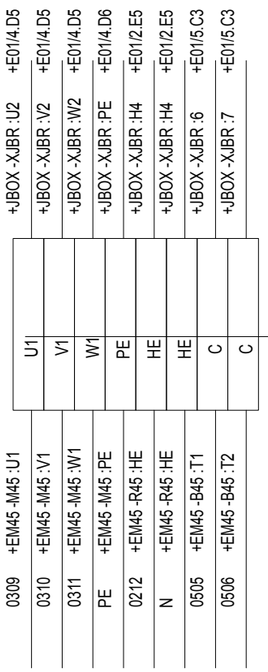
terminal _____

A	ISSUED FOR APPROVAL	DD	28/06/2012	TITLE: CAVOTEC NCC SYSTEM - 2 MOTORS, LINEAR CONTROL ENCODER TERMINAL	THIS DESIGN IS THE PROPERTY OF CAVOTEC SPECIMAS SPA AND IS ONLY TO BE USED WITH EXPRESS PERMISSION AND LICENCE FROM CAVOTEC SPECIMAS SPA	DRWN DD DATE 28/06/2012 CHKD APPD	+EM43
No.	REVISION	BY	DATE	CLIENT / PROJECT: ZPMC SRILANKA		SHEET No 14 OF 18	DWG No SC-121216-SE1 A

CAVOTEC
Connecting mobile equipment

terminal strip

-XM2



terminal _____

terminal _____

terminal _____

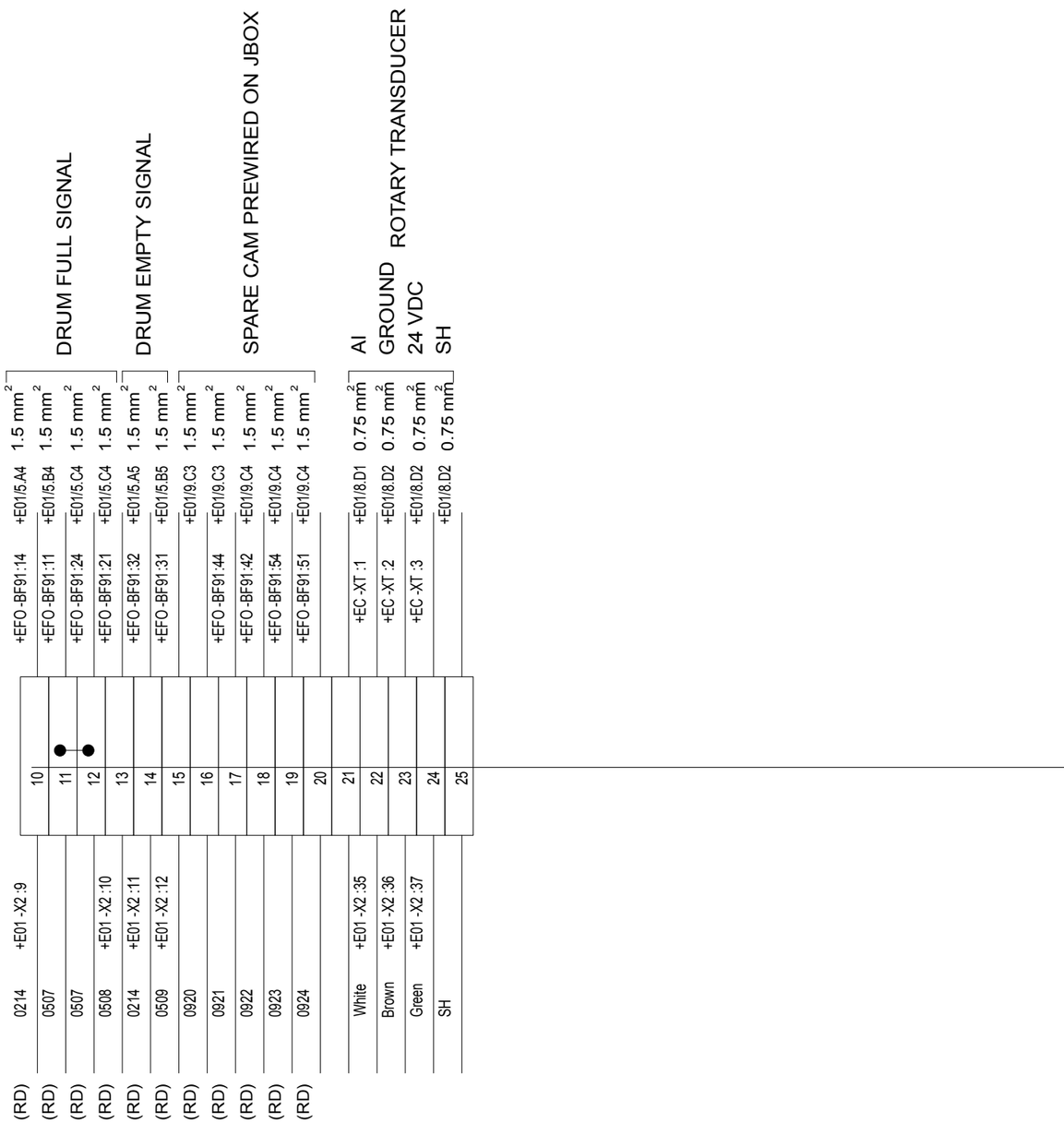
terminal _____

A	ISSUED FOR APPROVAL	DD	28/06/2012	TITLE: CAVOTEC NCC SYSTEM - 2 MOTORS, LINEAR CONTROL MOTOR TERMINAL	THIS DESIGN IS THE PROPERTY OF CAVOTEC SPECIMAS SPA AND IS ONLY TO BE USED WITH EXPRESS PERMISSION AND LICENCE FROM CAVOTEC SPECIMAS SPA	DRWN DD DATE 28/06/2012 CHKD APPD	+EM45
No.	REVISION	BY	DATE	CLIENT / PROJECT: ZPMC SRILANKA		SHEET No 15 OF 18	DWG No SC-121216-SE1 A

CAVOTEC
Connecting mobile equipment

terminal strip

-XJBL



terminal _____

terminal _____

terminal _____

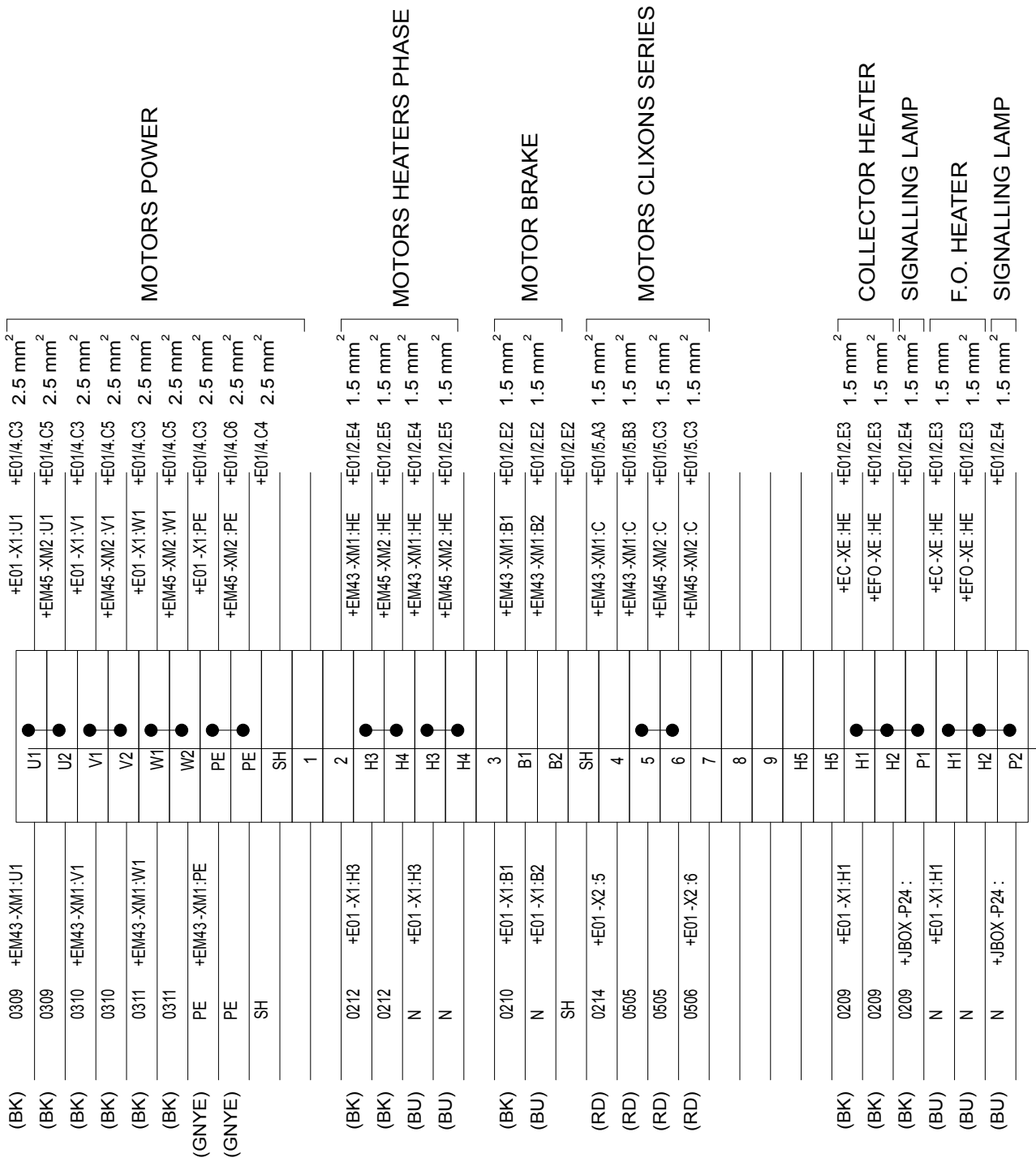
terminal _____

A	ISSUED FOR APPROVAL	DD	28/06/2012	TITLE: CAVOTEC NCC SYSTEM - 2 MOTORS, LINEAR CONTROL JUNCTION BOX LEFT SIDE	THIS DESIGN IS THE PROPERTY OF CAVOTEC SPECIMAS SPA AND IS ONLY TO BE USED WITH EXPRESS PERMISSION AND LICENCE FROM CAVOTEC SPECIMAS SPA	DRWN DD DATE 28/06/2012 CHKD APPD	+JBOX
No.	REVISION	BY	DATE	CLIENT / PROJECT: ZPMC SRILANKA		SHEET No 16 OF 18	DWG No SC-121216-SE1 A

CAVOTEC *Connecting mobile equipment*

terminal strip

-XJBR



terminal _____

terminal _____

terminal _____

terminal _____

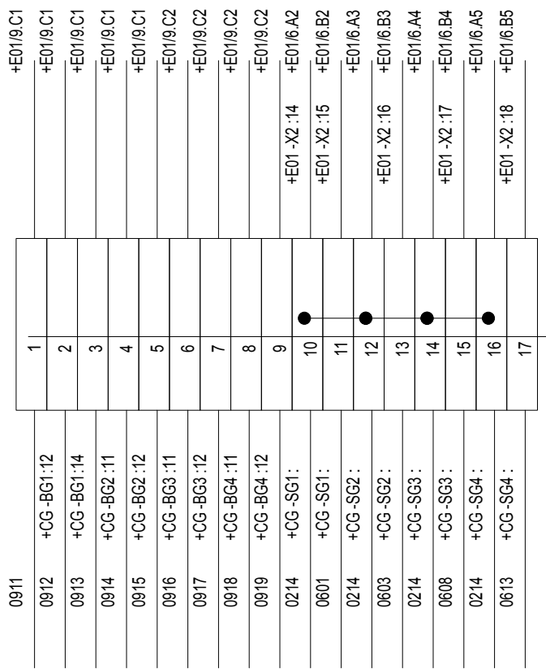
A	ISSUED FOR APPROVAL	DD	28/06/2012	TITLE	CAVOTEC NCC SYSTEM - 2 MOTORS, LINEAR CONTROL JUNCTION BOX RIGHT SIDE	DRWN DD DATE 28/06/2012 CHKD APPD	+JBOX
No.	REVISION	BY	DATE	CLIENT	PROJECT ZPMC SRILANKA	SHEET No 17 OF 18	DWG No SC-121216-SE1 A

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CAVOTEC
Connecting mobile equipment

terminal strip

-XG



terminal _____

terminal _____

terminal _____

terminal _____

A	ISSUED FOR APPROVAL	DD	28/06/2012		
No.	REVISION	BY	DATE		

TITLE
CAVOTEC NCC SYSTEM - 2 MOTORS, LINEAR CONTROL
EXTERNAL PUSHBUTTON TERMINAL

CLIENT

PROJECT
ZPMC SRILANKA

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DRWN DD
DATE 28/06/2012
CHKD
APPD

CAVOTEC
Connecting mobile equipment

+CG

SHEET No 18 OF 18

DWG No SC-121216-SE1

A

Functional Description – Linear control

Project:	ZPMC SRILANKA	Document No:	SC-121216-FD1
Client:		Revision:	B
Contract No:	121216	Date:	05 July 2012

Reference is to be made to drawing SC-121216-SE1

NOTATION

Q21	Circuit breaker
F34	Fuse
K31	Power contactor
RL34	Line reactor
A34	Frequency converter
M43, M45	Motors
R41	Braking resistor
Q23, Q26	Circuit breakers with thermal overloads
R27	Control panel heater
BF28	Thermostat
M28	Enclosure fan
XP27	Auxiliary power point
E27	Internal light
B41	Braking resistor Clixon
B43, B45	Motor Clixons
T43	Brake rectifier
U43	Brake
T26	Power supply 24Vdc
F26	Fuse
R23	Collector heater
R24	F.O. heater
R43, R45	Motor heaters
G43	Encoder
G81	Rotary transducer
Kxx	Relays
SG1	Manual/Auto selector switch mounted on cable guide
SG2	Manual reel in push button mounted on cable guide
SG3	Manual reel out push button mounted on cable guide
SG4	Fault reset push button mounted on cable guide
BF91 cam0	Anti-hysteresis rotary limit switch
BF91 cam1	Drum full rotary limit switch
BF91 cam2	Drum empty rotary limit switch
BF91 cam3-4	Spare rotary limit switches

Functional Description – Linear control

Project:	ZPMC SRILANKA	Document No:	SC-121216-FD1
Client:		Revision:	B
Contract No:	121216	Date:	05 July 2012

FUNCTIONAL SUMMARY

The reeling system is designed to recover and payout cable using 2 linear torque algorithms. This is achieved using an Vacon NXP flux vector drive together with some relay logic and feedback from a rotary transducer mounted on the geared rotary limit switch. The payout torque value is lower than the recovery torque value in order to compensate for mechanical inefficiencies within the drive system.

CLIENT SUPPLY

- Stable 400VAC, 230VAC 50Hz fused power supplies free from harmonic disturbances.
- All interconnecting cabling between:
    1. Reel motors/brake rectifier supply, signal of motor clixons to reel control center (RCC). Motors and brake cables should be screened.
    2. Collector & F.O. enclosure heaters and RCC.
    3. Geared rotary limit switch/transducer and RCC. Rotary transducer cable should be screened.
    4. Encoder on motor and RCC. Encoder cable must be screened.
    5. RCC and MCC.
    6. Cable guide and RCC/MCC.
- Machine moving signal **KC1**, which is to be activated whenever the machine is moving. This signal is to remain active for at least 6 seconds after completion of any machine movement except in the case of an emergency stop.
- Reeling out signal **KC2**, which is to be activated whenever the machine moves away from the feedpoint. This condition can be determined by comparing the status of the direction limit switch on the guide to the direction of travel of the machine. Note that this signal must be deactivated as soon as the machine stops.

Functional Description – Linear control

Project:	ZPMC SRILANKA	Document No:	SC-121216-FD1
Client:		Revision:	B
Contract No:	121216	Date:	05 July 2012

DETAILED FUNCTIONAL DESCRIPTION

The Vacon NXP drive acts to control the torque output of the reel motors using flux vector technology. An encoder mounted on one of the motors provides feedback to the drive, enabling accurate torque control even at zero motor speed. Since the cable is pulled at varying radii, and inertia effects are quite significant, the motors need to operate at multiple torque levels. These torque levels vary based on the amount of cable on the drum, whether the machine is travelling towards or away from the feedpoint. A rotary transducer mounted on the rotary limit switch is used to determine the amount of cable on the drum at any time, and enables the drive to control the torque output of the motor in a linear fashion. A brake mounted on one of the motors acts to prevent the cable unwinding when the drive isn't operating. Clixons mounted inside the motors act to prevent operation of the reeling system if any over-heating is detected. A braking resistor mounted near the RCC acts to dissipate the energy generated by the motors when the cable is pulled off the reel. Also braking resistor is protected by a clixon, which prevent operation of the cable reel in case of braking resistor overheating.

The reel acts to maintain adequate cable tension between itself and the cable guide during all modes of operation. The reeling out signal is used to maintain a step decrease of motors torque in order to compensate for mechanical inefficiencies when the cable is pulled off the drum. When the machine stops, the motors continue operating at the normal torque level (ie. no offsets applied) until the machine-moving signal is lost.

For installation, commissioning and maintenance purposes, it is possible to operate the reeling system in manual mode using controls on RCC or on cable guide. When the selector is in manual mode, Manual Reel In or Manual Reel Out pushbuttons will cause the reel to wind or unwind cable in speed control mode at a pre-set speed.

A geared rotary limit switch mounted in the collector housing acts to indicate reel full and reel empty conditions. The reel full switch is used to indicate that the drum is full, this signal is used to interlock Reel in in manual mode, and has to be used by machine logic to bypass slack cable signal from cable guide. The reel empty switch is used to indicate that there is insufficient cable on the drum, and this should initiate an emergency stop.

The anti-condensation heaters mounted in the collector and FO housing need to be energised at all times. The anti-condensation heater mounted in the motors need to be energised whenever they are not running.

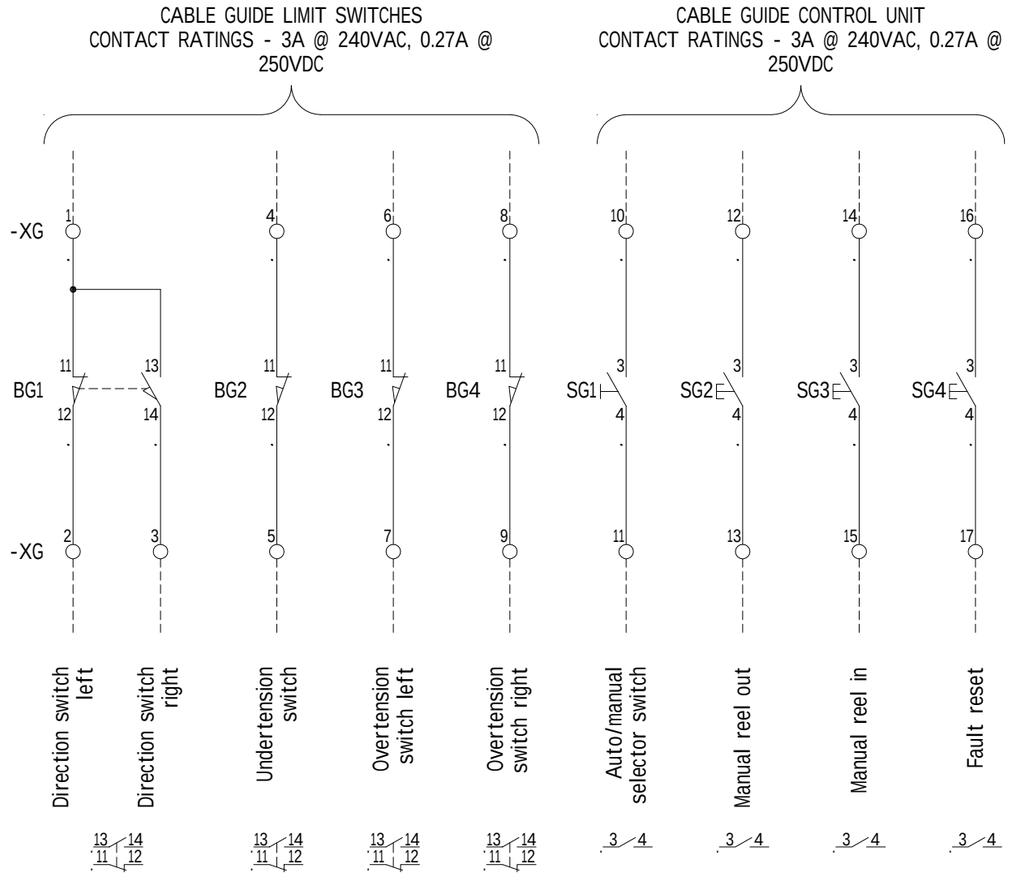
FEEDBACK

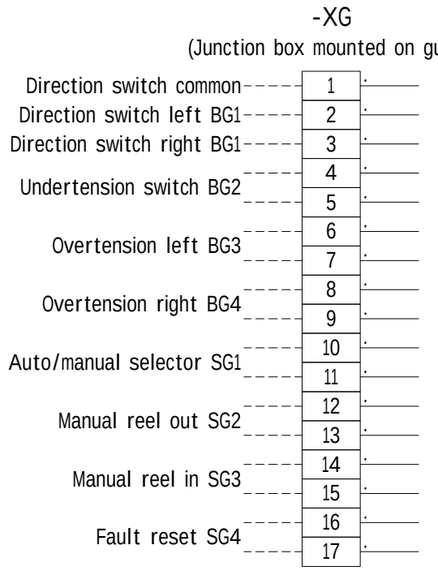
The following feedback signals are provided:

Reeler drive ready
Reeler drive running
Automatic mode selected (to be interlocked with machine movement to prevent movement while in manual mode)
Manual reeling out active
Manual reeling in active
Motor over-temperature fault
Drum empty fault
Drum full

NOTE

If the Vacon drive is switched off, please wait 1 minute before re-energising it. Not doing so may result in damage to the drive.





No.	REVISION	BY	DATE
C	CORRECTED CONTROL CONTACT TERMINAL NUMBERS	PR	14/9/04
B	ADDED CONTACT RATINGS	PR	15/7/04
A	ISSUED FOR INFORMATION	PR	17/2/04

TITLE: ELECTRICAL SCHEMATIC FOR CABLE GUIDE J-BOX WITH CONTROLS

CLIENT:

PROJECT:

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DRWN PR
DATE 17/2/04
CHKD PR
APRD PR

SHEET No 1 OF 1

DWG No STD-SE-002 C

CAVOTEC
Power supply and crane technology

Cavotec Specimas S.p.A.
A MANUFACTURING COMPANY OF THE CAVOTEC GROUP

Roller Guide Functional Description

Document No: STD-TD-003 Revision: D Date: 14 July 2006

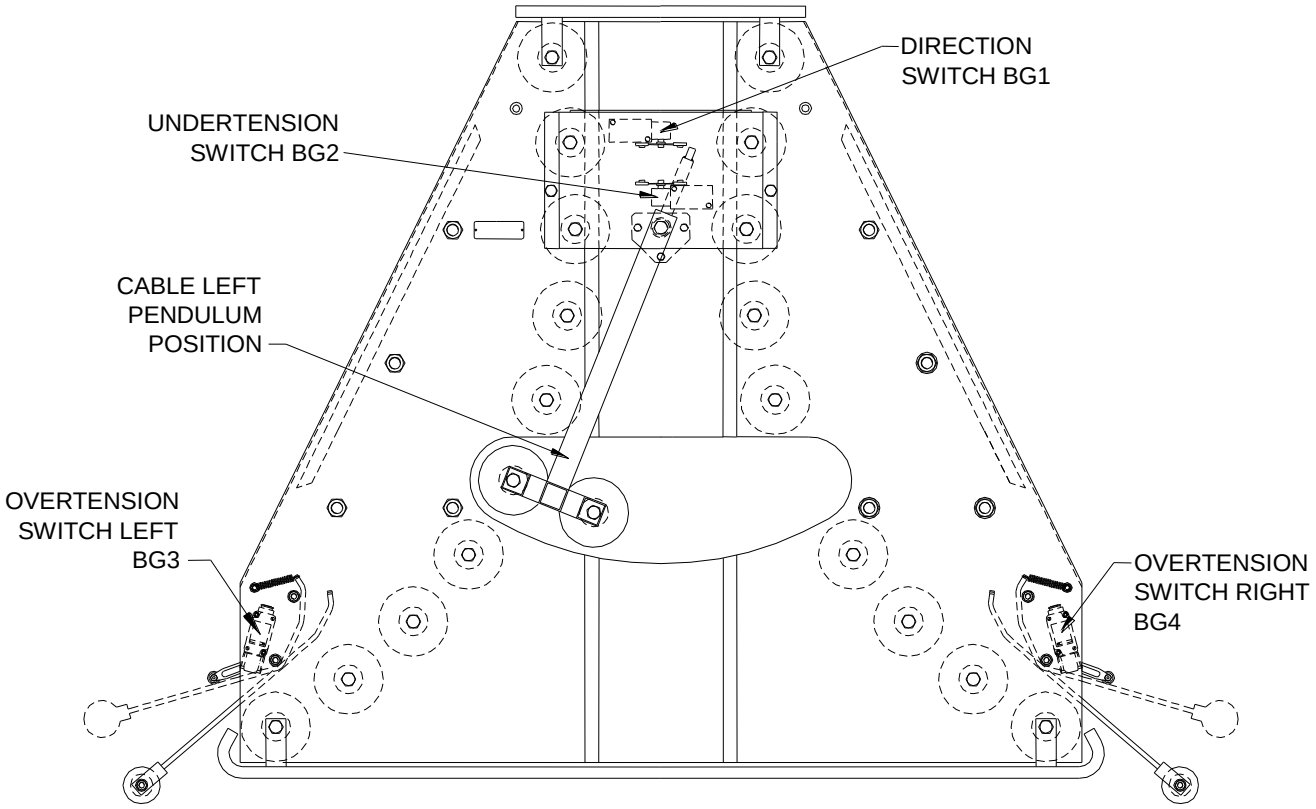


Figure 1 – Roller guide switches

UNDERTENSION SWITCH

The normally closed undertension switch BG2 is centrally located in the lower part of the roller guide (refer to Figure 1), and is actuated when the pendulum arm is in the vertical position. This switch is used to indicate a failure by the reeler to properly recover the cable/hose. Except in the case when the machine is travelling over the feedpoint, tripping of this switch should emergency stop machine travel to prevent damage to the cable/hose. Note that the undertension switch signal needs to be latched because the actual switch may only be activated briefly.

DIRECTION SWITCHES

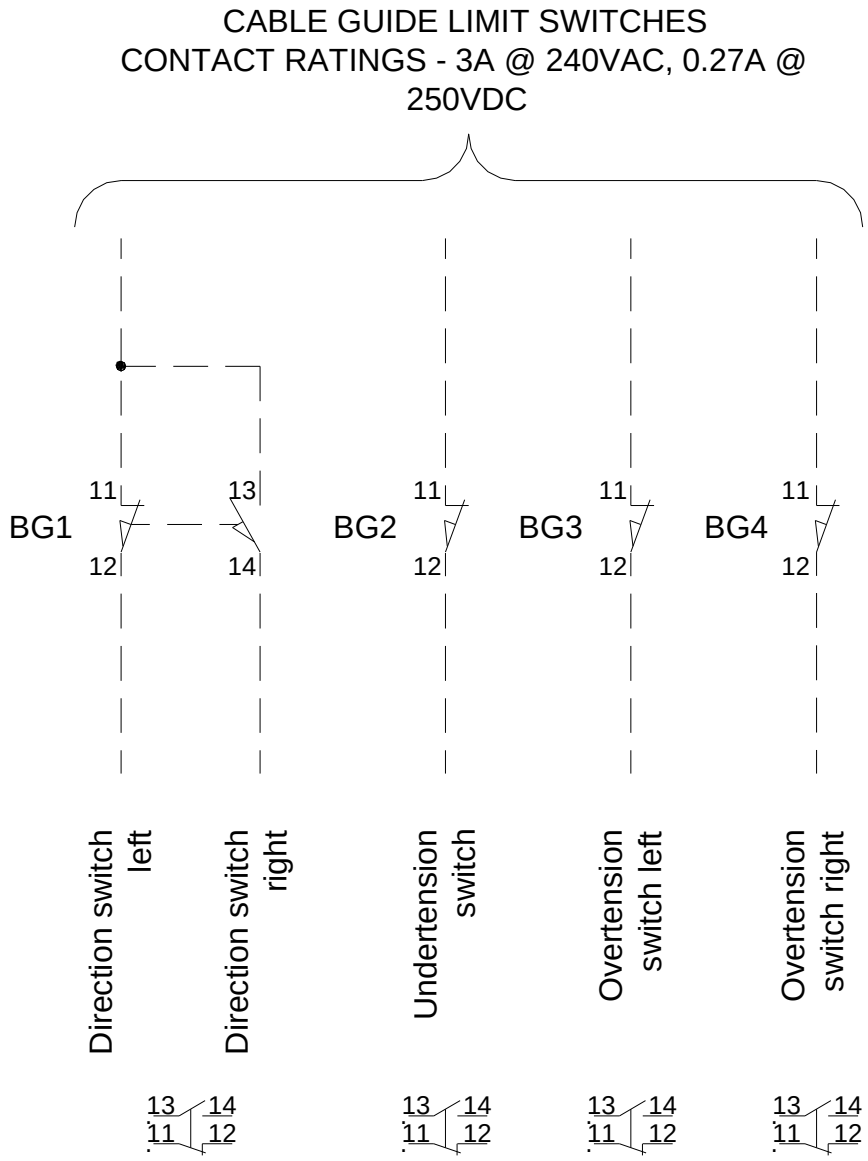
Two normally open cable/hose direction switches BG1 are located inside a common housing as shown in Figure 1. These are actuated by the pendulum arm and are used to indicate from which side the cable/hose is exiting the guide. This information is then used to determine how the reeler should operate. For example, if the cable/hose is exiting on the left and the machine is travelling left, then the reeler must recover cable/hose. On the other hand, if the cable/hose is exiting on the left and the machine is travelling right, then the reeler must payout cable/hose.

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OVERTENSION SWITCHES

Normally closed overtension switches are located on each side of the guide (refer to Figure 1), and are actuated when the spring loaded overtension arms are forced upwards beyond a preset limit by the cable/hose catenary. Tripping of either of these switches should emergency stop machine travel to prevent damage to the cable/hose. Note that the overtension switch signals need to be latched because the actual switches may only be activated briefly.

TYPICAL SCHEMATIC



Printed: 29/10/2012 11:54:00
Serialnumber: 13001696
Device: NXP
Saved: 24/10/2012 08:26:37
Firmware: 4.53
Application: Cavotec (C.121216_ZPMC_Gantry.par, APITQ224 3.20)
Parameter file: T:\Commesse\Corrente\121216 - CavChi ZPMC Sri Lanka\Disegni\Electrica

Index	Text	Value	Unit	Min	Max	ID
P 2.1.1	Max Frequency	60,00	Hz	n/a	n/a	102
P 2.1.2	Current Limit	14,0	A	n/a	n/a	107
P 2.1.3	Motor Nom Voltg	400	V	n/a	n/a	110
P 2.1.4	Motor Nom Freq	50,00	Hz	n/a	n/a	111
P 2.1.5	Motor Nom Speed	706	rpm	n/a	n/a	112
P 2.1.6	Motor Nom Currnt	12,0	A	n/a	n/a	113
P 2.1.7	Motor Cos Phi	0,69		n/a	n/a	120
P 2.1.8	MagnCurrent	5,1	A	n/a	n/a	612
P 2.2.1.1	Control Mode	1 / DI Logic		n/a	n/a	1551
P 2.2.1.2	Default Torque	0,0	%	n/a	n/a	1512
P 2.2.1.3	CONSTReel INMinT	36,0	%	n/a	n/a	1513
P 2.2.1.4	CONSTReel INMaxT	58,0	%	n/a	n/a	1514
P 2.2.1.5	DEC Reel IN MinT	0,0	%	n/a	n/a	1515
P 2.2.1.6	DEC Reel IN MaxT	0,0	%	n/a	n/a	1516
P 2.2.1.7	ACC Reel IN MinT	0,0	%	n/a	n/a	1517
P 2.2.1.8	ACC Reel IN MaxT	0,0	%	n/a	n/a	1518
P 2.2.1.9	CONSTReelOUTMinT	24,0	%	n/a	n/a	1519
P 2.2.1.10	CONSTReelOUTMaxT	38,0	%	n/a	n/a	1520
P 2.2.1.11	DECReel OUT MinT	0,0	%	n/a	n/a	1521
P 2.2.1.12	DECReel OUT MaxT	0,0	%	n/a	n/a	1522
P 2.2.1.13	ACCReel OUT MinT	0,0	%	n/a	n/a	1523
P 2.2.1.14	ACCReel OUT MaxT	0,0	%	n/a	n/a	1524
P 2.2.1.15	Min Torque Limit	0,0	%	n/a	n/a	1525
P 2.2.1.16	Max Torque Limit	100,0	%	n/a	n/a	1526
P 2.2.1.17	InertiaC IN ACC	0,00	%	n/a	n/a	1556
P 2.2.1.18	InertiaC IN DEC	0,00	%	n/a	n/a	1630
P 2.2.1.19	InertiaC OUT ACC	0,00	%	n/a	n/a	1554
P 2.2.1.20	InertiaC OUT DEC	0,00	%	n/a	n/a	1631
P 2.2.1.21	InertiaC Filter	0,10	s	n/a	n/a	1565
P 2.2.1.22	InertiaC Source	0 / MotorEncoder		n/a	n/a	1586
P 2.2.1.23	InertiaC@MaxDiam	100,0	%	n/a	n/a	1632
P 2.2.1.24	TorqueLevelsMode	1 / Absolute		n/a	n/a	1615
P 2.2.1.25	T Const-Linear	1 / Linear		n/a	n/a	1618
P 2.2.1.26	Torque Ramp Time	0,00	s	n/a	n/a	1616
P 2.2.1.27	PulseInAccTorque	0,0	%	n/a	n/a	1638
P 2.2.1.28	PulseIn Acc Time	2,00	s	n/a	n/a	1634
P 2.2.1.29	PulseInDecTorque	0,0	%	n/a	n/a	1639
P 2.2.1.30	PulseIn Dec Time	2,00	s	n/a	n/a	1635
P 2.2.1.31	PulsOutAccTorque	0,0	%	n/a	n/a	1640
P 2.2.1.32	PulseOut AccTime	2,00	s	n/a	n/a	1636
P 2.2.1.33	PulsOutDecTorque	0,0	%	n/a	n/a	1641
P 2.2.1.34	PulseOut DecTime	2,00	s	n/a	n/a	1637
P 2.2.1.35	Pulse@MaxDiamet	100,0	%	n/a	n/a	1642
P 2.2.2.1	Speed Window	0 / No Active		n/a	n/a	1527
P 2.2.2.2	Upper Limit	10	%	n/a	n/a	1558
P 2.2.2.3	Lower Limit	5	%	n/a	n/a	1560
P 2.2.2.4	Minim UpperLimit	5,00	Hz	n/a	n/a	1559
P 2.2.2.5	Minim LowerLimit	5,00	Hz	n/a	n/a	1629
P 2.2.2.6	UpperLim Action	1 / Limit+NoAct		n/a	n/a	1561
P 2.2.2.7	LowerLim Action	1 / Alarm+Alarm		n/a	n/a	1562
P 2.2.2.8	Alarm Delay	1,00	s	n/a	n/a	1563
P 2.2.2.9	Reference Select	0 / AIN2		n/a	n/a	1582
P 2.2.2.10	Enc Freq Mul	1		n/a	n/a	1583
P 2.2.2.11	Enc Ref Div	1		n/a	n/a	1584
P 2.2.2.12	Max Freq@MinDiam	50,00	Hz	n/a	n/a	1628
P 2.2.2.13	SpLimitMaxTorque	100,0	%	n/a	n/a	1643
P 2.2.2.14	Open WindowLogic	0 / Never		n/a	n/a	1644
P 2.2.2.15	OpenWindowLimit	0,00	%	n/a	n/a	1645
P 2.2.2.16	CloseWindowLimit	0,00	%	n/a	n/a	1646
P 2.2.2.17	Acc/Dec If Open	1 / Masked		n/a	n/a	1647
P 2.2.2.18	Speed Fault	1 / Yes		n/a	n/a	1648
P 2.2.2.19	Max Speed Error	5,00	Hz	n/a	n/a	1649
P 2.2.2.20	Check Delay	1,00	s	n/a	n/a	1650
P 2.2.3.1	Speed selection	0 / Jog Speed		n/a	n/a	1528
P 2.2.3.2	Jog speed	10,00	Hz	n/a	n/a	1529
P 2.2.3.3	AIN Min Speed	0,00	Hz	n/a	n/a	1530
P 2.2.3.4	AIN Max Speed	50,00	Hz	n/a	n/a	1531
P 2.2.3.5	Accel Time	3,0	s	n/a	n/a	502
P 2.2.3.6	Decel Time	3,0	s	n/a	n/a	503
P 2.2.3.7	Torque Limit	100,0	%	n/a	n/a	1533
P 2.2.3.8	Diameter Scaling	0 / No		n/a	n/a	1621
P 2.3.1.1	AI1 Signal Sel	AnIN:A.2		n/a	n/a	377
P 2.3.1.2	AI1 Signal Range	1 / 4mA/20%-100%		n/a	n/a	1501
P 2.3.1.3	AI1 Custom Min	0,00	%	n/a	n/a	1503
P 2.3.1.4	AI1 Custom Max	0,00	%	n/a	n/a	1504
P 2.3.1.5	AI1 Filter Time	0,10	s	n/a	n/a	1505
P 2.3.1.6	AI1 Signal Inv	0 / No Inversion		n/a	n/a	1506
P 2.3.2.1	AI2 Signal Sel	AnIN:A.2		n/a	n/a	388
P 2.3.2.2	AI2 Signal Range	0 / 0mA/0%-100%		n/a	n/a	1507
P 2.3.2.3	AI2 Custom Min	0,00	%	n/a	n/a	1508
P 2.3.2.4	AI2 Custom Max	0,00	%	n/a	n/a	1509

Index	Text	Value	Unit	Min	Max	ID
P 2.3.2.5	AI2 Filter Time	0,10	%	n/a	n/a	1510
P 2.3.2.6	AI2 Signal Inv	0 / No Inversion		n/a	n/a	1511
P 2.3.2.7	Speed Range	1 / Bidirect		n/a	n/a	1557
P 2.3.3.1	DI1 Moving	DigIN:A.1		n/a	n/a	1536
P 2.3.3.2	DI2 Reset	DigIN:A.2		n/a	n/a	1537
P 2.3.3.3	DI3 Reel OUT	DigIN:A.3		n/a	n/a	1538
P 2.3.3.4	DI4 Accel	DigIN:A.4		n/a	n/a	1539
P 2.3.3.5	DI5 Decel	DigIN:A.5		n/a	n/a	1540
P 2.3.3.6	DI6 Manual	DigIN:A.6		n/a	n/a	1541
P 2.3.3.7	Manual FWD	DigIN:A.4		n/a	n/a	1542
P 2.3.3.8	Manual REV	DigIN:A.5		n/a	n/a	1543
P 2.3.3.9	Ext Fault Close	DigIN:0.1		n/a	n/a	1544
P 2.3.3.10	Ext Fault Open	DigIN:0.2		n/a	n/a	1545
P 2.3.3.11	SpWindow Reverse	DigIN:0.1		n/a	n/a	1600
P 2.3.4.1	Minimum Diameter	2400	mm	n/a	n/a	1624
P 2.3.4.2	Maximum Diameter	6000	mm	n/a	n/a	1625
P 2.3.4.3	Signal@ Min Diam	20,00	%	n/a	n/a	1627
P 2.3.4.4	Signal@ Max Diam	100,00	%	n/a	n/a	1626
P 2.4.1	Iout 1 signal	AnOUT:A.1		n/a	n/a	464
P 2.4.2	Iout Content	1 / O/P Freq		n/a	n/a	307
P 2.4.3	Iout Filter Time	0,00	s	n/a	n/a	308
P 2.4.4	Iout Invert	0 / No Inversion		n/a	n/a	309
P 2.4.5	Iout Minimum	0 / 0 mA		n/a	n/a	310
P 2.4.6	Iout Scale	100	%	n/a	n/a	311
P 2.4.7	DO1 Content	0 / None		n/a	n/a	312
P 2.4.8	RO1 Content	1 / Ready		n/a	n/a	313
P 2.4.9	RO2 Content	16 / Brake		n/a	n/a	314
P 2.4.10	Freq Supv Lim 1	0 / No		n/a	n/a	315
P 2.4.11	Freq Supv Val 1	0,00	Hz	n/a	n/a	316
P 2.4.12	AIN1 Supv Lim 1	0 / No		n/a	n/a	1567
P 2.4.13	AIN1 Supv Val1	70,00	%	n/a	n/a	1568
P 2.4.14	AIN1 Supv Lim 2	0 / No		n/a	n/a	1569
P 2.4.15	AIN1 Supv Val 2	20,00	%	n/a	n/a	1570
P 2.4.16	Brake Open Delay	0,00	s	n/a	n/a	1579
P 2.4.17	Brake Close Freq	0,50	Hz	n/a	n/a	1580
P 2.4.18	Brk Steady Time	0,50	s	n/a	n/a	1597
P 2.4.19	Brk CloseTimeOut	0,50	s	n/a	n/a	1581
P 2.4.20	Hold Torque Time	0,50	s	n/a	n/a	1587
P 2.4.21	Opt RO3 Signal	DigOUT:D.1		n/a	n/a	1571
P 2.4.22	Opt RO3 Content	14 / AIN1 Superv1		n/a	n/a	1572
P 2.4.23	Opt RO4 Signal	DigOUT:D.2		n/a	n/a	1573
P 2.4.24	Opt RO4 Content	15 / AIN1 Superv2		n/a	n/a	1574
P 2.4.25	Opt RO5 Signal	DigOUT:D.3		n/a	n/a	1575
P 2.4.26	Opt RO5 Content	0 / None		n/a	n/a	1576
P 2.5.1	Brake Chopper	4 / On, No test		n/a	n/a	504
P 2.5.2	Stop Function	1 / Ramping		n/a	n/a	506
P 2.6.1	Motor Ctrl Mode	1 / Closed Loop		n/a	n/a	1500
P 2.6.2	Field WeakngPnt	50,00	Hz	n/a	n/a	602
P 2.6.3	Voltage at FWP	100,00	%	n/a	n/a	603
P 2.6.4	Switching Freq	10,0	kHz	n/a	n/a	601
P 2.6.5	Overvolt Contr	1 / On:NoRamping		n/a	n/a	607
P 2.6.6	Undervolt Contr	1 / Yes		n/a	n/a	608
P 2.6.7.1	U/f Optimization	0 / None		n/a	n/a	109
P 2.6.7.2	U/f Ratio Select	0 / Linear		n/a	n/a	108
P 2.6.7.3	U/f Mid Freq	50,00	Hz	n/a	n/a	604
P 2.6.7.4	U/f Mid Voltg	100,00	%	n/a	n/a	605
P 2.6.7.5	Zero Freq Voltg	1,50	%	n/a	n/a	606
P 2.6.8.1	Speed Control Kp	30		n/a	n/a	613
P 2.6.8.2	Speed Control Ti	30,0	ms	n/a	n/a	614
P 2.6.8.3	Slip Adjust	100	%	n/a	n/a	619
P 2.6.8.4	Encoder1FiltTime	0	ms	n/a	n/a	618
P 2.6.8.5	CurrentControlKp	40,00	%	n/a	n/a	617
P 2.6.9.1	TorqueControl Kp	150		n/a	n/a	1535
P 2.6.9.2	TorqueControl Ki	10		n/a	n/a	1534
P 2.6.9.3	SpeedLimAccTime	0,1	s	n/a	n/a	103
P 2.6.9.4	SpeedLimDecTime	0,2	s	n/a	n/a	104
P 2.7.1	Encoder Superv	0 / Active		n/a	n/a	1546
P 2.7.2	AI1 4mAFaultResp	2 / Fault		n/a	n/a	1547
P 2.7.3	AI1 4mAF Preset	4	mA	n/a	n/a	1549
P 2.7.4	AI2 4mAFaultResp	2 / Fault		n/a	n/a	1548
P 2.7.5	AI2 4mAF Preset	4	mA	n/a	n/a	1550
P 2.7.6	External Fault	2 / Fault		n/a	n/a	701
P 2.7.7	Input Phase Supv	2 / Fault		n/a	n/a	730
P 2.7.8	UVolt Fault Resp	2 / Fault		n/a	n/a	727
P 2.7.9	OutputPh. Superv	2 / Fault		n/a	n/a	702
P 2.7.10	Earth Fault	2 / Fault		n/a	n/a	703
P 2.7.11	Motor Therm Prot	2 / Fault		n/a	n/a	704
P 2.7.12	MotAmbTempFactor	0,0	%	n/a	n/a	705
P 2.7.13	MTP f0 Current	40,0	%	n/a	n/a	706
P 2.7.14	MTP Motor T	45	min	n/a	n/a	707
P 2.7.15	Motor Duty Cycle	100	%	n/a	n/a	708
P 2.7.16	Stall Protection	0 / No Action		n/a	n/a	709
P 2.7.17	Stall Current	15,6	A	n/a	n/a	710
P 2.7.18	Stall Time Lim	15,00	s	n/a	n/a	711
P 2.7.19	Stall Freq Lim	25,00	Hz	n/a	n/a	712
P 2.7.20	Underload Protec	0 / No Action		n/a	n/a	713
P 2.7.21	UP fnom Torque	50,0	%	n/a	n/a	714
P 2.7.22	UP f0 Torque	10,0	%	n/a	n/a	715
P 2.7.23	UP Time Limit	20,00	s	n/a	n/a	716
P 2.7.24	ThermistorF.Resp	2 / Fault		n/a	n/a	732
P 2.7.25	FBComm.FaultResp	2 / Fault		n/a	n/a	733
P 2.7.26	SlotComFaultResp	2 / Fault		n/a	n/a	734

Index	Text	Value	Unit	Min	Max	ID
P 2.7.27	PwrUnitTempSense	0 / Read		n/a	n/a	1577
P 2.7.28	Overspeed Protec	0 / No Action		n/a	n/a	1598
P 2.7.29	Overspeed Thresh	55,00	Hz	n/a	n/a	1599
P 2.8.1	Wait Time	0,50	s	n/a	n/a	717
P 2.8.2	Trial Time	30,00	s	n/a	n/a	718
P 2.8.3	Start Function	0 / Ramping		n/a	n/a	719
P 2.8.4	Undervolt. Tries	0	x	n/a	n/a	720
P 2.8.5	Overvolt. Tries	0	x	n/a	n/a	721
P 2.8.6	Overcurr. Tries	0	x	n/a	n/a	722
P 2.8.7	4mA Fault Tries	0	x	n/a	n/a	723
P 2.8.8	MotTempF Tries	0	x	n/a	n/a	726
P 2.8.9	Ext.Fault Tries	0	x	n/a	n/a	725
P 2.9.1	Commands Word	0 / Bus CW		n/a	n/a	1588
P 2.9.2	SpeedRef Word	0 / Bus Ref Word		n/a	n/a	1589
P 2.9.3	DigInputs Active	0 / No		n/a	n/a	1590
P 2.9.4	FB Data Out1 Sel	1		n/a	n/a	1593
P 2.9.5	FB Data Out2 Sel	3		n/a	n/a	1594
P 2.9.6	FB Data Out3 Sel	4		n/a	n/a	1595
P 2.9.7	FB Data Out4 Sel	37		n/a	n/a	1596
P 2.9.8	FB Data Out5 Sel	0		n/a	n/a	1610
P 2.9.9	FB Data Out6 Sel	0		n/a	n/a	1611
P 2.9.10	FB Data Out7 Sel	0		n/a	n/a	1612
P 2.9.11	FB Data Out8 Sel	0		n/a	n/a	1613
P 2.9.12	TorqueRef Word	1 / Bus PDI1		n/a	n/a	1614
P 2.10.1	License Key	0		n/a	n/a	1566
P 2.10.2	SerialNumberKey	52175		n/a	n/a	
P 3.1	Control Place	1 / I/O Terminal		n/a	n/a	125
P 3.3	Keypad Direction	0 / Forward		n/a	n/a	123
P 3.4	StopButtonActive	1 / Yes		n/a	n/a	114
P 3.5	String	1		n/a	n/a	1617
P 3.6	Auto Multimon	1 / Yes		n/a	n/a	1619
P 6.3.4	Autom. BackUp	0 / Yes		n/a	n/a	820
P 6.5.2	Parameter Lock	0 / ChangeEnable		n/a	n/a	819
P 6.5.3	Startup wizard	0 / No		n/a	n/a	826
P 6.5.4	Multimon. items	0 / ChangeEnable		n/a	n/a	822
P 6.5.5	OPTAF Remove	0		n/a	n/a	832
P 6.6.1	Default page	0.		n/a	n/a	
P 6.6.2	Default page/OM	1		n/a	n/a	
P 6.6.3	Timeout time	60	s	n/a	n/a	804
P 6.6.4	Contrast	18		n/a	n/a	805
P 6.6.5	Backlight time	10	min	n/a	n/a	818
P 6.7.1	InternBrakeRes	0 / Not conn.		n/a	n/a	821
P 6.7.2	Fan control	0 / Continuous		n/a	n/a	825
P 6.7.3	HMI ACK timeout	200	ms	n/a	n/a	823
P 6.7.4	HMI retry	5		n/a	n/a	824
P 6.7.5	Sine Filter	0 / Not conn.		n/a	n/a	
P 7.1.1.1	AI1 mode	2 / 4...20mA		n/a	n/a	
P 7.1.1.2	AI2 mode	2 / 4...20mA		n/a	n/a	
P 7.1.1.3	AO1 mode	1 / 0...20mA		n/a	n/a	
P 7.3.1.1	Pulse revolution	1024		n/a	n/a	
P 7.3.1.2	Invert direction	0 / No		n/a	n/a	
P 7.3.1.3	Reading rate	1 / 1 ms		n/a	n/a	
P 7.3.1.4	Encoder type	1 / A,B=Speed		n/a	n/a	
P 7.5.1.1	Slave Address	126		n/a	n/a	
P 7.5.1.2	Baud Rate	10 / Auto		n/a	n/a	
P 7.5.1.3	PPO Type	1 / PPO1		n/a	n/a	
P 7.5.1.4	Operate Mode	2 / ByPass		n/a	n/a	

CAVOTEC
Connecting mobile equipment

Commissioning Procedure

GANTRY CABLE REEL
LINEAR TORQUE LEVELS
VACON NXP DRIVE

INSTALLATION

Prepare and install cable reel as described in other chapters of this manual. Suggested steps are as follows:

1. Mount drive unit on crane.
2. Install control panel on crane, install cables (refer to Cable reel manual for cable details; see below for further details).
3. Check all cabling, especially encoder connections and brake
4. Verify motor direction using manual control mode (for drives with multiple motors it may be useful to connect each motor individually and test).
5. Install cable drum.
6. Install cable, using the manual mode and reeling in push button. While doing is possible to adjust the rotary transducer to give the mA output indicated in section SETTING CORRECT TORQUE LEVEL. mA output can be monitored by multimeter at the rotary transducer, or using the DTC panel (see initialize drive). Adjust drum full limit and drum empty switch(es) also.
7. Verify the reeling torque in all modes (see below).
8. Record final settings.

The DTC cable reel drive is delivered on AISI electrical enclosure mounted on a pedestal. The braking resistor is mounted on the same pedestal closer electrical panel.

The Vacon Installation and Start-Up manual and Programming manual are an integrated part of this manual and should be studied before installation.

CAVOTEC
Connecting mobile equipment

Commissioning Procedure

1. When the drive and all the auxiliary components have been mounted and wired as mechanical and electrical drawings and according to Vacon mounting and wiring instructions (see Vacon start-up and installation manual) the drive and the cable reel has to be tested and set to work with the correct torque.

2. Note the following important items during first installation:
   Cable to encoder should be a three pair twisted with screen round each pair and with a common screen, minimum 0.5 mm^2.
   Encoder module channels to be connected are: 0V (Terminal 9), +24V (Terminal 10), A+ (Terminal 1) and A- (Terminal 2), B+ (Terminal 3) and – B- (Terminal 4).
   Note: Signals Z+ and Z- are not used.

   Check electrical diagrams from detailed Encoder connection.
   Be very careful to connect 0V and +24V to the correct terminals – wrong connection can damage the encoder.
   Which channel is to be A and B can be identified when the reel is tested. Inverted channels will be noticed by encoder errors or abnormal speed at start-up. When switching cables from channel A to B ensure that the + and - signal lines are maintained. Screen must be wired at the encoder module terminal X2 where SH is indicated.

3. If cable is already on the drum, block the drum with a rope or remove the brake lines from the terminals on the panel before proceeding. This will prevent the drum turning and the cable being dropped and damaged during the set-up.

4. Power up the drive (see note below). Drive panel lights up and drive boots (approx. 10 sec).
   Crane will receive Cable Reel Ready signal from cable reel panel.

5. Drive has normally been initialised in the factory but it is recommended to do a new «first start» on site especially if cable lengths from drive to motor are long.

Commissioning Procedure

Notes:

Motors:
If two or three motors are powered from the same drive the motors must be connected with the correct phase sequence U / V / W so they run in the correct direction.

If two motors are used on the same drive and one has to be changed, then IT MUST BE REPLACED BY AN IDENTICAL MOTOR - or both motors must be replaced by identical motors. Drive must then be initialised again.
Note also that motor (one of the motors) must be fitted with a shaft to fix the encoder.

If two or three motors are used the motor values for motor Amperage given in the parameter group 2.1 must be the sum of the two or three motors values.

The DTC is always powered up if main circuit breaker is closed.

A motor overheating will powered off the drive.

Important! Pay attention not to power on the drive more the 5 time each 10 minutes.

CAVOTEC
Connecting mobile equipment

Commissioning Procedure

STARTING CABLE REEL

1. In the following, reference is made to the VACON drive manual and certain knowledge of the drive functions are presumed. For fast reference a few important pages from the drive manual are enclosed showing the main functions of the keyboard.

2. Keep cable reel drum blocked with a rope before correct torque level is set.
   The first time the drive is started it could default into doing a « first start ».
   This means the drive will for a few seconds not give torque, and will check the connection to motor, etc. To avoid any cable drop block the drum with a rope or remove lines to brake so it is still closed the first time the drive is started.

3. Torque level setting value is shown on the first line on the drive panel in mA.

4. Test motor rotation direction.
   Put drive in “manual mode” and press “Reel in” push button.
   Checks drum rotation direction.
   If not correct shut off drive and change over motor phase lines to get correct rotation direction.
   ATTENTION: if a change over motor phase line was necessary it is important to change channel A with B at encoder to avoid encoder fault.
   (A + ↔ B+ A - ↔ B -)

   WARNING: Check VACON manual for safety procedure as the drive has condensers and is under tension approx. 5 minutes after power off.

   If cable reel has two or more motors this has to be done with one motor at a time so you are sure the motors rotate the same way. If not switch one motor phase over so motors rotate in same direction.
   Check one motor at a time by switching off one motor protection at a time.

For Vertical retrieve cable reel (Spreader cable reel):

5. Set rotary transducer value.
   When working in linear mode it is important to set the value of the rotary transducer to have the correct torque input value.
   It is possible set the rotary transducer as follows:

   - Rotary transducer value decreasing reeling OUT (parameter 2.3.1.6=0 NO Inversion)

   Put drive in "manual mode" and reel in all the cable then stop and adjust rotating transducer by turning the plastic adjusting screw nearest to the rotary transducer. The signal may be monitored on the drive display, or directly with a multimeter.
   Note: if the black adjusting screw at the opposite end of the limit switch assembly is adjusted, the rotary transducer will need to be reset.

   The signal from the transducer must be 4.5 mA at drum full.
   Reel out using "manual mode" and check that rotating transducer is decreasing the value (see 3rd line on panel).
   If drum full position is not available use table on next page to set rotary transducer output at actual position (left graphic/table).

   - Rotary transducer value incresing reeling OUT (parameter 2.3.1.6=1 Ref Inverted)

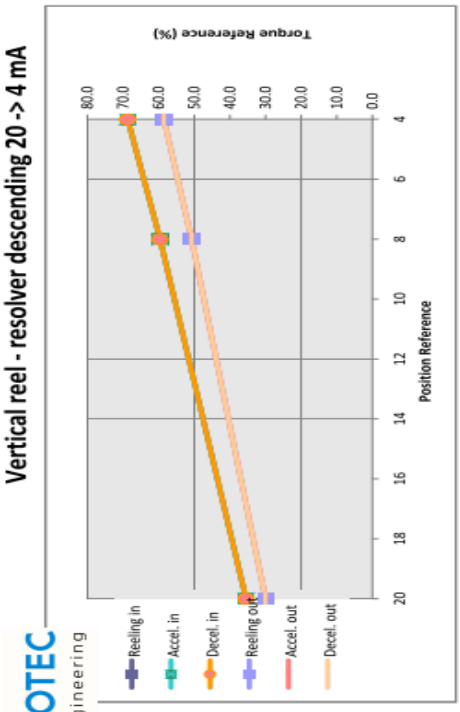
   Put drive in "manual mode" and reel in all the cable then stop and adjust rotating transducer by turning the plastic adjusting screw nearest to the rotary transducer. The signal may be monitored on the drive display, or directly with a multimeter.
   Note: if the black adjusting screw at the opposite end of the limit switch assembly is adjusted, the rotary transducer will need to be reset.

   The signal from the transducer must be 19.5 mA at drum full.
   Reel out using "manual mode" and check that rotating transducer is increasing the value (see 3rd line on panel).
   If drum full position is not available use table on next page to set rotary transducer output at actual position (right graphic/table).

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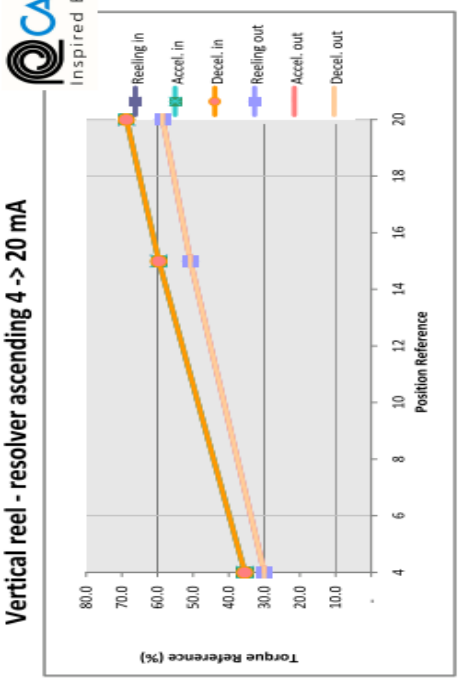
Commissioning Procedure

Table reference for rotary transducer set vertical



descending (20->4mA) Drive parameter N°: 13.10= YES

Reeled out lays from drum	mA per lay	Lays on drum	mA per lay
Drum full			
0	19.90		
1	19.32		
2	18.75		
3	18.17		
4	17.60		
5	17.02		
6	16.45		
7	15.87		
8	15.29		
9	14.72		
10	14.14		
11	13.57		
12	12.99		
13	12.42		
14	11.84		
15	11.26		
16	10.69		
17	10.11		
18	9.54		
19	8.96		
20	8.39		
21	dead lay		
22	dead lay		
23	dead lay		
Drum empty			



ascending (4->20mA) Drive parameter N°: 13.10 = NO

Reeled out lays from drum	mA per lay	Lays on drum	mA per lay
Drum full			
0	4.10		
1	4.68		
2	5.25		
3	5.83		
4	6.40		
5	6.98		
6	7.55		
7	8.13		
8	8.71		
9	9.28		
10	9.86		
11	10.43		
12	11.01		
13	11.58		
14	12.16		
15	12.74		
16	13.31		
17	13.89		
18	14.46		
19	15.04		
20	15.61		
21	dead lay		
22	dead lay		
23	dead lay		
Drum empty			

6. Connect cable end to hook / headlock.

7. Test the cable reel. When reeling out the first time be prepared to stop the hook lowering with the emergency stop if the torque is not sufficient when hook is at its lowest point.

8. Adjust torque as described in section “Setting the Correct Torque Levels”.

9. Test the crane emergency stop and check that the reel will stop and brake will apply if there is any emergency stop of the hoist.
   - or, if desired -
   The cable reel drive can remain powered up if the crane is emergency stopped.

10. Test a cable reel drive fault to check that the drive will send a fault signal to the crane and stop the crane hoist. The cable reel should stop with the brake applied (this can be tested by removing one cable from the encoder module).
    Reset drive by shutting down the system or by reset command on panel or by remote reset.

11. Check function at power loss, if the cable reel drive has a power loss also the logic’s for the brake should be without power. The power to the drive and the brake control circuit must be interlocked.

12. Check the manual drive function.
    1. Set automatic/manual selector switch in manual to enable manual control.
    2. Check the up and down functions with UP and DOWN buttons.
    3. Set the top limit switch and the bottom limit switch in the travel limit switch mounted in/on the collector on the cable reel. (Note: if the first cam, with black adjusting screw, is turned, the rotary transducer will need to be set again).

    To change manual reeling speed set value in the drive parameters 11.04 and 11.05 (same value).

For Horizontal cable reel (gantry movement):

5. Set rotary transducer value.
   When working in linear mode it is important to set the value of the rotary transducer to have the correct torque input value.
   It is possible set the rotary transducer as follows:

- Rotary transducer value increasing reeling IN (parameter 2.3.1.6=0 NO Inversion)

Put drive in "manual mode" and reel in the dead lays then stop and adjust rotating transducer by turning the plastic adjusting screw nearest to the rotary transducer. The signal may be monitored on the drive display, or directly with a multimeter.
Note: if the black adjusting screw at the opposite end of the limit switch assembly is adjusted, the rotary transducer will need to be reset.

The signal from the transducer must be 4.5 mA at drum empty.
Reel in using "manual mode" and check that rotating transducer is increasing the value (see 3rd line on panel).
If drum empty position is not available use table on next page to set rotary transducer output at actual position (left graphic/table).

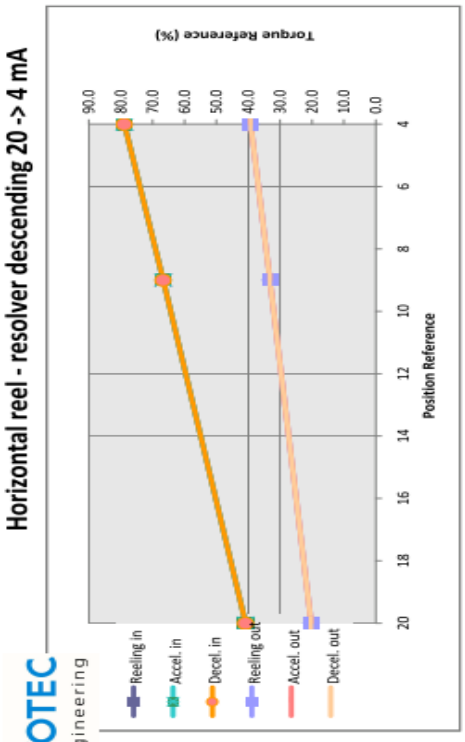
- Rotary transducer value decreasing reeling IN (parameter 2.3.1.6=1 Ref Inverted)

Put drive in "manual mode" and reel in the dead lays then stop and adjust rotating transducer by turning the plastic adjusting screw nearest to the rotary transducer. The signal may be monitored on the drive display, or directly with a multimeter.
Note: if the black adjusting screw at the opposite end of the limit switch assembly is adjusted, the rotary transducer will need to be reset.

The signal from the transducer must be 19.5 mA at drum empty.
Reel in using "manual mode" and check that rotating transducer is decresing the value (see 3rd line on panel).
If drum empty position is not available use table on next page to set rotary transducer output at actual position (right graphic/table).

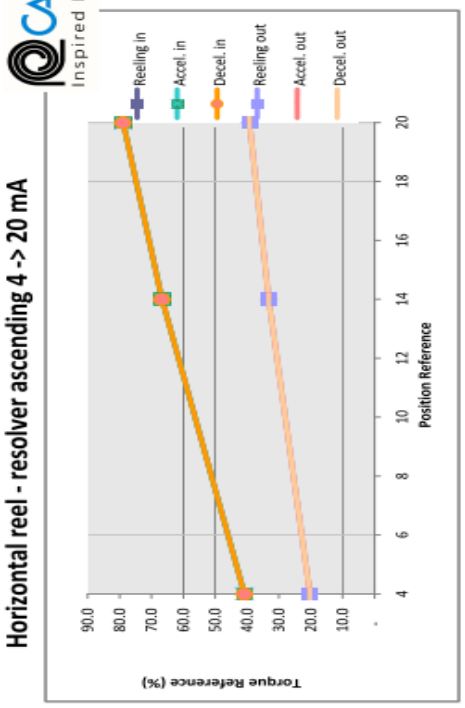
Commissioning Procedure

Table reference for rotary transducer set horizontal



descending (20->4mA) Drive parameter N°: 13.10= YES

Drum empty	Lays on drum	mA per lay	Lays on drum	mA per lay
	1	dead lay	35	8.42
	2	19.90		
	3	19.55		
	4	19.20		
	5	18.86		
	6	18.51		
	7	18.16		
	8	17.81		
	9	17.47		
	10	17.12		
	11	16.77		
	12	16.42		
	13	16.07		
	14	15.73		
	15	15.38		
	16	15.03		
	17	14.68		
	18	14.33		
	19	13.99		
	20	13.64		
	21	13.29		
	22	12.94		
	23	12.60		
	24	12.25		
	25	11.90		
	26	11.55		
	27	11.20		
	28	10.86		
	29	10.51		
	30	10.16		
	31	9.81		
	32	9.47		
Drum Full	33	9.12		
	34	8.77		



ascending (4->20mA) Drive parameter N°: 13.10 = NO

Drum Empty	Lays on drum	mA per lay	Lays on drum	mA per lay
	1	dead lay	35	15.58
	2	4.10		
	3	4.45		
	4	4.80		
	5	5.14		
	6	5.49		
	7	5.84		
	8	6.19		
	9	6.53		
	10	6.88		
	11	7.23		
	12	7.58		
	13	7.93		
	14	8.27		
	15	8.62		
	16	8.97		
	17	9.32		
	18	9.67		
	19	10.01		
	20	10.36		
	21	10.71		
	22	11.06		
	23	11.40		
	24	11.75		
	25	12.10		
	26	12.45		
	27	12.80		
	28	13.14		
	29	13.49		
	30	13.84		
	31	14.19		
	32	14.53		
Drum Full	33	14.88		
	34	15.23		

5. Connect cable end to centre feeding point.

6. Adjust torque as described in section “Setting the Correct Torque Levels”. Test drive reel in all working modes.

7. Remember that torque should be set as low as possible to minimise cable pull and increase cable life, but torque must be high enough to ensure that cable is always following the gantry movement properly without any tendency of slack.

9. Test the crane emergency stop.
   When crane is emergency stopped:
   Check that the reel will stop and brake will apply

   Note:
   If crane is travelling at full speed in reeling out direction and you emergency stop the crane the cable reel will probably give a slack cable signal.
   This is most likely because the stop time at emergency stop is faster than normal ramp-down time, so the forces to stop the cable are much bigger. The cable reel is not dimensioned for these extraordinary forces and will drop a little bit of cable. This is totally normal and not a malfunction.

   Reset any slack cable by hand using the manual speed control mode and the up / down push buttons, positioning the cable correctly in the guide before driving again.

10. Test the slack cable switch in the cable guide.
    The slack cable switch should stop the crane gantry motion and must be memorised by the crane control as the signal will appear and then, due to crane movement before it comes to a rest, disappear again. In such a case the cable will be on the wrong side of the guide so it is not allowed to drive on.

    The slack cable fault may ONLY be reset after checking the reason for the fault and after inspecting the cable position on cable guide.

    We recommend strongly having the slack cable reset switch located near the cable guide together with the cable guide Manual Operation switch which can be used to tension and reposition the cable in the cable guide again. After a slack cable fault the cable must be repositioned and tensioned manually.

    Do NOT have a slack cable reset in the driver cabin.

11. Check function of centre point bypass travel limit switch (drum full).
    The centre point by-pass switch is located in the travel limit switch in/on the collector. This switch is used to by-pass the slack cable fault signal when the crane moves over the centre point and the slack cable fault signal will indicate. For setting instructions see appropriate chapter in the manual.

12. Check function of the over tension switches.
    Check switches when accelerating and reeling out at empty drum. There should be some margin in switch movement before they indicate to eliminate false indications.

13. Check function of centre point right/left detection switch.
    This signal is normally not needed for the function of the DTC cable reel.
    This signal can be used together with gantry moving direction signals from the crane control PLC to calculate reeling IN, reeling OUT signal to cable reel.

14. Test a cable reel drive fault to check that the drive will send a fault signal to the crane and stop the crane movement. The cable reel should stop with the brake applied (this can be tested by removing one cable from the encoder module).

    Reset drive by shutting down the system or by reset command on panel or by remote reset.

15. Check function at power loss, if the cable reel drive has a power loss also the logic's for the brake should be without power. The power to the drive and the brake control circuit should be interlocked.

16. Check the manual operation.
    This function can be used to reset and tension the cable if there is a cable slack fault, or for cable maintenance.

CAVOTEC
Connecting mobile equipment

Commissioning Procedure

SETTING THE CORRECT TORQUE LEVEL
VERTICAL CABLE REEL

The drive has been shipped with parameters set in the drive according to the parameter list in this manual. The reeling torque is defined by parameters in the drive that must be adjusted on site according to local conditions.

The torque must be adjusted at two positions – at drum full condition, and also at drum empty condition. A third parameter set the offset for reeling out condition. Refer to table of parameters below.

To increase the torque, increase the value of the relevant parameter. Likewise, decrease the reeling torque by decreasing the appropriate parameter. Test for a given mode of operation, and change the relevant parameter up or down as required. Make small changes of a few %, and re-test, continuing this cycle until the reeling torque (cable tension) is correct for the given mode.

Suggested steps for setting torque

Parameter	**To be adjusted when**	**Initial setting**	**Final setting**
2.2.1.3	reeling in drum empty(% of max torque)		
2.2.1.4	reeling in drum full(% of max torque)		
2.2.1.9	reeling out drum empty (% of reeling in set)		
2.2.1.10	reeling out drum full (% of reeling in set)		

Set the rotary transducer output to the correct mA value as detailed in previous paragraph.

Start with drum empty condition.
At this location the parameters to be adjusted are 2.2.1.3 for reeling in and 2.2.1.9 for reeling out (%).

Before start to test the cable reel assure that the setting torque is sufficient to hold the cable when the spreader is in the lowest position (drum almost empty): if not, set the torque properly. To do it moves the spreader at the lowest position and reel in hoist up the cable at maximum speed. This is the most critical situation for the cable reel due to acceleration up + full speed and maximum weight of the cable hanging from the cable reel.

If the torque is not enough increase it by 2% - 5% each time, and re-test.

When the cable tension is correct for this mode of operation (reeling in, accelerating) at this location (drum empty), continue the test and note the cable tension once full speed has been reached (reeling out, constant speed).

Reel in all the cable on the drum, test and adjust when necessary for each mode of operation. In this location, the parameter 2.2.1.4 and 2.2.1.10 has to be adjusted.

CAVOTEC
Connecting mobile equipment

Commissioning Procedure

SETTING THE CORRECT TORQUE LEVEL
HORIZONTAL CABLE REEL

The drive has been shipped with parameters set in the drive according to the parameter list in this manual. The reeling torque is defined by parameters in the drive that must be adjusted on site according to local conditions.

The torque must be adjusted at two positions – at drum full condition, and also at drum empty condition. A third parameter set the offset for reeling out condition. Refer to table of parameters below.

To increase the torque, increase the value of the relevant parameter. Likewise, decrease the reeling torque by decreasing the appropriate parameter. Test for a given mode of operation, and change the relevant parameter up or down as required. Make small changes of a few %, and re-test, continuing this cycle until the reeling torque (cable tension) is correct for the given mode.

Important! Normally the drive runs in Remote control mode ("L" is not shown on drive display). If the drive has been set to Local control mode, ***always*** confirm that the reference value shown in the top line of the display is "0.0" before setting the drive back to Remote.

Suggested steps for setting torque

Parameter	**To be adjusted when**	**Initial setting**	**Final setting**
2.2.1.3	reeling in drum empty(% of max torque)	36	
2.2.1.4	reeling in drum full(% of max torque)	58	
2.2.1.9	reeling out drum empty (% of reeling in set)	24	
2.2.1.10	reeling out drum full (% of reeling in set)	38	

Set the rotary transducer output to the correct mA value as detailed in previous paragraph.

Start with drum full condition (crane at cable feed-point).
At this location the parameters to be adjusted are 2.2.1.4 for reeling in and 2.2.1.10 for reeling out.

Gantry away from the feed-point at full speed and full acceleration. Note how the cable acts during the acceleration phase – if it is too tight, decrease the parameter applicable to this mode of operation (reeling out offset→ Parameter 2.2.1.10); if cable is too slack, increase the value of the parameter.

The torque is correct when the cable is pulled against the rollers of the guide, but does not pull tightly against the two last (lowest) roller(s). The cable should form a single, continuous curve as it passes into, through, and from the guide.

Change the parameter value a little each time, e.g. 2%, and re-test.

When the cable tension is correct for this mode of operation (reeling out, accelerating) at this location (drum full), continue the test and note the cable tension once full speed has been reached (reeling out, constant speed).

Gantry the crane back to the starting point as required keeping the crane within several metres of the starting point.
Test and adjust when necessary for each mode of operation.
Gantry to the other limit of travel (drum empty) and repeat the tests. In this location, the parameter 2.2.1.9 and 2.2.1.10 must be adjusted.

Record the values obtained for future reference.

7. CONTROL KEYPAD

The control keypad is the link between the Vacon frequency converter and the user. The Vacon NX control keypad features an alphanumeric display with seven indicators for the Run status (RUN, , READY, STOP, ALARM, FAULT) and three indicators for the control place (I/O term/ Keypad/BusComm). There are also three Status Indicator LEDs (green - green - red), see Status LEDs (green – green – red) below. The control information, i.e. the number of menu, description of menu or the displayed value and the numeric information are presented on three text lines.

The frequency converter is operable through the nine push-buttons of the control keypad. Furthermore, the buttons serve the purposes of parameter setting and value monitoring.

The keypad is detachable and isolated from the input line potential.

7.1 Indications on the Keypad display

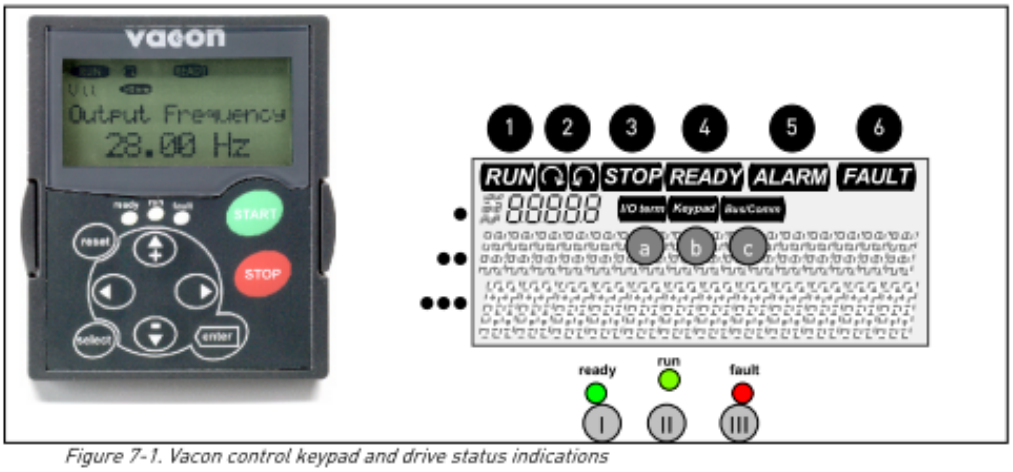


Figure 7-1. Vacon control keypad and drive status indications

7.1.1 Drive status indications (See control keypad)

The drive status indications tell the user what the status of the motor and the drive is and whether the motor control software has detected irregularities in motor or frequency converter functions.

1. RUN = Motor is running; Blinks when the stop command has been given but the frequency is still ramping down.
2. = Indicates the direction of motor rotation.
3. STOP = Indicates that the drive is not running.
4. READY = Lights when AC power is on. In case of a trip, the symbol will not light up.
5. ALARM = Indicates that the drive is running outside a certain limit and a warning is given.
6. FAULT = Indicates that unsafe operating conditions were encountered due to which the drive was stopped.

7.1.2 *Control place indications* (See control keypad)

The symbols *I/O term*, *Keypad* and *Bus/Comm* (see Figure 7-1) indicate the choice of control place made in the Keypad control menu (M3) (see chapter 7.3.3).

a *I/O term* = I/O terminals are the selected control place; i.e. START/STOP commands or reference values etc. are given through the I/O terminals.

b *Keypad* = Control keypad is the selected control place; i.e. the motor can be started or stopped, or its reference values etc. altered from the keypad.

c *Bus/Comm* = The frequency converter is controlled through a fieldbus.

7.1.3 *Status LEDs (green – green – red)* (See control keypad)

The status LEDs light up in connection with the READY, RUN and FAULT drive status indicators.

I ● = Illuminates with the AC power connected to the drive and no faults are active. Simultaneously, the drive status indicator READY is lit up.

II ● = Illuminates when the drive is running. Blinks when the STOP button has been pushed and the drive is ramping down.

III ● = Blinks when unsafe operating conditions were encountered due to which the drive was stopped (Fault Trip). Simultaneously, the drive status indicator FAULT blinks on the display and the fault description can be seen, see chapter 7.3.4, Active Faults.

7.1.4 *Text lines* (See control keypad)

The three text lines (•, ••, •••) provide the user with information on his present location in the keypad menu structure as well as with information related to the operation of the drive.

• = Location indication; displays the symbol and number of menu, parameter etc. Example: M2 = Menu 2 (Parameters); P2.1.3 = Acceleration time

•• = Description line; Displays the description of menu, value or fault.

••• = Value line; Displays the numerical and textual values of references, parameters etc. and the number of submenus available in each menu.

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7.2 Keypad push-buttons

The Vacon alphanumeric control keypad features 9 push-buttons that are used for the control of the frequency converter (and motor), parameter setting and value monitoring.

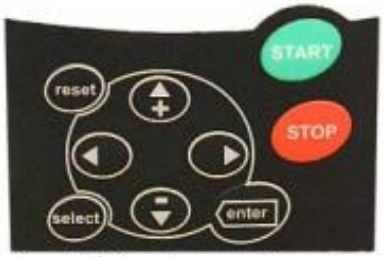


Figure 7-2. Keypad push-buttons

7.2.1 Button descriptions

reset = This button is used to reset active faults (see Chapter 7.3.4).

select = This button is used to switch between two latest displays. May be useful when you want to see how the changed new value influences some other value.

enter = The Enter button serves for:
1) confirmation of selections
2) fault history reset (2...3 seconds)

▲ + = Browser button up
Browse the main menu and the pages of different submenus.
Edit values.

- ▼ = Browser button down
Browse the main menu and the pages of different submenus.
Edit values.

◄ = Menu button left
Move backward in menu.
Move cursor left (in parameter menu).
Exit edit mode.
Hold down for 3 seconds to return to main menu.

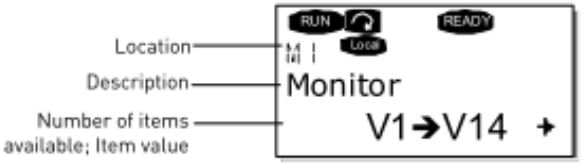
► = Menu button right
Move forward in menu.
Move cursor right (in parameter menu).
Enter edit mode.

start = Start button.
Pressing this button starts the motor if the keypad is the active control place. See Chapter 7.3.3.

stop = Stop button.
Pressing this button stops the motor (unless disabled by parameter R3.4/R3.6).
See Chapter 7.3.3.

7.3 Navigation on the control keypad

The data on the control keypad are arranged in menus and submenus. The menus are used for example for the display and editing of measurement and control signals, parameter settings (chapter 7.3.2), reference values and fault displays (chapter 7.3.4). Through the menus, you can also adjust the contrast of the display (page 100).



The first menu level consists of menus M1 to M7 and is called the *Main menu*. The user can navigate in the main menu using the *Browser buttons* up and down. The desired submenu can be entered from the main menu using the *Menu buttons*. When there still are pages to enter under the currently displayed menu or page, you can see an arrow [→] in the lower right corner of the display and by pressing the *Menu button right*, you can reach the next menu level.

The control keypad navigation chart is shown on the next page. Please note that the menu ***M1*** is located in the lower left corner. From there you will be able to navigate your way up to the desired menu using the menu and browser buttons.

More detailed descriptions of the menus you will find later in this Chapter.

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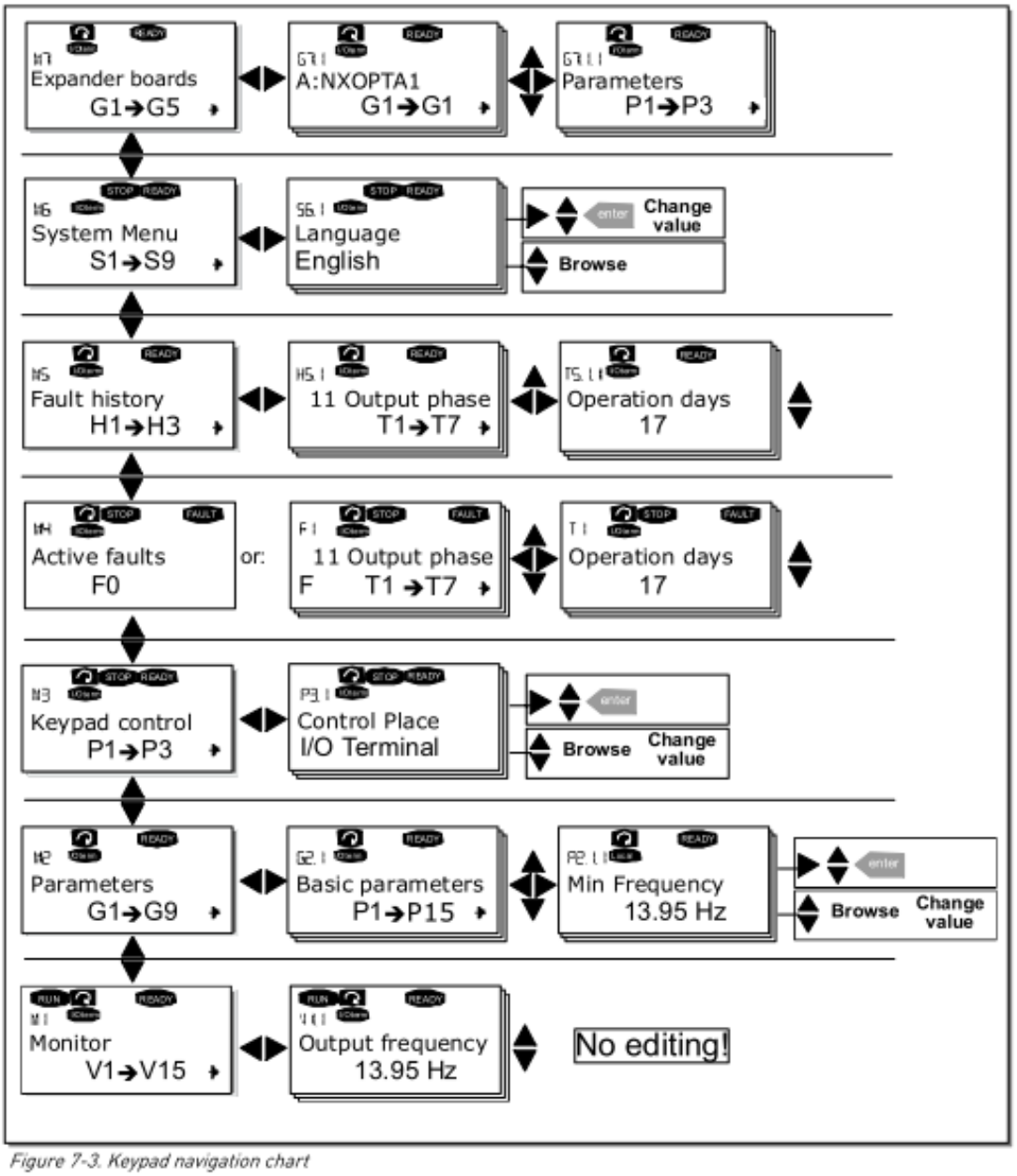


Figure 7-3. Keypad navigation chart

7.3.1 Monitoring menu (M1)

You can enter the Monitoring menu from the Main menu by pushing the *Menu button right* when the location indication **M1** is visible on the first line of the display. How to browse through the monitored values is presented in Figure 7-4.
The monitored signals carry the indication **V#.#** and they are listed in Table 7-1. The values are updated once every 0.3 seconds.

This menu is meant only for signal checking. The values cannot be altered here. For changing values of parameters see Chapter 7.3.2.

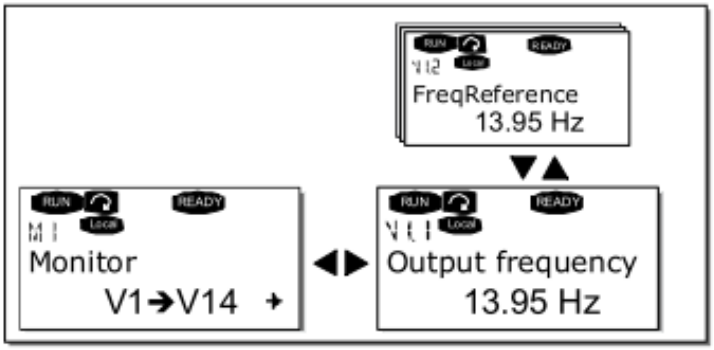


Figure 7-4. Monitoring menu

Code	Signal name	Unit	Description
V1.1	Output frequency	Hz	Frequency to the motor
V1.2	Frequency reference	Hz	
V1.3	Motor speed	rpm	Calculated motor speed
V1.4	Motor current	A	Measured motor current
V1.5	Motor torque	%	Calculated motor shaft torque
V1.6	Motor power	%	Calculated motor shaft power
V1.7	Motor voltage	V	Calculated motor voltage
V1.8	DC-link voltage	V	Measured DC-link voltage
V1.9	Unit temperature	°C	Heat sink temperature
V1.10	Motor temperature	%	Calculated motor temperature. See All in One application manual.
V1.11	Voltage input	V	AI1*
V1.12	Current input	mA	AI2*
V1.13	DIN1, DIN2, DIN3		Digital input statuses
V1.14	DIN4, DIN5, DIN6		Digital input statuses
V1.15	DO1, RO1, RO2		Digital and relay output statuses
V1.16	Analogue output	mA	AO1
M1.17	Multimonitoring items		Displays three selectable monitoring values. See chapter 7.3.6.5.

Table 7-1. Monitored signals

* If the unit is supplied with +24V only (for control board power-up) this value is not reliable.

The All in One applications embody more monitoring values.

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7.3.2 Parameter menu (M2)

Parameters are the way of conveying the commands of the user to the frequency converter. The parameter values can be edited by entering the *Parameter Menu* from the *Main Menu* when the location indication **M2** is visible on the first line of the display. The value editing procedure is presented in Figure 7-5.

Push the *Menu button right* once to move into the *Parameter Group Menu (G#)*. Locate the parameter group desired by using the *Browser buttons* and push the *Menu button right* again to enter the group and its parameters. Use again the *Browser buttons* to find the parameter *(P#)* you want to edit. From here you can proceed in two different ways: Pushing the *Menu button right* takes you to the edit mode. As a sign of this, the parameter value starts to blink. You can now change the value in two different manners:

1. Just set the new desired value with the *Browser buttons* and confirm the change with the *Enter button*. Consequently, the blinking stops and the new value is visible in the value field.
2. Push the *Menu button right* once again. Now you will be able to edit the value digit by digit. This editing manner may come in handy, when a relatively greater or smaller value than that on the display is desired. Confirm the change with the *Enter button*.

The value will not change unless the Enter button is pushed. Pressing the *Menu button left* takes you back to the previous menu.

Several parameters are locked, i.e. uneditable, when the drive is in RUN status. If you try to change the value of such a parameter the text **Locked** will appear on the display. The frequency converter must be stopped in order to edit these parameters.
The parameters values can also be locked using the function in menu **M6** (see Chapter Parameter lock (P6.5.2)).

You can return to the *Main menu* anytime by pressing the *Menu button left* for 3 seconds.

The basic application package "All in One+" includes seven applications with different sets of parameters. You will find the parameter lists in the Application Section of this manual.

Once in the last parameter of a parameter group, you can move directly to the first parameter of that group by pushing the *Browser button up*.

See the diagram for parameter value change procedure on page 80.

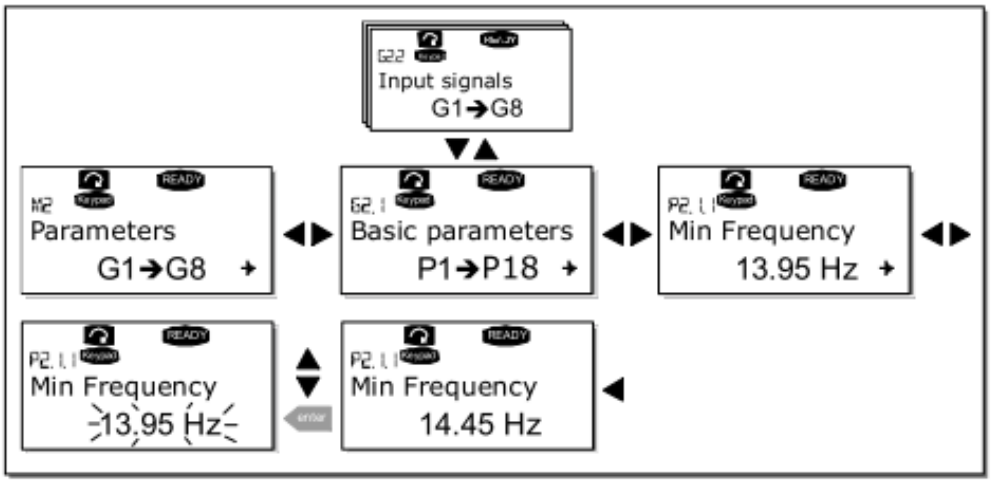


Figure 7-5. Parameter value change procedure

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7.3.4 Active faults menu (M4)

The *Active faults menu* can be entered from the *Main menu* by pushing the *Menu button right* when the location indication **M4** is visible on the first line of the keypad display.

When a fault brings the frequency converter to a stop, the location indication F1, the fault code, a short description of the fault and the **fault type symbol** (see Chapter 7.3.4.1) will appear on the display. In addition, the indication FAULT or ALARM (see Figure 7-1 or Chapter 7.1.1) is displayed and, in case of a FAULT, the red led on the keypad starts to blink. If several faults occur simultaneously, the list of active faults can be browsed with the *Browser buttons*.

The memory of active faults can store the maximum of 10 faults in the order of appearance. The display can be cleared with the *Reset button* and the read-out will return to the same state it was before the fault trip. The fault remains active until it is cleared with the *Reset button* or with a reset signal from the I/O terminal or fieldbus.

Note! Remove external Start signal before resetting the fault to prevent unintentional restart of the drive.

Normal state, no faults:



7.3.4.1 Fault types

In the NX_ frequency converter, there are four different types of faults. These types differ from each other on the basis of the subsequent behaviour of the drive. See Table 7-3.

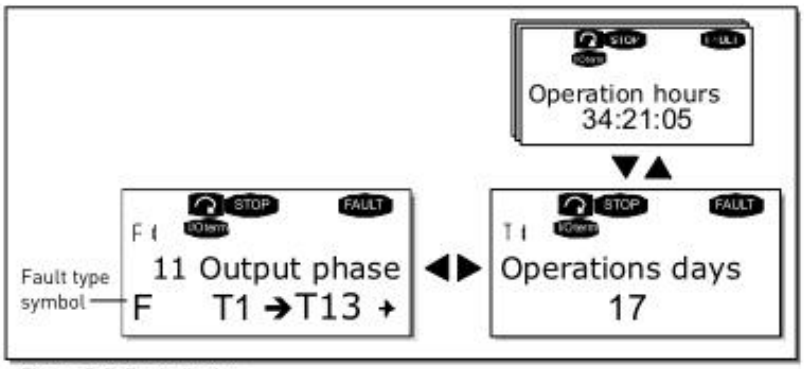


Figure 7-7. Fault display

Fault type symbol	Meaning
A (Alarm)	This type of fault is a sign of an unusual operating condition. It does not cause the drive to stop, nor does it require any special actions. The 'A fault' remains in the display for about 30 seconds.
F (Fault)	An 'F fault' is a kind of fault that makes the drive stop. Actions need to be taken in order to restart the drive.
AR (Fault Autoreset)	If an 'AR fault' occurs the drive will also stop immediately. The fault is reset automatically and the drive tries to restart the motor. Finally, if the restart is not successful, a fault trip (FT, see below) occurs.
FT (Fault Trip)	If the drive is unable to restart the motor after an AR fault an FT fault occurs. The effect of the 'FT fault' is basically the same as that of the F fault: the drive is stopped.

Table 7-3. Fault types

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7.3.4.2 Fault codes

The fault codes, their causes and correcting actions are presented in the table below. The shadowed faults are A faults only. The items written in white on black background present faults for which you can program different responses in the application. See parameter group Protections.
Note: When contacting distributor or factory because of a fault condition, always write down all texts and codes on the keypad display.

Fault code	Fault	Possible cause	Correcting measures
1	Overcurrent	Frequency converter has detected too high a current (>4*I_H) in the motor cable: – sudden heavy load increase – short circuit in motor cables – unsuitable motor	Check loading. Check motor. Check cables.
2	Overvoltage	The DC-link voltage has exceeded the limits defined in Table 4-7. – too short a deceleration time – high overvoltage spikes in supply	Make the deceleration time longer. Use brake chopper or brake resistor (available as options)
3	Earth fault	Current measurement has detected that the sum of motor phase current is not zero. – insulation failure in cables or motor	Check motor cables and motor.
5	Charging switch	The charging switch is open, when the START command has been given. – faulty operation – component failure	Reset the fault and restart. Should the fault re-occur, contact the distributor near to you. Please see Vacon contact info on the Web
6	Emergency stop	Stop signal has been given from the option board.	
7	Saturation trip	Various causes: – defective component – brake resistor short-circuit or overload	Cannot be reset from the keypad. Switch off power. DO NOT RE-CONNECT POWER! Contact factory. If this fault appears simultaneously with Fault 1, check motor cables and motor
8	System fault	- component failure - faulty operation Note exceptional fault data record, see 7.3.4.3. S1 = Feedback of motor voltage S2 = Reserved S3 = Reserved S4 = ASIC trip S5 = Disturbance in VaconBus S6 = Feedback of charging switch S7 = Charging switch S8 = No power to driver card S9 = Power unit communication (TX) S10 = Power unit communication (Trip) S11 = Power unit comm. (Measurement)	Reset the fault and restart. Should the fault re-occur, contact the distributor near to you. Please see Vacon contact info on the Web

Fault code	Fault	Possible cause	Correcting measures
9	Undervoltage	DC-link voltage is under the voltage limits defined in Table 4-7. – most probable cause: too low a supply voltage – frequency converter internal fault	In case of temporary supply voltage break reset the fault and restart the frequency converter. Check the supply voltage. If it is adequate, an internal failure has occurred. Contact the distributor near to you. Please see Vacon contact info on the Web
10	Input line supervision	Input line phase is missing.	Check supply voltage and cable.
11	Output phase supervision	Current measurement has detected that there is no current in one motor phase.	Check motor cable and motor.
12	Brake chopper supervision	– no brake resistor installed – brake resistor is broken – brake chopper failure	Check brake resistor. If the resistor is ok, the chopper is faulty. Contact the distributor near to you. Please see Vacon contact info on the Web
13	Frequency converter under-temperature	Heatsink temperature is under –10°C	
14	Frequency converter overtemperature	Heatsink temperature is over 90°C (or 77°C, NX_6, FR6). Overtemperature warning is issued when the heatsink temperature exceeds 85°C (72°C).	Check the correct amount and flow of cooling air. Check the heatsink for dust. Check the ambient temperature. Make sure that the switching frequency is not too high in relation to ambient temperature and motor load.
15	Motor stalled	Motor stall protection has tripped.	Check motor.
16	Motor over-temperature	Motor overheating has been detected by frequency converter motor temperature model. Motor is overloaded.	Decrease the motor load. If no motor overload exists, check the temperature model parameters.
17	Motor underload	Motor underload protection has tripped.	
22	EEPROM checksum fault	Parameter save fault – faulty operation – component failure	
24	Counter fault	Values displayed on counters are incorrect	
25	Microprocessor watchdog fault	– faulty operation – component failure	Reset the fault and restart. Should the fault re-occur, contact the distributor near to you. Please see Vacon contact info on the Web
26	Start-up prevented	Start-up of the drive has been prevented.	Cancel prevention of start-up.
29	Thermistor fault	The thermistor input of option board has detected increase of the motor temperature	Check motor cooling and loading Check thermistor connection (If thermistor input of the option board is not in use it has to be short circuited)
31	IGBT temperature (hardware)	IGBT Inverter Bridge overtemperature protection has detected too high a short term overload current	Check loading. Check motor size.

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Fault code	Fault	Possible cause	Correcting measures
32	Fan cooling	Cooling fan of the frequency converter does not start, when ON command is given	Contact the distributor near to you. Please see Vacon contact info on the Web
34	CAN bus communication	Sent message not acknowledged.	Ensure that there is another device on the bus with the same configuration.
35	Application	Problem in application software	Contact your distributor. If you are application programmer check the application program.
36	Control unit	NXS Control Unit can not control NXP Power Unit and vice versa	Change control unit
37	Device changed (same type)	Option board or control unit changed. Same type of board or same power rating of drive.	Reset **Note:** No fault time data record!
38	Device added (same type)	Option board or drive added. Drive of same power rating or same type of board added.	Reset **Note:** No fault time data record!
39	Device removed	Option board removed. Drive removed.	Reset **Note:** No fault time data record!
40	Device unknown	Unknown option board or drive.	Contact the distributor near to you. Please see Vacon contact info on the Web
41	IGBT temperature	IGBT Inverter Bridge overtemperature protection has detected too high a short term overload current	Check loading. Check motor size.
42	Brake resistor overtemperature	Brake resistor overtemperature protection has detected too heavy braking	Set the deceleration time longer. Use external brake resistor.
43	Encoder fault	Note the exceptional Fault data record. See 7.3.4.3. Additional codes: 1 = Encoder 1 channel A is missing 2 = Encoder 1 channel B is missing 3 = Both encoder 1 channels are missing 4 = Encoder reversed	Check encoder channel connections. Check the encoder board.
44	Device changed (different type)	Option board or control unit changed. Option board of different type or different power rating of drive.	Reset **Note:** No fault time data record! **Note:** Application parameter values restored to default.
45	Device added (different type)	Option board or drive added. Option board of different type or drive of different power rating added.	Reset **Note:** No fault time data record! **Note:** Application parameter values restored to default.
50	Analogue input I_{in} < 4mA (sel. signal range 4 to 20 mA)	Current at the analogue input is < 4mA. – control cable is broken or loose – signal source has failed	Check the current loop circuitry.
51	External fault	Digital input fault.	
52	Keypad communication fault	The connection between the control keypad and the frequency converter is broken.	Check keypad connection and possible keypad cable.
53	Fieldbus fault	The data connection between the fieldbus Master and the fieldbus board is broken	Check installation. If installation is correct contact the nearest Vacon distributor. Please see Vacon contact info on the Web

Fault code	Fault	Possible cause	Correcting measures
54	Slot fault	Defective option board or slot	Check board and slot. Contact the nearest Vacon distributor. Please see Vacon contact info on the Web
56	PT100 board temp. fault	Temperature limit values set for the PT100 board parameters have been exceeded	Find the cause of temperature rise

Table 7-4. Fault codes

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7.3.4.3 Fault time data record

When a fault occurs the information described above in 7.3.4 is displayed. By pushing the *Menu button right* here you will enter the *Fault time data record menu* indicated by **T.1→T.13**. In this menu, some selected important data valid at the time of the fault are recorded. This feature is intended to help the user or the service person to determine the cause of fault.

The data available are:

T.1	Counted operation days *(Fault 43: Additional code)*	d
T.2	Counted operation hours *(Fault 43: Counted operation days)*	hh:mm:ss *(d)*
T.3	Output frequency *(Fault 43: Counted operation hours)*	Hz *(hh:mm:ss)*
T.4	Motor current	A
T.5	Motor voltage	V
T.6	Motor power	%
T.7	Motor torque	%
T.8	DC voltage	V
T.9	Unit temperature	°C
T.10	Run status	
T.11	Direction	
T.12	Warnings	
T.13	0-speed*	

Table 7-5. Fault time recorded data

* Tells the user if the drive was at zero speed (< 0.01 Hz) when the fault occurred

Real time record

If real time is set to run on the frequency converter the data items **T1** and **T2** will appear as follows:

T.1	Counted operation days	yyyy-mm-dd
T.2	Counted operation hours	hh:mm:ss,sss

7.3.5 Fault history menu (M5)

The *Fault history menu* can be entered from the *Main menu* by pushing the *Menu button right* when the location indication **M5** is visible on the first line of the keypad display. Find the fault codes in Table 7-4.

All faults are stored in the *Fault history menu* in which you can browse through them using the *Browser buttons*. Additionally, the *Fault time data record* pages (see Chapter 7.3.4.3) are accessible at each fault. You can return to the previous menu anytime by pushing the *Menu button left*.

The memory of the frequency converter can store a maximum of 30 faults in the order of appearance. The number of faults currently in the fault history is shown on the value line of the main page (**H1→H#**). The order of the faults is indicated by the location indication in the upper left corner of the display. The latest fault carries the indication F5.1, the second latest F5.2 etc. If there are 30 uncleared faults in the memory the next occurring fault will erase the oldest from the memory.

Pressing the *Enter button* for about 2 to 3 seconds resets the whole fault history. Then, the symbol **H#** will change to **0**.

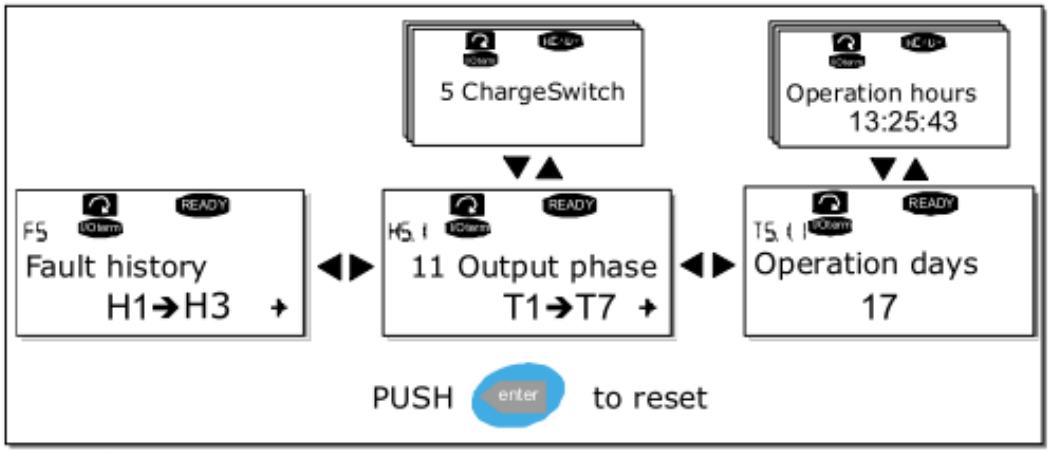


Figure 7-8. Fault history menu

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7.3.6.1 Language selection

The Vacon control keypad offers you the possibility to control the frequency converter through the keypad in the language of your choice.

Locate the language selection page under the *System menu*. Its location indication is **S6.1**. Press the *Menu button right* once to enter the edit mode. As the name of the language starts to blink you are able to choose another language for the keypad texts. Confirm the selection by pushing the *Enter button* The blinking stops and all textual information on the keypad is presented in the language you chose.

You can return to the previous menu anytime by pushing the *Menu button left*.

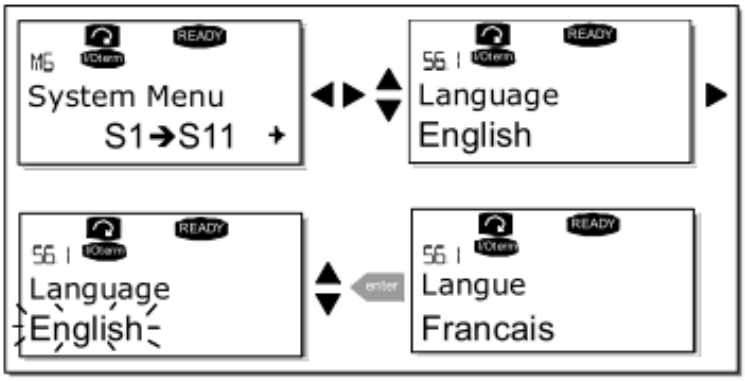


Figure 7-9. Selection of language

7.3.6.2 Application selection

The user can select the application desired by entering the *Application selection page (S6.2)*. This is done by pushing the *Menu button right* when on the first page of the *System menu*. Change then the application by pushing the *Menu button right* once again. The name of the application starts to blink. Now you can browse through the applications with the *Browser buttons* and select another application with the *Enter button*.

Changing application will reset all parameters. After application change, you will be asked if you want the parameters of the **new** application to be uploaded to the keypad. If you wish this to happen push the *Enter button* Pushing any other button leaves the parameters of the **previously used** application saved in the keypad. For more information, see Chapter 7.3.6.3.

For more information about the Application Package, see Vacon NX Application Manual.

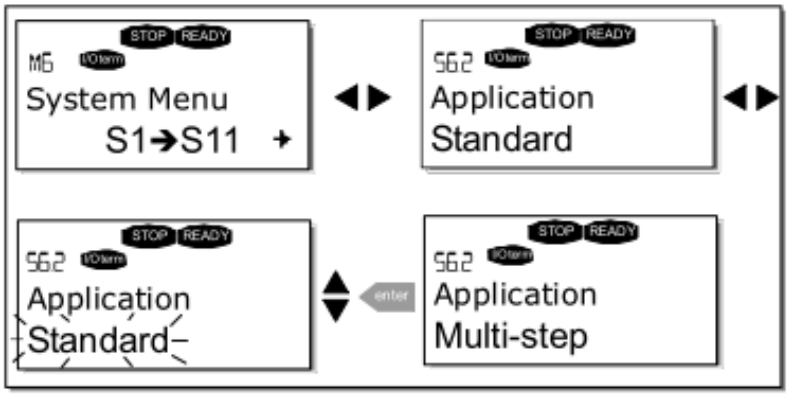


Figure 7-10. Change of application

7.3.6.3 Copy parameters

The parameter copy function is used when the operator wants to copy one or all parameter groups from one drive to another or to store parameter sets in the internal memory of the converter. All the parameter groups are first *uploaded* to the keypad, then the keypad is connected to another drive and then the parameter groups are *downloaded* to it (or possibly back to the same drive).

Before any parameters can successfully be copied from one drive to another the **drive** has to be **stopped** when the parameters are downloaded to it:

The parameter copy menu **(S6.3)** embodies four functions:

Parameter sets (S6.3.1)

The Vacon NX_ frequency converter features a possibility for the user to load back the factory default parameter values and to store and load two customised parameter sets (all parameters included in the application).

On *Parameter sets* page **(S6.3.1)**, push the *Menu button right* to enter the *Edit menu*. The text *LoadFactDef* begins to blink and you can confirm the loading of factory defaults by pushing the *Enter button*. The drive resets automatically.
Alternatively you can choose any other of the storing or loading functions with the *Browser buttons*. Confirm with the *Enter button*. Wait until 'OK' appears on the display.

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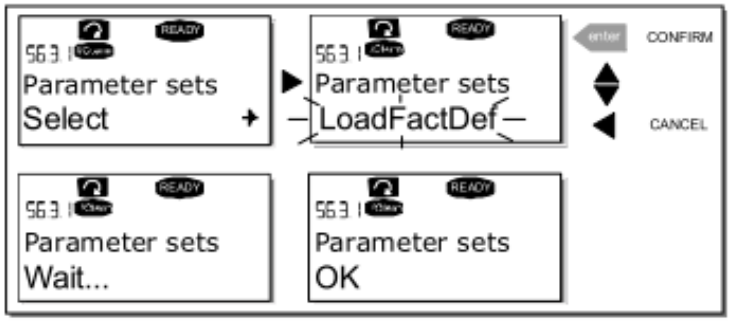


Figure 7-11. Storing and loading of parameter sets

Upload parameters to keypad (To keypad, S6.3.2)

This function uploads **all** existing parameter groups to the keypad provided that the drive is stopped.

Enter the *To keypad* page (S6.3.2) from the *Parameter copy menu*. Push the *Menu button right* to enter the edit mode. Use the *Browser buttons* to select the option *All parameters* and press the *Enter button*. Wait until 'OK' appears on the display.

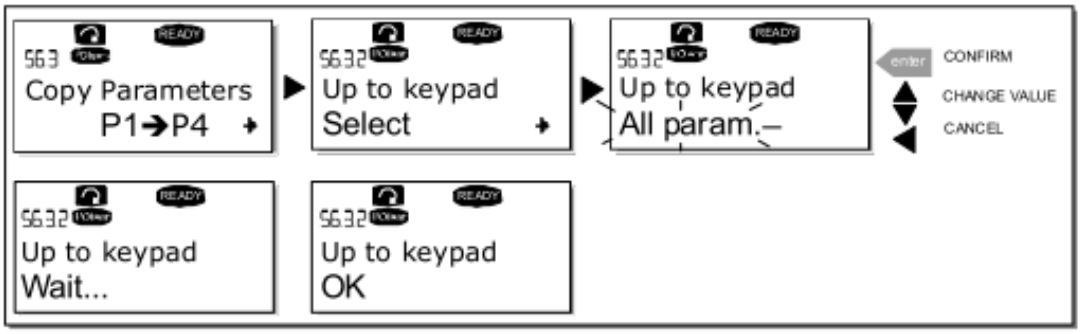


Figure 7-12. Parameter copy to keypad

Download parameters to drive (From keypad, S6.3.3)

This function downloads **one** or **all parameter** groups uploaded to the keypad to a drive provided that the drive is in STOP status.

Enter the *From keypad* page (S6.3.3) from the *Parameter copy menu*. Push the *Menu button right* to enter the edit mode. Use the *Browser buttons* to select either the option *All parameters* or *Application parameters* and press the *Enter button*. Wait until 'OK' appears on the display.

The procedure to download the parameters from keypad to drive is similar to that of from drive to keypad. See above.

Automatic parameter backup (P6.3.4)

On this page you can activate or inactivate the parameter backup function. Enter the edit mode by pressing the *Menu button right*. Choose *Yes* or *No* with the *Browser buttons*.

When the Parameter backup function is activated Vacon NX control keypad makes a copy of the parameters of the presently used application. Every time a parameter is changed the keypad backup is automatically updated.

When applications are changed, you will be asked if you wish the parameters of the **new** application to be uploaded to the keypad. For this to happen, push the *Enter button*. If you wish to keep the copy of the parameters of the **previously used** application saved in the keypad push any other button. Now you will be able to download these parameters to the drive following the instructions given in chapter 7.3.6.3.

If you want the parameters of the new application to be automatically uploaded to the keypad you have to do this for the parameters of the new application once on page 6.3.2 as instructed. **Otherwise the panel will always ask for the permission to upload the parameters.**

Note: Parameters saved in the parameter settings on page **S6.3.1** will be deleted when applications are changed. If you want to transfer the parameters from one application to another you have to upload them first to the keypad.

7.3.6.4 Parameter comparison

In the *Parameter comparison* submenu **(S6.4)**, you can compare the **actual parameter values** to the values of your customised parameter sets and those loaded to the control keypad.

The comparison is performed by pushing the *Menu button right* when in the *Compare parameters submenu*. The actual parameter values are first compared to those of the customised parameter Set1. If no differences are detected a '0' is displayed on the lowermost line. But if any of the parameter values differ from those of the Set1 the number of the deviations is displayed together with symbol **P** (e.g. P1→P5 = five deviating values). By pressing the *Menu button right* once again you can still enter the pages where you can see both the actual value and the value it was compared to. In this display, the value on the Description line (in the middle) is the default value and the one on the value line (lowermost) is the edited value. Furthermore, you can also edit the actual value with the *Browser buttons* in the *edit mode* that you can reach by pushing the *Menu button right* once again.
In the same way, you can perform the comparison of the actual values to *Set2*, *Factory Settings* and *Keypad Set*.

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VACON NXP
FREQUENCY CONVERTERS

Application

CAVOTEC SPECIMAS

Application CAVOTEC SPECIMAS (Software ASITQ224_v320)

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1. Description

The software for the frequency converter control provides two ways of working: automatic and manual.
Manual control is activated through the digital input named DI6.

Automatic
In automatic state, the motor is controlled by torque reference; torque level is selected through combination of four digital inputs (named DI1, DI3, DI4 e DI5) and one analog input (named AIN1).
At power on, without run command (DI1) active, torque value is constant and equal to parameter “Default” (P2.2.1.2); if that parameter is equal to zero, then frequency converters does not drive the motor.
When digital input DI1 get an high value, a “Reeling IN Const Speed” torque level, with programmable minimum (P2.2.1.3) and maximum (P2.2.1.4) values, is activated; actual torque value is linearly adjusted between those limits by the analog input AIN1.

Five more torque levels are available, activated by signals DI3,4,5. Usually these signals are related to Reel OUT, Acceleration and Deceleration conditions.
Depending on parameter P2.2.1.24, those levels are either used as absolute torque values or are added to Reel IN level (Offset logic). They are adjusted by means of analog input AIN1 (parameters from P2.2.1.5 to P2.2.1.14).

It is also possible to have constant torque values (= minimum), by programming parameter P2.2.1.25.

“Offset” correction can also assume negative values; absolute minimum (P2.2.1.15) and maximum (P2.2.1.16) torque limits are anyway present, which will not be exceed with any “Offset” combination.

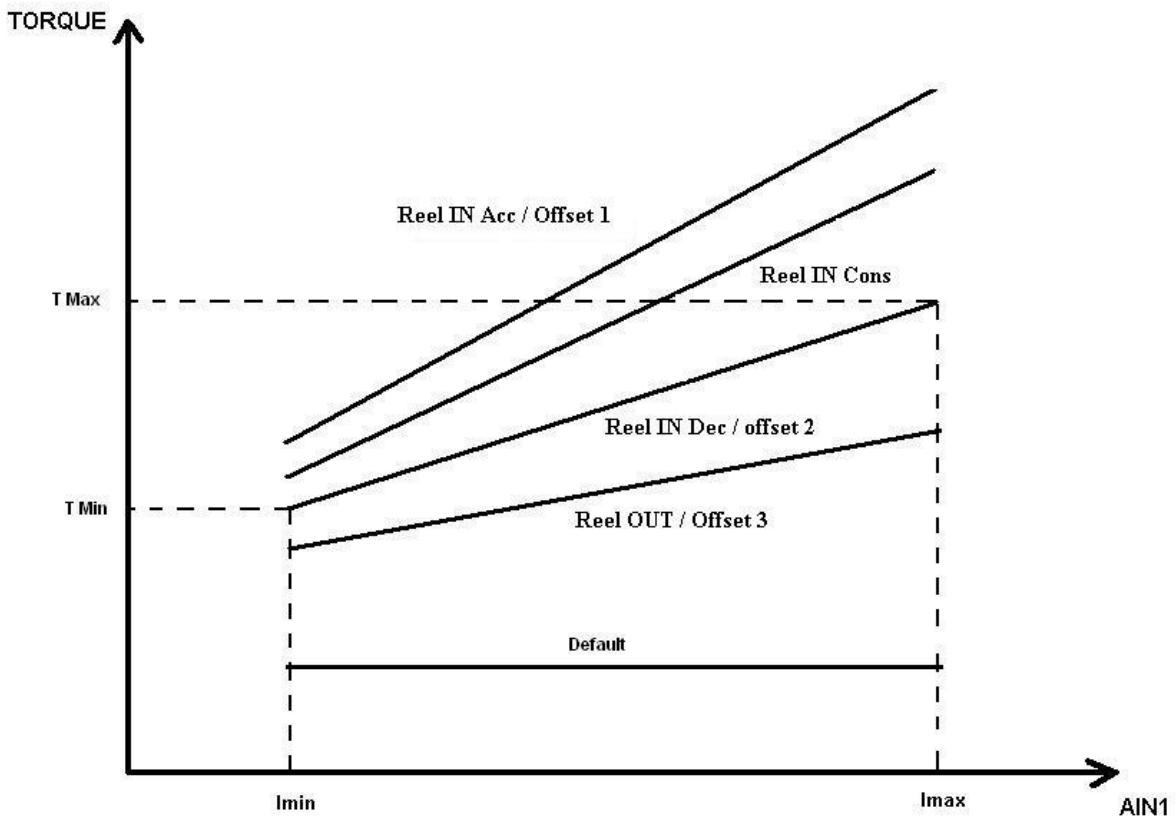


Figure 1-1. Example of torque levels.

The logic for torque level selection features three different ways of operating, selected by parameter (P2.2.1.1)

All the digital and analog control signals can be assigned to drive input/output channels in flexible way, through parameters (group G2.3).

Usually the maximum speed achievable by the motor under torque control is corresponding to maximum frequency (P2.1.1), in both directions.
A function called “Speed window” is available, which uses a second analog signal (named AIN2) as a theoretical speed reference and check if actual motor speed is inside a certain range as to the reference.
Through parameterization and depending on direction, it is possible to define if motor speed must be kept inside range limits (control mode will change automatically to speed control) or if it can exceed limits, with signaling on digital output.

A logic for automatic recognizing of the motor speed variation is available, which adds corrective terms (P2.2.1.17-20) to the torque reference for inertia compensation during acceleration and deceleration. This logic needs the speed reference.

Manual
Manual control mode is activated rising digital input DI6 at high level.
In this state, signals DI4 and DI5 act as commands for reeling in and reeling out operations.
Motor is under speed control; speed reference can be either a constant value (P2.2.2.2) or a value set on panel, or it can be adjusted by an analog input.
Motor torque can be anyway limited (P2.2.2.7).
Speed reference can also be internally scaled as function of actual diameter of the reel (measure from analogue input AI1), to have a coarsely constant cable speed.

NOTE: application software is conceived as that a positive torque reference causes the motor run in cable rewinding direction.
If at commissioning time the direction is wrong, it is necessary to exchange two motor power phases and to reverse encoder counting by parameter P7.3.1.2 (see also chapter 6 Expansion boards).

Fieldbuses support: command signals and analogue references can be replaced by fieldbus communication. An optional fieldbus board must be installed on the NXP drive for this purpose.
Fully supported buses are: Profibus, CANOpen, Modbus (both RTU and TCP/IP). Constraints exist for DeviceNet.

2. Control I/O

NXOPTA1			
Terminal		Signal	Description
1	$+10V_{ref}$	Reference output	Voltage for potentiometer, etc.
2	AI1+	Analogue input 1 Range: 0 - 10V / 0 –20 mA	Torque control reference
3	AI1-		
4	AI2+	Analogue input 2 Range: 0 –10V / 0 – 20 mA	Speed control reference
5	AI2-		
6	+24V	Control voltage output	Voltage for switches etc.; max 0.1A
7	GND	I/O ground	Ground for reference and controls
8	DIN1	Run	Contact closed = run
9	DIN2	Fault reset	Contact closed = reset
10	DIN3	Offset 3	Contact closed = offset 3 activation
11	CMA	Common for DIN 1—DIN 3	Connect to GND o +24V
12	+24V	Control voltage output	Voltage for switches etc.; max 0.1A
13	GND	I/O ground	Ground for reference and controls
14	DIN4	Offset 2	Contact closed = offset 2 / reel in manual
15	DIN5	Offset 1	Contact closed = offset 1 / reel out manual
16	DIN6	Manual	Contact closed = manual activation
17	CMB	Common for DIN 4—DIN 6	Connect to GND o +24V
18	AO1+	Analogue output	Output frequency Range 0—20 mA/R_L, max. 500Ω
19	AO1-		
20	DO1	Digital output	Run signaling Open collector, I≤50mA, U≤48 VDC
NXOPTA2			
21	RO1	 Relay output 1	Ready signaling
22	RO1		
23	RO1		
24	RO2	 Relay output 2	Fault signaling
25	RO2		
26	RO2		

Table 1- 2. Standard I/O configuration .

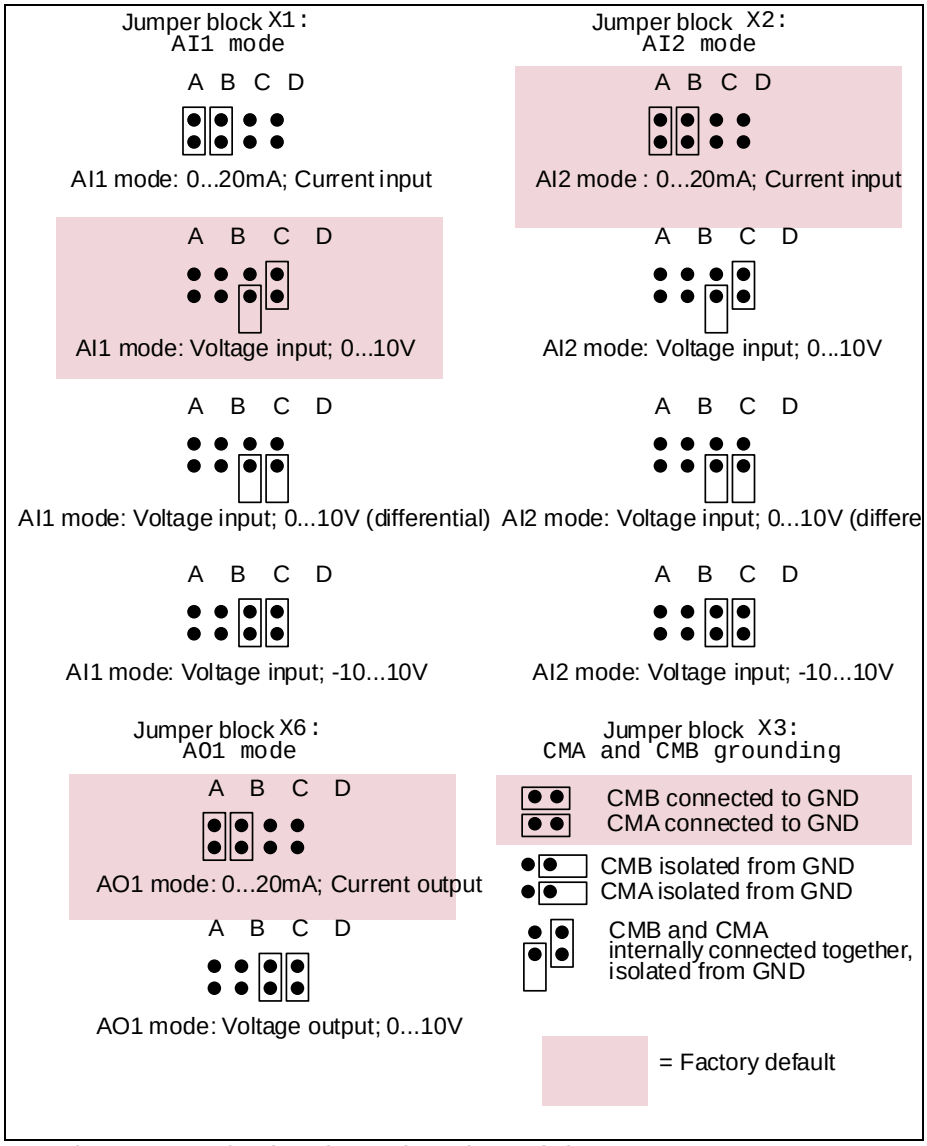


Figure 1-2. I/0 signals setting through jumpers.

3. Monitoring values (Menu M1)

The monitoring values are the actual values of parameters and signals, as well as statuses and measurements. Monitoring values cannot be edited.
See Vacon NX User's manual, Chapter 7 for more information.

.

Code	Parameter	Unit	Description
V1.1	Output frequency	Hz	Output frequency to motor
V1.2	Frequency reference	Hz	Frequency reference to motor control
V1.3	Torque reference	%	Torque reference to motor control
V1.4	Motor speed	rpm	Motor speed in rpm
V1.5	Motor current	A	Measured motor current
V1.6	Motor torque	%	In % of motor nominal torque
V1.7	Motor power	%	Motor shaft power
V1.8	Motor voltage	V	Motor voltage
V1.9	DC link voltage	V	DC link voltage
V1.10	Unit temperature	°C	Heatsink temperature
V1.11	Motor temperature	°C	Calculated motor temperature
V1.12	Analogue input 1	%	AIN1
V1.13	Analogue input 2	%	AIN2
V1.14	DIN1, DIN2, DIN3		Digital inputs status
V1.15	DIN4, DIN5, DIN6		Digital inputs status
V1.16	DO1,RO1,RO2		Digital output status
V1.17	Analogue output	mA	AO1
V1.18	Control mode		Active control mode
V1.19	Inertia compensation	%	Torque correction for acceleration/deceleration
V1.20	Reference for speed window	Hz	Theoretical speed reference (as Hz) used for the speed window
V1.21	SW Upper limit	Hz	Upper limit when speed window is active
V1.22	SW Lower limit	Hz	Lower limit when speed window is active
V1.23	Encoder frequency	Hz	Actual speed, as Hz, read from encoder
V1.24	Actual diameter	%	Actual reel diameter, read from AI1 input
V1.25	Mov Out Acc Dec		Status of most relevant digital inputs (it follows DigIn programming)
G1.27	Multimonitor items		Displays three selectable monitoring values.

Table 1-1. Monitoring values

Note: Multimonitor page is automatically shown at power on and after 1 minute of inactivity on the keypad (depends on P3.6). It is also immediately shown when pressing Reset button.

4. Parameters (Menu M2)

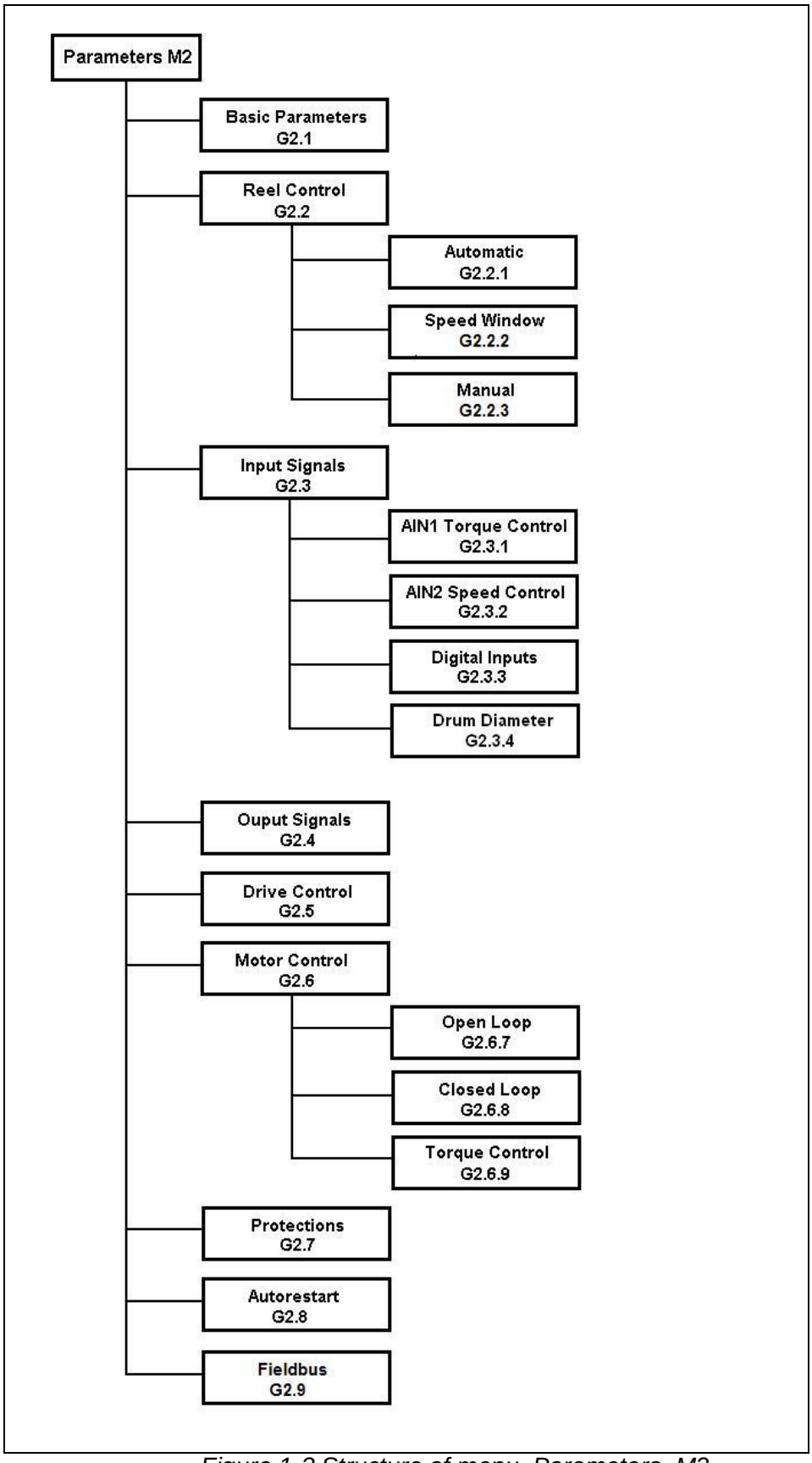


Figure 1-3.Structure of menu Parameters M2

4.1. Basic Parameters (Menu M2 → G2.1)

Code	Parameter	Min	Max	Unit	Default	Note
P2.1.1	Max Frequency	0	320,00	Hz	50,00	**NOTE**: If f_{max} > than motor synchronous speed, check suitability for motor and drive system.
P2.1.2	Current Limit	0,2 x I_H	2 x I_H	A	I_H	**NOTE**: This applies for frequency converters up to FR7. For greater sizes, consult the factory.
P2.1.3	Nominal Motor Voltage	180	690	V	NX2: 230 NX5:400 NX6:690	Check the rating plate of the motor.
P2.1.4	Nominal Motor Frequency	30,00	320,00	Hz	50,00	Check the rating plate of the motor.
P2.1.5	Nominal Motor Speed	30	20000	rpm	1440	The default applies for a 4 pole motor and a nominal size frequency converter.
P2.1.6	Nominal Motor Current	0,2 x I_H	2 x I_H	A	I_L	Check the rating plate of the motor.
P2.1.7	Motor cosφ	0,3	1,0		0,85	Check the rating plate of the motor.
P2.1.8	Magnetizing Current	0,2 x I_H	2 x I_H	A	0,5 x I_L	Check the rating plate of the motor.

Table 1-3. Basic Parameters, G2.1

2.1.1 Max frequency

Defines the higher operating frequency of the motor, corresponding to the maximum speed of the crane. It is very important not to set this parameter to a value lower than the actually requested: see also chapter 4.2.1.1 Speed Window.

2.1.2 Current limit

This parameter determines the maximum current from the frequency converter. The current limit is 1.5 times the rated current (I_L) by default.

2.1.3 Nominal motor voltage

Find this value on the rating plate of the motor. This parameter sets the voltage at the field weakening point (parameter 2.6.3) to 100% x U_{nmotor}

2.1.4 Nominal motor frequency

Find this value on the rating plate of the motor. This parameter sets the field weakening point (parameter 2.6.2) to the same value.

2.1.5 Nominal motor speed

Find this value n_n on the rating plate of the motor.

2.1.6 Nominal motor current

Find this value I_n on the rating plate of the motor.

2.1.7 Motor cosφ

Find this value cosφ on the rating plate of the motor.

2.1.8 Motor magnetizing current

Find this value on the rating plate of the motor (no-load current).

4.2. Reel Control Parameters

4.2.1. Automatic (Menu M2 → G2.2.1)

Code	Parameter	Min	Max	Unit	Default	Note
P2.2.1.1	Control mode	0	2		0	0= Basic 1= DI logic 2= Auto Reverse
P2.2.1.2	Default Torque	-300,0	300,0	%	20,0	Torque without run command
P2.2.1.3	Reel IN Const Speed TMin	0,0	300,0	%	40,0	Level standard minimum
P2.2.1.4	Reel IN Const Speed TMax	0,0	300,0	%	60,0	Level standard maximum
P2.2.1.5	Reel IN Decel TMin	-300,0	300,0	%	0,0	Level 1 minimum
P2.2.1.6	Reel IN Decel TMax	-300,0	300,0	%	0,0	Level 1 maximum
P2.2.1.7	Reel IN Accel TMin	-300,0	300,0	%	0,0	Level 2 minimum
P2.2.1.8	Reel IN Accel TMax	-300,0	300,0	%	0,0	Level 2 maximum
P2.2.1.9	Reel OUT Const Speed TMin	-300,0	300,0	%	0,0	Level 3 minimum
P2.2.1.10	Reel OUT Const Speed TMax	-300,0	300,0	%	0,0	Level 3 maximum
P2.2.1.11	Reel OUT Decel TMin	-300,0	300,0	%	0,0	Level 4 minimum
P2.2.1.12	Reel OUT Decel Tmax	-300,0	300,0	%	0,0	Level 4 maximum
P2.2.1.13	Reel OUT Accel TMin	-300,0	300,0	%	0,0	Level 5 minimum
P2.2.1.14	Reel OUT Accel TMax	-300,0	300,0	%	0,0	Level 5 maximum
P2.2.1.15	Min Torque Limit	-100.0	300,0	%	0,0	Resultant torque min limit
P2.2.1.16	Max Torque Limit	P2.2.1.15	300,0	%	100,0	Resultant torque max limit
P2.2.1.17	Inertia Comp. IN Acc	0	20,00	%	0,00	Torque for reeling in acc
P2.2.1.18	Inertia Comp. IN Dec	0	20,00	%	0,00	Torque for reeling in dec
P2.2.1.19	Inertia Comp. IN Acc	0	20,00	%	0,00	Torque for reeling out acc
P2.2.1.20	Inertia Comp. IN Dec	0	20,00	%	0,00	Torque for reeling out dec
P2.2.1.21	Inertia Comp. Filter	0,00	5,00	s	0,20	Filter on torque correction
P2.2.1.22	Signal source for Inertia Comp.	0	1		0	0= motor encoder 1= external speed reference
P2.2.1.23	Inertia Comp. Filter	0,00	5,00	s	0,20	Filter on torque correction
P2.2.1.24	Inertia Comp at Max Diameter	0	500,0	%	100,0	Diameter depending factor
P2.2.1.25	Torque Constant/Linear	0	1		1	0= constant 1= linear
P2.2.1.26	Torque Ramp Time	0,00	10,00	s	0,00	Time to reach 100% torque

Code	Parameter	Min	Max	Unit	Default	Note
P2.2.1.27	Pulse IN ACC Torque	0	100,0	%	0,0	Impulsive additional torque
P2.2.1.28	Pulse IN ACC Time	0	30,00	s	2,00	Pulse time
P2.2.1.29	Pulse IN DEC Torque	-100,0	0	%	0,0	Impulsive subtractive torque
P2.2.1.30	Pulse IN DEC Time	0	30,00	s	2,00	Pulse time
P2.2.1.31	Pulse OUT ACC Torque	-100,0	0	%	0,0	Impulsive subtractive torque
P2.2.1.32	Pulse OUT ACC Time	0	30,00	s	2,00	Pulse time
P2.2.1.33	Pulse OUT DEC Torque	0	100,0	%	0,0	Impulsive additional torque
P2.2.1.34	Pulse OUT DEC Time	0	30,00	s	2,00	Pulse time
P2.2.1.35	Pulse Torque at Max Diameter	0	500,0	%	100,0	Diameter depending factor

Table 1-4. Automatic parameters , G2.2.1

2.2.1.1 Control Mode

Defines the modality for the selection of the torque level, as function of digital inputs state:
0: Basic
1: DI Logic
2: Auto Reverse

The following tables represent the logics for selection; digital signals DI_ can be associated in flexible manner to the digital inputs of the drive.
Torque levels can linearly vary as function of analogue signal AIN1.
Auto Reverse mode works with automatic recognizing of the motor direction.

BASIC CONTROL

DI 1	DI 5	DI 4	DI 3	TORQUE LEVEL
-	-	-	-	Default
-	X	-	-	Default
-	-	X	-	Default
-	-	-	X	Default
X	-	-	-	Reel IN Const
X	X	-	-	Reel IN Dec
X	-	X	-	Reel IN Acc
X	-	-	X	Reel OUT Const
X	-	X	X	Reel IN Acc or Reel OUT Const (the first activated)

DI LOGIC CONTROL

DI 1	DI 5	DI 4	DI 3	TORQUE LEVEL
-	-	-	-	Default
-	X	-	-	Default
-	-	X	-	Default
-	-	-	X	Default
X	-	-	-	Reel IN Const
X	X	-	-	Reel IN Dec
X	-	X	-	Reel IN Acc
X	-	-	X	Reel OUT Const
X	X	-	X	Reel OUT Dec
X	-	X	X	Reel OUT Acc
X	X	X	-	NOT ALLOWED
X	X	X	X	NOT ALLOWED

AUTO REVERSE CONTROL

DI 1	DI 5	DI 4	DI 3	REV	TORQUE LEVEL
-	-	-	-	-	Default
-	X	-	-	-	Default
-	-	X	-	-	Default
-	-	-	X	-	Default
X	-	-	-	-	Reel IN Const
X	X	-	-	-	Reel IN Dec
X	-	X	-	-	Reel IN Acc
X	-	-	X	-	Reel OUT Const
X	-	-	-	X	Reel OUT Dec
X	X	-	-	X	Reel IN Dec + Reel OUT Dec
X	-	X	-	X	Reel IN Acc + Reel OUT Dec
X	-	-	X	X	Reel OUT Const + Reel OUT Dec

REV : motor on reverse direction

2.2.1.2 Default Torque

Defines DEFAULT torque level.

2.2.1.3 Reel IN Const Speed Torque Min

Defines STANDARD torque level, corresponding to minimum of analogue signal AIN1.

2.2.1.4 Reel IN Const Speed Torque Max

Defines STANDARD torque level, corresponding to maximum of analogue signal AIN1.

2.2.1.5 Reel IN Dec / Offset 1 TMin

Defines torque level 1, corresponding to minimum of analogue signal AIN1.

2.2.1.6 Reel IN Dec / Offset 1 TMax

Defines torque level 1, corresponding to maximum of analogue signal AIN1.

2.2.1.7 Reel IN Acc / Offset 2 TMin

Defines torque level 2, corresponding to minimum of analogue signal AIN1.

2.2.1.8 Reel IN Acc / Offset 2 TMax

Defines torque level 2, corresponding to maximum of analogue signal AIN1.

.

2.2.1.9 Reel OUT Const Speed / Offset 3 TMin

Defines torque level 3, corresponding to minimum of analogue signal AIN1.

2.2.1.10 Reel OUT Const Speed / Offset 3 TMax

Defines torque level 3, corresponding to maximum of analogue signal AIN1.

2.2.1.11 Reel OUT Dec / Offset 4 TMin

Defines torque level 4, corresponding to minimum of analogue signal AIN1.

2.2.1.12 Reel OUT Dec / Offset 4 TMax

Defines torque level 4, corresponding to maximum of analogue signal AIN1.

2.2.1.13 Reel OUT Acc / Offset 5 TMin

Defines torque level 5, corresponding to minimum of analogue signal AIN1.

2.2.1.14 Reel OUT Acc / Offset 5 TMax

Defines torque level 5, corresponding to maximum of analogue signal AIN1.

2.2.1.15 Min Torque Limit

Defines minimum limit for the resultant torque, when DI1 (run) is active.

2.2.1.16 Max Torque Limit

Defines maximum limit for the resultant torque, when DI1 (run) is active.

2.2.1.17 Inertia Compensation for reeling IN Accel

Defines a corrective term for torque level during acceleration on reeling in, proportional to theoretical motor speed derivative.
Software recognizes acceleration sign, so this term is added to torque reference.
The parameter represents torque variation for a 1rad/ s^2 acceleration.
NOTE: the function needs the crane speed signal (P2.2.1.22= 1).

2.2.1.18 Inertia Compensation for reeling IN Decel
2.2.1.19 Inertia Compensation for reeling OUT Accel
2.2.1.20 Inertia Compensation for reeling OUT Decel

Similar to P2.2.1.18, but activated in different conditions. Software recognizes acceleration sign, so these positive terms are then automatically added or subtracted to torque reference.

2.2.1.21 Inertia Compensation Filter

Defines the time constant of the low-pass filter acting on inertia compensation; an higher value provides a more progressive torque correction, but it also introduces a delay on system response.

2.2.1.22 Signal Source for Inertia Compensation

Defines the source of the speed signal used for computation of inertia compensation.
0: motor encoder (not suggested)
1: theoretical speed reference, from P2.2.8

2.2.1.23 Inertia Compensation at Max Diameter

Values in P2.2.1.17-20 are applied with motor speed derivative calculated at minimum drum diameter. The inertia compensation can then vary linearly with actual diameter. This parameter is the multiplying factor at maximum diameter (100,0% means compensation not influenced by diameter value).
It is suggested to tune parameters P2.2.1.17-20 with the reel at minimum diameter and then to adjust P2.2.1.23 at maximum.

2.2.1.24 Torque Levels Mode

Defines how torque levels 1-5 are managed.
0: offsets (added to standard Reel IN Const)
1: absolute

2.2.1.25 Torque Constant / Linear

Defines if torque level is adjusted through analogue signal
0: constant (= T minimum)
1: linearly scaled

2.2.1.26 Torque Ramp Time

If greater then 0, changes in torque reference are smoothened with linear ramp. The parameter is the time to change from 0 to 100%.

2.2.1.27 Pulse reeling IN Accel Torque (only DI Logic)
2.2.1.28 Pulse reeling IN Accel Time

When DI Logic is used, an additional torque term can be automatically and temporary added when the crane changes its speed, to help the initial speed change of the drum. This pulse lasts the time programmed in P2.2.1.28, unless the activating DI falls before.

2.2.1.29 Pulse reeling IN Decel Torque (only DI Logic)
2.2.1.30 Pulse reeling IN Decel Time
2.2.1.31 Pulse reeling OUT Accel Torque (only DI Logic)
2.2.1.32 Pulse reeling OUT Accel Time
2.2.1.33 Pulse reeling OUT Decel Torque (only DI Logic)
2.2.1.34 Pulse reeling OUT Decel Time

Similar to P2.2.1.27, but activated in different conditions.
Values in P2.2.1.29 and P2.2.1.31 must be negative, because the pulling torque has to be lower to facilitate the speed change in these conditions.

2.2.1.35 Pulse torque at Max Diameter

Values in P2.2.1.27-29-31-33 are applied at minimum drum diameter. They can then vary linearly with actual diameter. This parameter is the multiplying factor at maximum diameter (100,0% means pulse torque not influenced by diameter value).
It is suggested to tune parameters P2.2.1.27-34 with the reel at minimum diameter and then to adjust P2.2.1.35 at maximum.

4.2.2. Speed Window (Menu M2 → G2.2.2)

Code	Parameter	Min	Max	Unit	Default	Note
P2.2.2.1	Speed Window	0	1		0	0= Not active 1= Active
P2.2.2.2	Upper Limit	0	100	%	10	Upper limit %
P2.2.2.3	Lower Limit	0	100	%	5	Lower limit %
P2.2.2.4	Min Upper Limit	0	50,00	Hz	5,00	Minimum value of upper limit
P2.2.2.5	Min Lower Limit	0	50,00	Hz	5,00	Minimum value of lower limit
P2.2.2.6	Upper Lim Action	0	3		1	0: no action 1: limit (rewind and unwind) 2: limit (rewind) + alarm (unwind) 3: limit (rewind) + limit (unwind)
P2.2.2.7	Lower Lim Action	0	3		1	0: no action 1: alarm (rewind and unwind) 2: limit (rewind) + alarm (unwind) 3: limit (rewind) + limit (unwind)
P2.2.2.8	Alarm Delay		10,00	s	1,00	Filter time for alarm
P2.2.2.9	Speed Reference Selection	0	2		0	0= analog input AIN2 1= encoder 2 2= fieldbus
P2.2.2.10	Encoder Multiplier	1	32000		1	Multiplication factor for encoder frequency
P2.2.2.11	Encoder Divider	1	32000		1	Division factor for encoder frequency
P2.2.2.12	Max Frequency at Minimum Diameter	0	P2.1.1		50,00	Motor speed at max crane speed and minimum drum diameter
P2.2.2.13	Max Torque for speed limit	50,0	300,0	%	100,00	Maximum produced torque
P2.2.2.14	Open window logic	0	2		0	0: never 1: resolver above threshold (and reel IN) 2: resolver below threshold (and Reel IN)
P2.2.2.15	Open window limit	0,0	100,0	%	0	Threshold in resolver signal, to open the window limit in Reel In condition
P2.2.2.16	Close window limit	0,0	100,0	%	0	Threshold in resolver signal, to close the window limit in Reel OUT condition
P2.2.2.17	Acc/Dec if open window	0	1		1	0: Acc/Dec signals are active 1:Acc/Dec signals are masked if window is open
P2.2.2.18	Speed fault	0	1		1	0: actual speed is not checked 1: speed error is checked
P2.2.2.19	Max speed error	1,00	100,00	Hz	5,00	Max speed error allowed (out of window condition)
P2.2.2.20	Speed check delay	0,00	10,00	s	1,00	Delay in speed error check, after window control activation

Table 1-5. Speed Window parameters, G2.2.2

If function “Speed window” is not active, the torque controlled motor can reach a maximum positive (rewinding direction) speed corresponding to maximum frequency P2.1.1, and a maximum negative (unwinding) speed corresponding to the frequency: - (P2.1.1 +5Hz).

IMPORTANT: when motor speed reaches those limits, control mode changes to speed control and the frequency converter can drive the motor with up to the maximum current to avoid limits overtaking. It is thus necessary to set correctly P2.1.1 so that it is not lower than the actual operating frequency, requested when the crane moves at its maximum speed.
During unwinding operation there is a 5Hz extension for the limit, to assure that an excessive tension on the cable will not take place because of the speed limit.

When Speed window is active (P2.2.2.1 =1) analogue signal AIN2 is considered to represent the speed of the crane. This value is then internally scaled accordingly to the drum diameter, to determine a theoretical speed reference for the motor.
AIN2 can be configured as “bi-directional” (P2.3.2.7 =1) when the analogue signal range is divided between positive and negative speed (recommended solution), or “unidirectional” (P2.3.2.7 =0) when it indicates the speed absolute value.

In the latter case the sense of movement is discriminated by the rotation direction of the motor.
It is also possible to get the actual direction from a digital input (defined in P2.3.3.11) if P2.3.2.7=2.

As alternative to AIN2, the speed reference can be acquired from a second encoder (OPTA7 board required) or from a fieldbus. In these cases the signal is always bi-directional.

It is then considered a “window” around the theoretical speed, based on upper and lower limits, expressed as percentage; there is also a minimum limit, expressed as Hz, which allows to have a significant window also at very low speed.
Parameters P2.2.2.6 and P2.2.2.7 allow to define if drive must avoid overtaking of the speed limits or simply to give an alarm signaling overtaking by digital output.

The implemented logic is:

Rewinding: the motor never exceeds upper limit (P2.2.2.6 = 1,2 or 3); it is possible either to avoid lower limit overtaking (P2.2.2.7 = 2 or 3) or to signal whenever it happens (P2.2.2.7 = 1).

Unwinding: if P2.2.2.6= 1 or 2 the motor can always reach a positive speed equal to minimum limit; when P2.2.2.6 = 2, there is also alarm if speed exceeds upper limit.
If P2.2.2.6= 3, motor speed is not allowed to be under upper limit.
There is always alarm if speed overtakes lower limit (P2.2.2.7 = 1 o 2); maximum negative speed is given by the sum of P2.1.1, the corresponding lower limit and a 5Hz margin.
With P2.2.2.7 = 3, motor speed is not allowed to exceed lower limit.
The following figures shows graphically how Speed window works, under some different configurations.

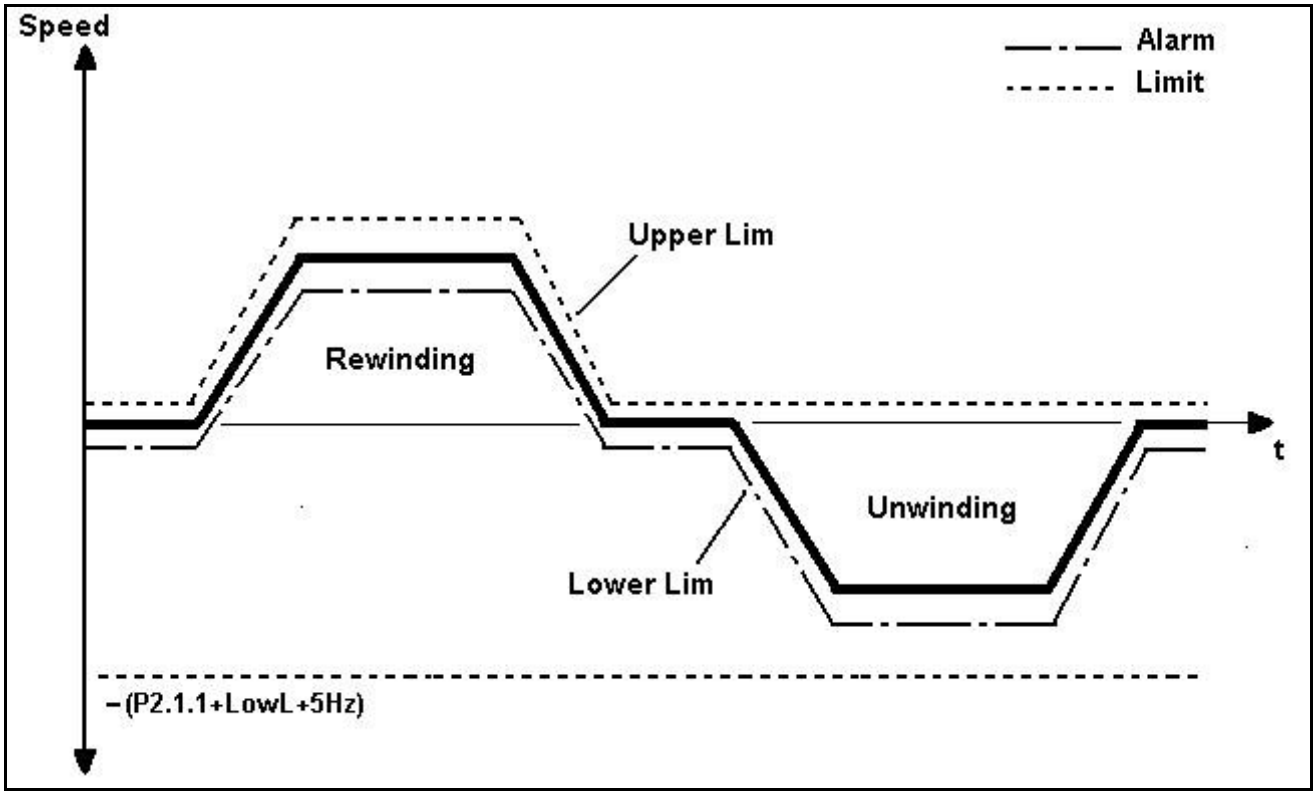


Figure 1-4. Speed window with P2.2.2.6=1 (limit) and P2.2.2.7=1 (alarm).

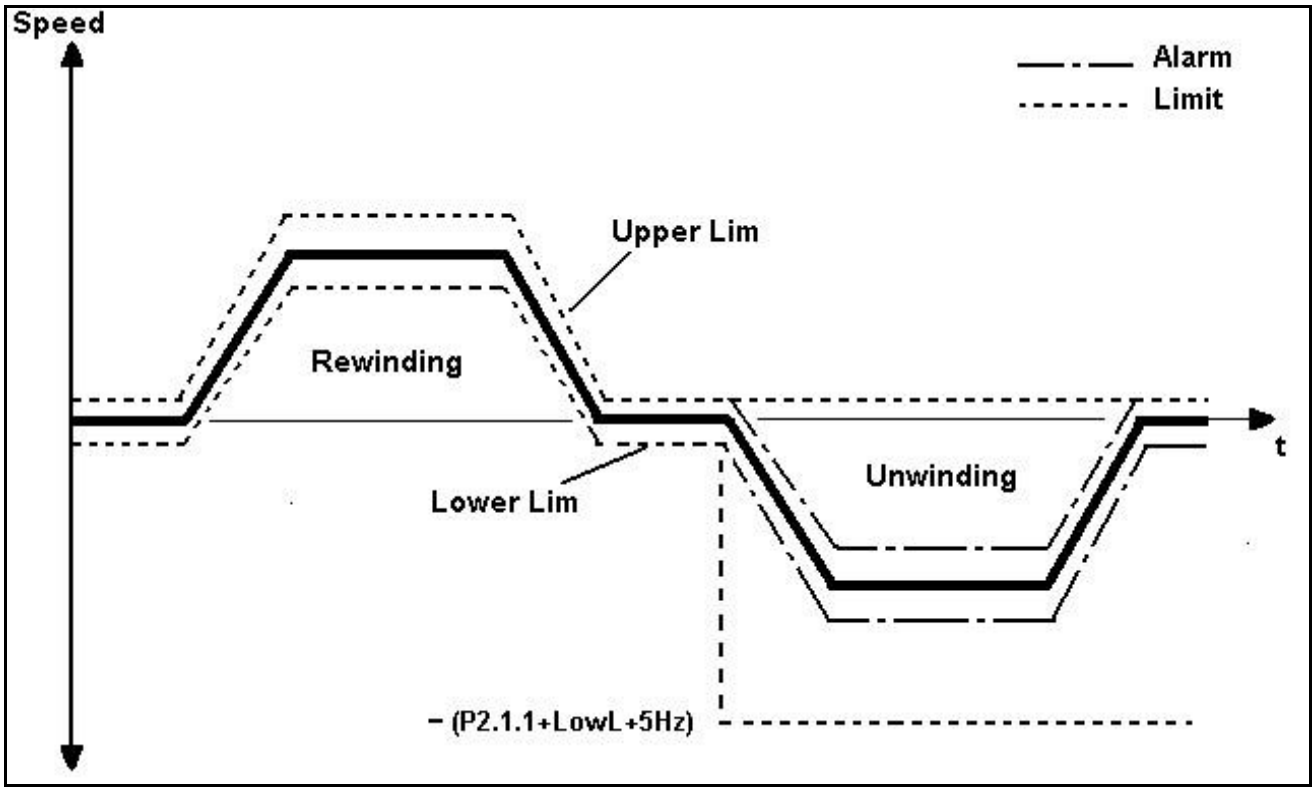


Figure 1-5. Speed window with P2.2.2.6 and P2.2.2.7=2 (limit + alarm).

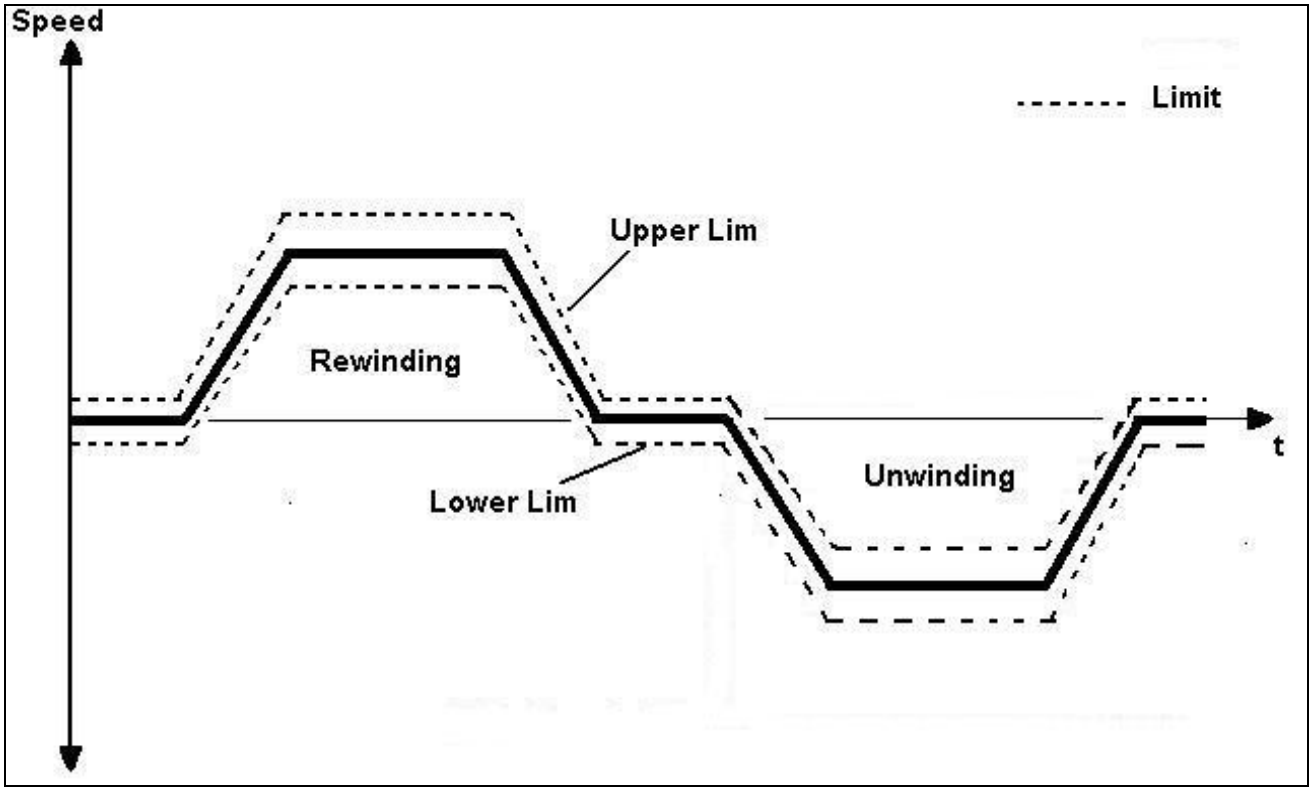


Figure 1-6. Speed window with P2.2.2.6 and P2.2.2.7=3 (limit + limit).

2.2.2.1 Speed Window

Activates speed control function.
0: Not active
1: Active

2.2.2.2 Upper Limit

This parameter represents upper limit of speed window; it is expressed as % of theoretical speed (signal AIN2).

2.2.2.3 Lower Limit

This parameter represents lower limit of speed window; it is expressed as % of theoretical speed (signal AIN2).

2.2.2.4 Minimum Upper Limit

This parameter represents the minimum value for upper limit of the window, to assure a suitable width also at very low speed; it is expressed as Hz.

2.2.2.5 Minimum Lower Limit

This parameter represents the minimum value for lower limit of the window, to assure a suitable width also at very low speed; it is expressed as Hz.

2.2.2.6 Upper Limit Action

Defines how frequency converter must work at reaching of upper limit.
0: No action (as P2.2.1.20.1=0)
1: Limit (rewinding) + No Action (unwinding) *
2: Limit (rewinding) + Alarm (unwinding) *
3: Limit (rewinding and unwinding)

***** Positive speed of the motor in unwinding is anyway limited to minimum positive limit.

2.2.2.7 Lower limit Action

Defines how frequency converter must work at reaching of lower limit.
0: No action (as P2.2.1.20.1=0)
1: Alarm (rewinding and unwinding)
2: Limit (rewinding) + Alarm (unwinding)
3: Limit (rewinding and unwinding)

2.2.2.8 Alarm Delay

This parameter allows to set a temporal filter on speed alarm.
Alarm will be activated only if overtaking of speed limit persists more than the programmed time.

Digital outputs can be set for signaling of speed error. An alarm text will be also showed on display, with code 61 when upper limit is overtaken (unwinding only) and code 62 when lower limit is overtaken (both unwinding and rewinding).

2.2.2.9 Speed Reference Source

Defines the signal for theoretical speed reference.
0: analog input AIN2
1: encoder 2, read by board OPTA7
2: fieldbus channel, configured by P2.9.2

Note: the low-pass filter programmed in P2.3.2.5 is active also on speed reference from encoder.

2.2.2.10 Encoder Multiplier
2.2.2.11 Encoder Divider

These parameters allow to define the kinematic ratio between the external encoder and the motor shaft.
Example: if the external encoder rotates at 1000rpm when the motor rotates at 1500rpm, parameters should be programmed as P2.2.2.10= 3 and P2.2.2.11=2.

2.2.2.12 Max Frequency at Minimum Diameter

This parameter sets the theoretical frequency of the motor when the crane is moving at its maximum speed and the diameter of the reeling drum is at minimum value.
It depends on mechanical ratio and polarity of the motor.

2.2.2.13 Max Torque for Speed Limit

This parameter sets the maximum torque that the motor will develop to respect the speed limits.
It is active when P2.2.2.1= 1 and (P2.2.2.6 >0 or P2.2.2.7>1).

This value is considered also when the motor is running within the speed window, so it must be higher than the maximum torque reference.
Speed limits can be overtaken if the torque limit is low.

2.2.2.14 Open window logic

The window limits can be automatically opened to maximum value, when in Reel IN condition, depending on the resolver signal level (analogue input set in P2.3.1.1).
The limits will be restored to normal value when in Reel OUT condition, still considering the resolver signal value.
IN or OUT condition is determined by motor encoder speed.
The window limits are also restored at any start of the machine, if the resolver has the proper signal.

0: never
1: resolver above threshold
2: resolver below threshold

2.2.2.15 Open window limit (in Reel IN)

Threshold in resolver signal, to open the speed window limits as from P2.2.2.14.

2.2.2.16 Close window limit (in Reel OUT and start of operations)

Threshold in resolver signal, to close the speed window limits as from P2.2.2.14.

2.2.2.17 Acc/Dec signals if window is open

0: Acc/Dec signals are considered when the window is open (specific torque levels are applied)
1: Acc/Dec signals are masked when the window is open (torque levels for constant speed are applied).

2.2.2.18 Speed fault

2.2.2.19 Max Speed Error

2.2.2.20 Speed check delay

Fault F64 can be activated with P2.2.2.18, in case the actual speed (motor encoder) does not respect the window (this can happen in case of fault on transducers or signals).
The maximum speed error (from window limits) is set in P2.2.2.19.
Checking of actual speed starts when the time in P2.2.2.20 has elapsed, after the drive has entered Run state and the window has been set to normal limits (not open).

4.2.3. Manual (Menu M2 → G2.2.3)

Code	Parameter	Min	Max	Unit	Default	Note
P2.2.3.1	Speed Selection	0	3		0	0 = jog speed 1= AIN2 2= AIN1 3= Panel
P2.2.3.2	Jog Speed	0,00	P2.1.1	Hz	10,00	Preset speed
P2.2.3.3	AIN Min Speed	0,0	P2.1.1	Hz	0,00	Minimum speed from AIN
P2.2.3.4	AIN Max Speed	P2.2.2.3	P2.1.1	Hz	50,00	Maximum speed from da AIN
P2.2.3.5	Acceleration Time	0,1	3000,0	s	3,0	Time from 0 to P2.1.1
P2.2.3.6	Deceleration Time	0,1	3000,0	s	3,0	Time from 0 to P2.1.1
P2.2.3.7	Torque Limit	0	400,0	%	100,0	Torque limit for manual commands
P2.2.3.8	Diameter Scaling	0	1		0	0= no scaling 1= scaling with reel diameter

Table 1-6 Manual parameters, G2.2.3

2.2.3.1 Speed Selection

Defines speed reference source for manual command.
0: preset speed P2.2.3.2
1: reference from AIN2
2: reference from AIN1
3: reference from panel

2.2.3.2 Jog Speed

It is preset speed reference.

2.2.3.3 AIN Min Speed

Defines speed reference corresponding to the minimum of analogue input.

2.2.3.4 AIN Max Speed

Defines speed reference corresponding to the maximum of analogue input.

2.2.3.5 Acceleration Time

Defines the time required for accelerate from zero frequency to value P2.1.1 .

2.2.3.6 Deceleration Time

Defines the time required for decelerate to zero frequency from value P2.1.1 .

2.2.3.7 Torque Limit

Defines torque limit, active when manual commands are on.

2.2.3.8 Diameter Scaling

0= reel speed does not depend on actual diameter
1= reel speed is scaled depending on actual diameter.

This function allows to keep the cable speed coarsely constant, also when the reel diameter varies.
Diameter measure is taken from AIN1 analogue signal (resolver used also for torque modulation).

4.3. Input Signals Parameters

4.3.1. AIN1 Torque Control (Menu M2 → G2.3.1)

Code	Parameter	Min	Max	Unit	Default	Note
P2.3.1.1	Signal AIN1Selection	0.1	E.9		A.1	Selection of analogue channel
P2.3.1.2	Signal AIN1 Range	0	2		1	0: 0-20mA/ 0-10V 1: 4-20mA/ 2-10V 2: custom values
P2.3.1.3	AIN1 Custom Min	-300,00	300,00	%	0,00	Signal minimum value
P2.3.1.4	AIN1 Custom Max	-300,00	300,00	%	0,00	Signal maximum value
P2.3.1.5	AIN1 Filter Time	0	10,0	s	0,1	Filter time constant
P2.3.1.6	AIN1 Signal Inversion	0	1		0	0: no inversion 1: inverted signal

Table 1-7. AIN1 Torque Control parameters, G2.3.1

2.3.1.1 Signal AIN1 Selection

Defines analogue input channel connected to signal AIN1, for torque control.
The same signal is also used for drum diameter measurement.
Parameter carries a letter corresponding to slot into which I/O board is inserted, and a number which describes the ordinal position of that input on the board.
NOTE: D.1 or E.1 selection means that the torque reference is read by fieldbus communication (see parameter P2.9.12).
In this case, A.1 input is still used for diameter measurement.
Fieldbus board can be placed either in slot D or E.

2.3.1.2 Signal AIN1 Range

Defines range of the analogue signal
0: range 0…10V / 0… 20mA
1: range 2…10V / 4… 20mA
2: custom range defined by parameters P2.3.1.3 e P2.3.1.4
NOTE: parameter should be set to 0 when torque reference from fieldbus is active and resolver is not connected to AI1.

2.3.1.3 AIN1 Custom Min
2.3.1.4 AIN1 Custom Max

Define minimum and maximum levels of signal; active only when P2.3.1.2 = 2.

2.3.1.5 AIN1 Filter Time

When this parameter is given a value greater than zero, it activates the function that filters out disturbances from the incoming analogue signal.

2.3.1.6 AIN1 Signal Inversion

Signal inversion allows to have maximum value of controlled quantity when signal is at minimum and vice versa.
0: No inversion
1: Inverted signal

4.3.2. AIN2 Speed Reference (Menu M2 → G2.3.2)

Code	Parameter	Min	Max	Unit	Default	Note
P2.3.2.1	Signal AIN2Selection	0.1	E.9		A.2	Selection of analogue channel
P2.3.2.2	Signal AIN2 Range	0	2		1	0: 0-20mA/ 0-10V 1: 4-20mA/ 2-10V 2: custom values
P2.3.2.3	AIN2 Custom Min	-300,00	300,00	%	0,00	Signal minimum value
P2.3.2.4	AIN2 Custom Max	-300,00	300,00	%	0,00	Signal maximum value
P2.3.2.5	AIN2 Filter Time	0	10,0	s	0,1	Filter time constant
P2.3.2.6	AIN2 Signal Inversion	0	1		0	0: no inversion 1: inverted signal
P2.3.2.7	Speed Range	0	1		1	0: unidirectional 1: bidirectional 2: direction from digital input

Table 1-8. AIN2 Speed Reference, G2.3.2

2.3.2.1 Signal AIN2 Selection

Defines analogue input channel connected to signal AIN2, for speed reference.
Parameter carries a letter corresponding to slot into which I/O board is inserted, and a number which describes the ordinal position of that input on the board.

2.3.2.2 Signal AIN2 Range

Defines range of the analogue signal
0: range 0…10V / 0… 20mA
1: range 2…10V / 4… 20mA
2: custom range defined by parameters P2.3.2.3 e P2.3.2.4

2.3.2.3 AIN2 Custom Min
2.3.2.4 AIN2 Custom Max

Define minimum and maximum levels of signal; active only when P2.3.2.2 = 2.

2.3.2.5 AIN2 Filter Time

When this parameter is given a value greater than zero, it activates the function that filters out disturbances from the incoming analogue signal.

2.3.2.6 AIN2 Signal Inversion

Signal inversion allows to have maximum value of controlled quantity when signal is at minimum and vice versa.
0: No inversion
1: Inverted signal

2.3.2.7 Speed range (for window)

Defines if analogue signal indicates also direction of movement.
0: Unidirectional: signal represents absolute value of speed. Direction is from motor encoder
1: Bi-directional: signal is divided between positive and negative speed. As default, upper semi-interval indicates positive speed (rewinding direction).
2: Analogue signal represents absolute value of speed. Direction is got from digital input (defined in P2.3.3.11).

4.3.3. Digital Inputs (Menu M2 → G2.3.3)

Code	Parameter	Min	Max	Default	Note
P2.3.3.1	DI1 Moving	0.1	E.9	A.1	
P2.3.3.2	DI2 Fault Reset	0.1	E.9	A.2	
P2.3.3.3	DI3 Reel OUT	0.1	E.9	A.3	
P2.3.3.4	DI4 Accel	0.1	E.9	A.4	
P2.3.3.5	DI5 Decel	0.1	E.9	A.5	
P2.3.3.6	DI6 Manual	0.1	E.9	A.6	
P2.3.3.7	Manual Reel In	0.1	E.9	A.4	
P2.3.3.8	Manual Reel Out	0.1	E.9	A.5	
P2.3.3.9	External Fault Closed	0.1	E.9	0.1	
P2.3.3.10	External Fault Open	0.1	E.9	0.2	
P2.3.3.11	Speed Window Reverse	0.1	E.9	0.1	

Table 1- 9. Digital Inputs parameter , G2.3.3

Parameter carries a letter corresponding to slot into which I/O board is inserted, and a number which describes the ordinal position of that input on the board
Setting 0.2 forces at high level the corresponding logic signal.

2.3.3.1 DI1 Moving

Defines digital input connected to “run” signal DI1.

2.3.3.2 DI2 Fault Reset

Defines digital input connected to reset signal DI2.

2.3.3.3 DI3 Reel OUT / Offset 3

2.3.3.4 DI4 Accel / Offset 2

2.3.3.5 DI5 Decel / Offset 1

Define digital inputs connected to signals for torque selection DI3, DI4, DI5.

2.3.3.6 DI6 Manual

Defines digital input which enables manual commands.

2.3.3.7 Manual Reel In

2.3.3.8 Manual Reel Out

Define digital inputs connected to forward and reverse commands, active only when DI6=1.

2.3.3.9 External Fault Closed

Defines digital input connected to signal for external activation of fault state; fault is activated with high signal.

2.3.3.10 External Fault Open

Defines digital input connected to signal for external activation of fault state; fault is activated with low signal (default value 0.2).

2.3.3.11 Speed Window Reverse

Reverse for speed reference, active only if P2.3.2.7=2.

4.3.4. Drum Diameter (Menu M2 → G2.3.4)

Code	Parameter	Min	Max	Unit	Default	Note
P2.3.4.1	Minimum Diameter	100	10000	mm	2400	Value of drum minimum diameter
P2.3.4.2	Maximum Diameter	100	10000	mm	6000	Value of drum maximum diameter
P2.3.4.3	Signal at Min Diameter	0	100,00	%	0	Analogue signal AI1 at minimum diameter
P2.3.4.4	Signal at Max Diameter	0	100,00	%	0	Analogue signal AI1 at maximum diameter

Table 1-8.1. Drum Diameter, G2.3.4

Actual diameter of the drum is measured through analogue signal AIN1 (P2.3.1.1).

The measure is used to calculate the theoretical speed reference for speed window function and speed reference for manual control. Takes also part in additive torque terms (inertia compensation and DI Logic pulses).

In case the torque reference is controlled by Fieldbus, the first analogue input AI1 is anyway considered for diameter measure.

2.3.4.1 Minimum Diameter

Value of drum minimum diameter.

2.3.4.2 Maximum Diameter

Value of drum maximum diameter.

2.3.4.3 Signal at Minimum Diameter

Value of analogue signal AI1 corresponding to minimum diameter (can be read on monitor V1.12).

2.3.4.4 Signal at Maximum Diameter

Value of analogue signal AI1 corresponding to maximum diameter (can be read on monitor V1.12).

4.4. Output Signals Parameters (Menu M2 → G2.4)

Code	Parameter	Min	Max	Unit	Default	Note
P2.4.1	Analogue Output Selection	0.1	E.10		A.1	Selection of the analogue output on I/O boards
P2.4.2	Analogue Output Function	0	10		1	**0**=Not used **1**=Output frequency (0—f_{max}) **2**=Frequency reference (0—f_{max}) **3**=Motor speed (0—Nominal motor speed) **4**=Motor current (0—I_{nMotor}) **5**=Motor torque (0—T_{nMotor}) **6**=Motor power (0—P_{nMotor}) **7**=Motor voltage (0--U_{nMotor}) **8**=DC link voltage (0—1000V) **9**=Input AI1 (0-100%) **10**= Input AI2 (0-100%)
P2.4.3	Analogue Output Filter Time	0,00	10,00	s	1,00	
P2.4.4	Analogue Output Inversion	0	1		0	0 = No inversion 1 = Inverted signal
P2.4.5	Analogue Output Minimum	0	1		0	**0** = 0 mA **1** = 4 mA
P2.4.6	Analogue Output Scale	10	1000	%	100	
P2.4.7	Digital Output Function	0	17		2	**0**=Not used **1**=Ready **2**=Run **3**=Fault **4**=Fault Inverted **5**=Drive over-temperature **6**=External fault **7**=Fault/Warning 4mA **8**=Warning **9=** Reverse **10**=At speed **11**=Frequency limit supervision **12=**Motor over-temperature **13=**Speed window overtaking **14=**Superv. limit 1 AIN1 **15=**Superv. limit 2 AIN1 **16**= Brake control **17=** Overspeed
P2.4.8	Relay 1 Output Function	0	17		1	As parameter 2.4.6
P2.4.9	Relay 2 Output Function	0	17		3	As parameter 2.4.6
P2.4.10	Frequency Limit Supervision Function	0	2		0	**0** = No limit **1** = Low limit superv. **2** = High limit superv.
P2.4.11	Frequency Limit Supervision Value	0,00	320,00	Hz	0,00	Threshold for P2.4.9
P2.4.12	Input AIN1 Limit 1 Supervision	0	2		0	**0** = No limit **1** = Low limit superv. **2** = High limit superv.
P2.4.13	Input AIN1 Supervision Value 1	0,00	100,00	%	70,00	Threshold for P2.4.12
P2.4.14	Input AIN1 Limit 2 Supervision	0	2		0	**0** = No limit **1** = Low limit superv. **2** = High limit superv.

P2.4.15	Input AIN1 Supervision Value 2	0,00	100,00	Hz	20,00	Threshold for P2.4.14
P2.4.16	Brake open delay	0,00	10,00	s	0,00	Delay on open brake command
P2.4.17	Min freq for brake closing	0,00	320,00	Hz	0,50	Minimum frequency for brake closing
P2.4.18	Stabilization Time	0,00	100,00	s	1,00	Time for speed stabilization, beore brake closing
P2.4.19	Brake Closing Time Out	0,00	100,00	s	5,00	Brake closing time outt. If = 0, the function is not active
P2.4.20	Hold Torque Time	0,00	100,00	s	1,00	Time for torque hold after brake closing, in automatic mode
P2.4.21	Channel for opt. relay RO3	0.1	E.10		D.1	Selection of the channel on expander board
P2.4.22	Opt. Relay RO3 Function	0	16		14	As parameter P2.4.7
P2.4.23	Channel for opt. relay RO4	0.1	E.10		D.2	Selection of the channel on expander board
P2.4.24	Opt. Relay RO4 Function	0	16		15	As parameter P2.4.7
P2.4.25	Channel for opt. relay RO5	0.1	E.10		D.3	Selection of the channel on expander board
P2.4.26	Opt. Relay RO5 Function	0	16		0	As parameter P2.4.7

Table 1- 10. Output Signals parameters, G2.4

2.4.1 Analog output channel

Position of the analog output on boards, normally A.1.

2.4.2 Analogue Output Function

Defines the variable to be connected to analogue output signal.

2.4.3 Analogue Output Filter Time

Defines the filtering time of the analogue output signal.

2.4.4 Analogue Output Inversion

If active, inverts the analogue output signal: max output when variable is at minimum.

2.4.5 Analogue Output Minimum

Defines the minimum signal to either 0 mA or 4 mA (living zero).
0: set minimum to 0 mA
1: set minimum to 4 mA

2.4.6 Analogue Output Scale

Scaling factor for analogue output.

2.4.7 Digital Output Function

2.4.8 Relay 1 Output Function

2.4.9 Relay 2 Output Function

Setting Value	Signal Content
0 = Not used	Not used
	Digital output DO1 sinks the current and relay (RO1,RO2) is activated when:
1 = Ready	The frequency converter is ready to operate
2 = Run	The frequency converter operates (motor is running)
3 = Fault	A fault trip has occurred
4 = Fault Inverted	A fault trip has not occurred
5 = Drive over-temperature	Heat-sink temperature exceeds +70°C
6 = External fault/warning	Fault or warning depending on P2.7.6
7 = Reference fault/warning	Fault or warning if analogue reference is 4—20 mA and signal is <4mA
8 = Warning	Always if an alarm exists
9 = Reverse	The reverse command has been selected
10 = At speed	The output frequency has reached the set reference
11 = Frequency limit supervision	The output frequency goes outside the set supervision low/high limit (see parameters 2.4.9 and 2.4.10)
12 = Motor over-temperature	The thermistor input of option board indicates over-temperature.
13 = Speed window overtaking	See chapter 4.2.1.1
14 = Analog input AIN1 supervision 1	Analog input AIN1 (torque reference) exceeds the first supervision threshold
15 = Analog input AIN1 supervision 2	Analog input AIN1 (torque reference) exceeds the second supervision threshold
16 = Brake control	All the conditions for brake opening are verified
17 = Overspeed	Motor speed exceeds the value programmed in P2.7.29. The signal is kept active for 2 seconds at least.

Table 1- 11. Output signal by DO1 and output relays RO1 and RO2.

2.4.10 Frequency Limit Supervision Function

0: No supervision
1: Low limit supervision
2: High limit supervision

If the output frequency goes under/over the set limit (P 2.4.11), this function generates a warning message via digital output

2.4.11 Frequency Limit Supervision Value

Select the frequency value supervised by parameter 2.4.10.

2.4.12 Supervision function 1 of analog input AIN1

0: No supervision
1: Low limit supervision
2: High limit supervision

When the analog torque reference is under/over the limit fixed in P 2.4.13, this function activates the digital output DO1 or a realy output.

2.4.13 Threshold 1 of AIN1 supervision

Threshold 1 value.

2.4.14 Supervision function 2 of analog input AIN1

0: Nessuna supervisione
1: Supervisione limite inferiore
2: Supervisione limite superiore

When the analog torque reference is under/over the limit fixed in P 2.4.15, this function activates the digital output DO1 or a realy output.

2.4.15 Threshold 2 of AIN1 supervision

Threshold 2 value.

2.4.16 Delay on opening of mechanical brake

2.4.17 Minimum frequency for brake closing

2.4.18 Stabilization time before brake closing

2.4.19 Brake closing time out

2.4.20 Holding torque time after closing

The brake control logic is activated when one of the output is configured for this function.

When the drive works on automatic mode, brake is opened when signal DI1 is active and the drive enters Run status, possibly with a small delay programmed in P2.4.16.
The brake is closed when, with DI1 fallen, the motor rotates at a speed below the threshold programmed in P2.4.17, for the stabilization time defined in P2.4.18.
During this phase the motor is still controlled by torque (as DI1 was still active), and it is released when the time in P2.4.20 ha elapsed after brake closing.
If different than zero, parameter P2.4.19 sets a time-out, after that the brake is closed independently by the actual speed.
When a Default torque value is set, this is indefinitely kept after brake closing.

When the drive works on manual mode, brake is opened when entering Run status and is closed when, with run commands fallen, the minimum frequency is reached.
Time out logic is not active.

2.4.21 Channel optional relay RO3

2.4.22 Function relay RO3

2.4.23 Channel optional relay RO4

2.4.24 Function relay RO4

2.4.25 Channel optional relay RO5

2.4.26 Function relay RO5

These parameters define the position on expander boards and the function (see table 1-11) of three optional relays.

4.5. Drive Control Parameters (Menu M2 → G2.5)

Code	Parameter	Min	Max	Unit	Default	Note
P2.5.1	Brake Chopper	0	4		4	0: not used 1: used and tested when running 2: used external 3: used and tested when stopped/running 4: used, no testing
P2.5.2	Stop Function	0	1		1	0: coasting 1: ramp

Table 1- 12. Drive Control parameters, G2.5

2.5.1 Brake Chopper

0 = No brake chopper used
1 = Brake chopper used and tested when running
2 = External brake chopper (no testing)
3 = Brake chopper used and tested in READY state and when running
4 = Brake chopper used when running, no testing

2.5.2 Stop Function

0: The motor coasts to a halt without control from the frequency converter after the Stop command.

1: After the stop command, the speed of the motor is decelerated according to the set deceleration parameters.

4.6. Motor Control Parameters (Menu M2 → G2.6)

Code	Parameter	Min	Max	Unit	Default	Note
P2.6.1	Motor Control Mode	0	1		1	0: open loop 1: closed loop
P2.6.2	Field Weakening Point	8,00	320,00	Hz	50,00	
P2.6.3	Voltage at field weakening point	10,00	200,00	%	100,00	
P2.6.4	Switching Frequency	1,0	16,0	kHz	10,0	
P2.6.5	Overvoltage Controller	0	1		1	0: not active 1: active
P2.6.6	Undervoltage Controller	0	1		1	0: not active 1: active

Table 1- 13. Motor Control parameters, G2.6

2.6.1 Motor Control Mode

0: "sensorless" vector regulation.
1: closed loop vector regulation, with encoder.

2.6.2 Field Weakening Point

The field weakening point is the output frequency at which the output voltage reaches the set maximum value.

2.6.3 Voltage at field weakening point

Above the frequency at the field weakening point, the output voltage remains at the set maximum value.
When parameters 2.1.3 e 2.1.4 (nominal voltage and frequency of the motor) are set, the parameters 2.6.2 e 2.6.3 are automatically given the corresponding values. If you need different values for the field weakening point and the maximum output voltage, change these parameters **after** setting parameters.

2.6.9 Switching Frequency

Motor noise can be minimized using a high switching frequency. Increasing the switching frequency reduces the capacity of the frequency converter unit.
The range of this parameter depends on of the size of the converter:
up to NX 0061: 1…16 kHz
> NX 0072: 1…10 kHz

2.6.10 Overvoltage Controller

2.6.11 Undervoltage Controller

These parameters allow the under/over-voltage controllers to be switched out of operation. This may useful, for example, if the mains supply voltage varies more than 15% to 10% and the application will not tolerate this variation.
These regulator control the output frequency taking the supply fluctuations into account.

Note: Over/under-voltage trips may occur when controllers are switched out of operation.

0: Controller switched off
1: Controller switched on

4.6.1. Open Loop (Menu M2 → G2.6.7)

Code	Parameter	Min	Max	Unit	Default	Note
P2.6.7.1	U/f Optimization	0	1		0	0: not used 1: automatic torque boost
P2.6.7.2	U/f Ratio Selection	0	3		0	0: linear 1: squared 2: programmable
P2.6.7.3	U/f curve Middle Point Frequency	0,00	par. P2.6.2	Hz	50,00	
P2.6.7.4	U/f curve Middle Point Voltage	0,00	100,00	%	100,00	n% x U_{nmot}
P2.6.7.5	Output Voltage at Zero Frequency	0,00	40,00	%	0,00	n% x U_{nmot}

Table 1- 14. Open Loop parameters, G2.6.7

2.6.7.1 U/f Optimization

0: not used
1: automatic torque "boost": the voltage to the motor changes automatically, which makes the motor produce sufficient torque to start and run at low frequency. The voltage increase depends on the motor type and power.

2.6.7.2 U/f Ratio Selection

0: Linear: motor voltage changes linearly with the frequency in the constant flux area from 0 Hz to the field weakening point where the nominal voltage is supplied to the motor.

1: Squared: motor voltage changes following a squared curve from 0 Hz to the field weakening point where the nominal voltage is supplied to the motor.

2: Programmable U/f curve: the U/f curve can be programmed with three different points, corresponding to frequencies 0, P2.6.7.3 and P2.6.2.

2.6.7.3 U/f curve, Mid Point Frequency

If the programmable U/f curve has been selected, this parameter defines the middle point frequency of the curve.

2.6.7.4 U/f curve, Mid Point Voltage

If the programmable U/f curve has been selected, this parameter defines the middle point voltage of the curve.

2.6.7.5 Output Voltage at Zero Frequency

If the programmable U/f curve has been selected, this parameter defines the zero frequency voltage of the curve.

4.6.2. Closed Loop (Menu M2 → G2.6.8)

Code	Parameter	Min	Max	Unit	Default	Note
P2.6.8.1	Speed Control Kp	0	1000		30	Proportional gain for speed control
P2.6.8.2	Speed Control Ti	0,0	500,0	ms	30,0	Integral time constant for speed control
P2.6.8.3	Slip Adjust	0	500	%	100	Nominal slip adjust
P2.6.8.4	Encoder Filter Time	0	1000	ms	0	Constant time for filtering encoder signal
P2.6.8.5	Current Control Kp	0,00	100,00	%	40,00	Proportional gain for current control

Table 1- 15. Closed Loop parameters, G2.6.8

2.6.8.1 Speed Control Kp

2.6.8.2 Speed Control Ti

Gains for PI speed regulator; they allow to optimize dynamic response of the motor.

2.6.8.3 Slip Adjust

This value is used to adjust the voltage of the motor when loaded. The name plate speed is sometimes a little inaccurate and this parameter can therefore be used to trim the slip.

2.6.8.4 Encoder Filter Time

Sets the filter time constant for speed measurement. This parameter can be used to eliminate encoder signal noise.

2.6.8.5 Current Control Kp

Proportional gain for current regulator.

4.6.3. Torque Control (Menu M2 → G2.6.9)

Code	Parameter	Min	Max	Unit	Default	Note
P2.6.9.1	Torque Control Kp	0	1000		150	Proportional gain for torque control (open loop only)
P2.6.9.2	Torque Control Ti	0	1000		10	Integral time constant for speed control (open loop only)
P2.6.9.3	Speed Limit Acc Time	0,1	3000,0	s	1,0	Accel ramp time on speed limit
P2.6.9.4	Speed Limit Dec Time	0,1	3000,0	s	1,0	Decel ramp time on speed limit

Tabella 1- 16. Torque Control parameters, G2.6.9

2.6.9.1 Torque Control Kp (only open loop)

2.6.9.2 Torque Control Ti (only open loop)

Gains for PI speed regulator; they allow to optimize dynamic response of the motor when torque reference varies. Parameters are not active in closed loop motor control mode.

2.6.9.3 Speed Limit Acc Time

Defines rising time for speed limit. Value must be lower than crane acceleration time.

2.6.9.4 Speed Limit Dec Time

Defines falling time for speed limit. Value must be lower than crane deceleration time.

4.7. Protection (Menu M2 → G2.7)

Code	Parameter	Min	Max	Unit	Default	Note
P2.7.1	Encoder Supervision	0	1		1	**0=** Active **1**= Not active
P2.7.2	Response to Reference AIN1 Fault 4mA	0	2		0	**0**= No response **1**= Warning + preset P2.7.3 **2**= Fault
P2.7.3	Preset Reference for AIN1 fault 4mA	4	20	mA	4	Value set if 4mA fault on AIN1 reference
P2.7.4	Response to Reference AIN2 Fault 4mA	0	2		0	**0**= No response **1**= Warning + preset P2.7.5 **2**= Fault
P2.7.5	Preset Reference for AIN2 fault 4mA	4	20	mA	4	Value set if 4mA fault on AIN2 reference
P2.7.6	Response to External Fault	0	3		2	**0**=No response **1**=Warning **2**=fault, stop according to P2.5.2 **3**=fault, stop by coasting
P2.7.7	Input Phase Supervision	0	3		0	
P2.7.8	Response to Undervoltage Fault	1	3		2	
P2.7.9	Output Phase Supevision	0	3		2	
P2.7.10	Earth Fault Protection	0	3		2	
P2.7.11	Thermal Protection of the Motor	0	3		2	
P2.7.12	Motor Ambient Temperature Factor	–100,0	100,0	%	0,0	
P2.7.13	Motor Cooling Factor at Zero Speed	0,0	150,0	%	40,0	
P2.7.14	Motor Thermal Time Constant	1	200	min	10	
P2.7.15	Motor Duty Cycle	0	100	%	100	
P2.7.16	Stall Protection	0	3		0	**0**=No response **1**=Warning **2**=fault, stop according to P2.5.2 **3**=fault, stop by coasting
P2.7.17	Stall Current	0,1	6000,0	A	10,0	
P2.7.18	Stall Time Limit	1,00	120,00	s	15,00	
P2.7.19	Stall Frequency Limit	1,0	Par. 2.1.2	Hz	25,0	

Table 1-17/1. Protection parameters, G2.7

Code	Parameter	Min	Max	Unit	Default	Note
P2.7.20	Underload Protection	0	3		0	**0**=No response **1**=Warning **2**=fault, stop according to P2.5.2 **3**=fault, stop by coasting
P2.7.21	Underload Curve at Nominal Frequency	10	150	%	50	
P2.7.22	Underload Curve at Zero Frequency	5,0	150,0	%	10,0	
P2.7.23	Underload Protection Time Limit	2	600	s	20	
P2.7.24	Response to Thermistor Fault	0	3		2	See P 2.7.20
P2.7.25	Response to Fieldbus Fault	0	3		2	See P 2.7.20
P2.7.26	Response to Slot Fault	0	3		2	See P 2.7.20
P2.7.27	Temperature sensor on power unit	0	1		1	**0**= Read **1**= Ignore
P2.7.28	Overspeed Protection	0	3			See P 2.7.20
P2.7.29	Overspeed threshold	1,00	320,00	Hz	60,00	Threshold for alarm or signaling

Table 1-17/2. Protection parameters, G2.7

2.7.1 Encoder Supervision

This parameter allows to disable some encoder supervision functions; it can be useful in case of unsuitable trips under particular conditions.
0 = Supervision active
1 = Supervision not active

2.7.2 Response to Reference AIN1 Fault 4mA

2.7.4 Response to Reference AIN2 Fault 4mA

0 = No response
1 = Warning, preset value P2.7.3 (or P2.7.5) is set as reference
2 = Fault, stop mode according to parameters P2.5.2

A warning or a fault action and message is generated if the 4…20 mA reference signal is used and the signal falls below 3.5 mA for 5 seconds or below 0.5 mA for 0.5 seconds. Fault information can also be programmed into digital output DO1 or the relay outputs RO1 and RO2.

2.7.3 Preset Reference for AIN1 fault 4mA

2.7.5 Preset Reference for AIN2 fault 4 mA

If the value of parameter P2.7.2 (or P2.7.4) is set to 1 and the 4mA fault occurs, then a signal is internally imposed as reference equal to this parameter.

2.7.6 Response to External Fault

0 = No response
1 = Warning
2 = Fault, stop mode according to parameter P2.5.2
3 = Fault, stop mode always by coasting

A warning or a fault action is generated from the external fault signal on the digital input selected.

2.7.7 Input Phase Supervision

0 = No response
1 = Warning
2 = Fault, stop mode according to parameter P2.5.2
3 = Fault, stop mode always by coasting

The input phase supervision ensures that the input phases of the frequency converter have an approximately equal current.

2.7.8 Response to Undervoltage Fault

1 = Warning
2 = Fault, stop mode according to parameter P2.5.2
3 = Fault, stop mode always by coasting

2.7.9 Output Phase Supervision

0 = No response
1 = Warning
2 = Fault, stop mode according to parameter P2.5.2
3 = Fault, stop mode always by coasting

The output phase supervision ensures that the motor phases have an approximately equal current.
.

2.7.10 Earth Fault Protection

0 = No response
1 = Warning
2 = Fault, stop mode according to parameter P2.5.2
3 = Fault, stop mode always by coasting

Earth fault protection ensures that the sum of the motor phase currents is zero. The overcurrent protection is always working and protects the frequency converter from earth faults with high currents.

Parameters 2.7.11—2.7.15, Motor Thermal Protections: General

The motor thermal protection is to protect the motor from overheating. If the load requires a current higher than nominal, there is a risk that the motor will be thermally overloaded. At low frequencies the cooling effect of the motor is reduced as well as its capacity. If the motor is equipped with an external fan the load reduction at low speed is small.

The motor thermal protection is based on a calculated model and it uses the output current of the drive to determine the load on the motor.

The motor thermal protection can be adjusted with parameters.

Estimated thermal state of the motor can be monitored on panel display.



CAUTION! *The calculated model does not protect the motor if the airflow to the motor is reduced by blocked air intake grill.*

2.7.11 Motor Thermal Protection

0 = No response
1 = Warning
2 = Fault, stop mode according to parameter P2.5.2
3 = Fault, stop mode always by coasting
If tripping is selected, the drive will stop and activate the fault stage.
Deactivating the protection, i.e. setting parameter to 0, will reset the thermal stage of the motor to 0%.

2.7.12 MTP: Motor Ambient Temperature Factor

The value of the factor can be set between -100.0% and +100.0%. To set positive values if the ambient in which motor is working is hot, or negative values if ambient is cooled.

2.7.13 MTP: Zero Frequency Cooling Power

The current can be set between 0 and 150.0% x $I_{n\ Motor}$. This parameter defines the value of sustainable thermal current at zero frequency. The default value is set assuming that there is no external fan cooling the motor. If such fan is used, this parameter can be set to 90% or even higher.

2.7.14 MTP: Time Constant

This time can be set between 1 and 200 minutes.
This is the thermal time constant of the motor. The bigger the motor, the bigger the time constant. The time constant is the time within the calculated thermal stage has reached 63% of its final value.
The motor thermal time is specific to the motor design; to keep default value if exact data are not available.

2.7.15 MTP: Duty Cycle

Defines how much of the nominal motor is applied.
The value can be set to 0%…100%.

Parameters 2.7.16—2.7.19, Stall Protection:
General

The motor stall protection protects the motor from short time overload situations. The reaction time of the stall protection can be set shorter than that of the motor thermal protection.
The stall state is defined with two parameters P2.7.16 (Stall current) and P2.7.19 (Stall Frequency). If the current is higher than the set limit and output frequency is lower than the set limit, the stall state is true. Stall protection is a type of overcurrent protection.

2.7.16 Stall Protection

0 = No response
1 = Warning
2 = Fault, stop mode according to parameter P2.5.2
3 = Fault, stop mode always by coasting
Setting the parameter to 0 will deactivate the protection and reset the stall time counter.

2.7.17 Stall Current Limit

For a stall stage to occur, the current must have exceeded this limit.
The value is set as percent of motor nominal current.

2.7.18 Stall Time

This time can be set between 1.0 and 120.0s.
This is the maximum time allowed for a stall stage. The stall time is counted by an internal up/down counter. If the stall time counter goes above this limit, the protection will cause a trip.

2.7.19 Maximum Stall Frequency

The frequency can be set between 1-f_{max} (par. P2.1.1).
For a stall state to occur, the output frequency must have remained below this limit.

Parameters 2.7.20—2.7.23, Underload Protection:
General

The purpose of the motor underload protection is to censure that there is load on the motor when the drive is running. If the motor loses its load, there might be a problem in the process, e.g. a broken belt.
Motor underload protection can be adjusted by setting the underload curve with parameters P2.7.21 (field weakening area load) and P2.7.22 (zero frequency load). The underload curve is a squared curve set between the zero frequency and the field weakening point. The protection is not active below 5Hz (the underload time counter is stopped).

2.7.20 Underload Protection

0 = No response
1 = Warning
2 = Fault, stop mode according to parameter P2.5.2
3 = Fault, stop mode always by coasting

If tripping is selected, the drive will stop and activate the fault stage.
Deactivating the protection, i.e. setting parameter to 0, will reset the underload time counter.

2.7.21 Field Weakening Area Load

The torque limit can be set between 0 and 150.0 % x $T_{n\ Motor}$.
This parameter gives the value for the minimum torque allowed when the output frequency is above the field weakening point.

2.7.22 Zero Frequency Load

The torque limit can be set between 0 and 150.0 % x $T_{n\ Motor}$.
This parameter gives the value for the minimum torque allowed with zero frequency.

2.7.23 Underload Time

This time can be set between 2.0 and 600.0 s.
This is the maximum time allowed for an underload state to exist. An internal up/down counter counts the accumulated underload time. If the counter value goes above this limit, the protection will cause a trip.
If the drive is stopped the underload counter is reset to zero.

2.7.24 Response to Thermistor Fault

0 = No response
1 = Warning
2 = Fault, stop mode according to parameter P2.5.2
3 = Fault, stop mode always by coasting

Set here the response mode for motor overheating by thermistor (option)

2.7.25 Response to Fieldbus Fault

0 = No response
1 = Warning
2 = Fault, stop mode according to parameter P2.5.2
3 = Fault, stop mode always by coasting

Set here the response mode for fieldbus bus if a fieldbus board is used.
.

2.7.26 Response to slot fault

0 = No response
1 = Warning
2 = Fault, stop mode according to parameter P2.5.2
3 = Fault, stop mode always by coasting

Set here the response for a board slot fault due to missing or broken board.

2.7.27 Temperature sensor on power unit

This parameter can by-pass the temperature sensor on power unit, allowing the drive to operate in case of malfunction of the sensor.
0 = read
1 = ignore

Note: by-pass can also be activated by pressing pushbuttons Select and Enter together for at least 5 seconds, regardless the page that is shown on panel.

2.7.28 Overspeed protection

0 = No response
1 = Warning
2 = Fault, stop mode according to parameter P2.5.2
3 = Fault, stop mode always by coasting

The protection diagnoses if the motor exceeds, also when drive is not in run status, the threshold in P2.7.29.

2.7.29 Overspeed threshold

Threshold for fault and also for signaling on digital output.

4.8. Auto Restart (Menu M2 → G2.8)

Code	Parameter	Min	Max	Unit	Default	Note
P2.8.1	Wait Time	0,10	10,00	s	0,50	
P2.8.2	Trial Time	0,00	60,00	s	30,00	
P2.8.3	Start Function	0	1		0	0=Ramp 1=Flying start
P2.8.4	Number of Tries after Undervoltage Trip	0	10		0	
P2.8.5	Number of Tries after Overvoltage Trip	0	10		0	
P2.8.6	Number of Tries after Overcurrent Trip	0	3		0	
P2.8.7	Number of Tries after Reference Trip	0	10		0	
P2.8.8	Number of Tries after Motor Temperature Trip	0	10		0	
P2.8.9	Number of Tries after External Fault Trip	0	10		0	

Table 1- 18. Auto Restart parameters, G2.8

2.8.1 Wait Time

Defines the time after which the frequency converter tries to automatically restart after the fault has disappeared.

2.8.2 Trial Time

The automatic restart function restarts the frequency converter when the faults selected with parameters P2.8.4 to P2.8.10 have disappeared and the waiting time has elapsed. Parameters P2.8.4 to P2.8.10 determine the maximum number of automatic restarts during the trial time set by parameter P2.8.2. The time count starts from the first autorestart. If the number of faults occurring during the trial time exceeds the values of parameters P2.8.4 to P2.8.10, the fault state becomes active. Otherwise the fault is cleared after the trial time has elapsed and the next fault starts the trial count again.

2.8.3 Start Function

The start function for automatic restart is selected with this parameter.

0 = Start with ramp
1 = Flying start

2.8.4 Number of Tries after Undervoltage Trip

This parameter determines how many automatic restarts can be made during the trial time after an undervoltage trip.

2.8.5 Number of Tries after Overvoltage Trip

This parameter determines how many automatic restarts can be made during the trial time after an overvoltage trip.
.

2.8.6 Number of Tries after Overcurrent Trip

This parameter determines how many automatic restarts can be made during the trial time after an overcurrent trip.
NOTE! IGBT temperature fault also included.

2.8.7 Number of Tries after Reference Fault Trip

This parameter determines how many automatic restarts can be made during the trial time after an reference fault (4mA) trip.

2.8.8 Number of Tries after Motor Temperature Trip

This parameter determines how many automatic restarts can be made during the trial time after motor overtemperature trip.

2.8.9 Number of Tries after External fault Trip

This parameter determines how many automatic restarts can be made during the trial time after external fault trip.

4.9. Fieldbus (Menu M2 → G2.9)

Code	Parameter	Min	Max	Unit	Default	Note
P2.9.1	Commands Word	0	1		0	Commands word mapping 0= CW 1= PDI1
P2.9.2	Speed reference word	0	2		0	Speed reference mapping 0= REF 1= PDI1 2= PDI2 3= PDI3
P2.9.3	Digital inputs active	0	1		0	0= inputs active with FB 1= inputs not active
P2.9.4	PDO1 Selection	0	2000		1	ID of the monitor variable mapped on PDO1
P2.9.5	PDO2 Selection	0	2000		3	ID of the monitor variable mapped on PDO2
P2.9.6	PDO3 Selection	0	2000		4	ID of the monitor variable mapped on PDO3
P2.9.7	PDO4 Selection	0	2000		37	ID of the monitor variable mapped on PDO4
P2.9.8	PDO5 Selection	0	2000		0	ID of the monitor variable mapped on PDO5
P2.9.9	PDO6 Selection	0	2000		0	ID of the monitor variable mapped on PDO6
P2.9.10	PDO7 Selection	0	2000		0	ID of the monitor variable mapped on PDO7
P2.9.11	PDO8 Selection	0	2000		0	ID of the monitor variable mapped on PDO8
P2.9.12	Torque reference word	0	2		0	Torque reference mapping 0= REF 1= PDI1 2= PDI2 3= PDI3

Table 1- 19. Fieldbus, G2.9

Drive control by fiedbus requires programming of parameter P3.1 Control Place as Fieldbus. Among parameters of the specific fieldbus board (menu M7), modality By-pass must be selected, when available.

Drive control is implemented by means of a Commands Word, whose bits reply the hard wired digital inputs.

.

bit	meaning
0	Crane moving DI1
1	Fault reset DI2
2	Offset 3 DI3
3	Offset 2 DI4
4	Offset 1 DI5
5	Manual DI6
6	Jog forward
7	Jog backward
8	Reverse speed reference

Commands word can be mapped into fields CW or PDI1 of the message structure

On the bus it is possible to transmit the speed reference, used for the speed window or for inertia compensation. This reference is bi-directional: direction of movement is controlled by the reference sign OR by means of bit8 of commands word.
Numerical value is expressed as percent of maximum speed
(-10000..+10000 : -100%..+100%).

Speed reference from bus is activated by programming P2.2.2.8= 2. It is recognized also when the drive is controlled by I/O terminals, and can be mapped into fields REF, PDI1, PDI2 or PDI3.

On the bus it is also possible to transmit the torque reference. The signal has the same meaning of the analogue input for torque scaling between minimum and maximum values.
Numerical range is 0..+10000 : 0%..+100%.

Torque reference from bus is activated by programming P2.3.1.1= D.1 or E.1. It is recognized also when the drive is controlled by I/O terminals, and can be mapped into fields REF, PDI1, PDI2 or PDI3.

Drive working condition can be diagnosed by means of a Status Word, with this coding:

bit	Significato
0	Ready
1	Run
2	Direction (1= unreel)
3	Fault
4	Alarm
5	AIN1 supervision 1
6	AIN2 supervision 2
7	Manual commands active
8	Brake control

Status word is always mapped into field SW of the message; can also be duplicated on one of the four PDO available.

The field ACT shows the motor actual speed, expressed as percent of maximum frequency
(-10000..+10000 : -100%..+100%).

Up to eight programmable PDO (process data output) are available, for reading of monitor variables.

2.9.1 Commands word mapping

0= field CW defined by fieldbus protocol
1= field PDO1 defined by fieldbus protocol

2.9.2 Speed reference mapping

0= field REF defined by fieldbus protocol
1= field PDI1 defined by fieldbus protocol
2= field PDI2 defined by fieldbus protocol
3= field PDI3 defined by fieldbus protocol

2.9.3 Digital inputs active

Defines if terminal digital inputs are active also when drive is controlled by bus.
In this case, all the functions are controlled by the logic sum (OR) of the signal and the correspondent bit.

2.9.4 Selection of variable on PDO1
2.9.5 Selection of variable on PDO2
2.9.6 Selection of variable on PDO3
2.9.7 Selection of variable on PDO4
2.9.8 Selection of variable on PDO5
2.9.9 Selection of variable on PDO6
2.9.10 Selection of variable on PDO7
2.9.11 Selection of variable on PDO8

Selection is done by writing into parameter the value ID (identifier) of the variable to monitor (see table below).

2.9.12 Torque reference mapping

0= field REF defined by fieldbus protocol
1= field PDI1 defined by fieldbus protocol
2= field PDI2 defined by fieldbus protocol
3= field PDI3 defined by fieldbus protocol

Monitoring variables

ID	Meaning	Format
1	Output frequency Hz	1 count= 0.01Hz
25	Frequency reference Hz	1 count= 0.01Hz
1552	Torque reference %	1 count= 0.1% of motor nominal torque
2	Motor speed rpm	1 count= 1 rpm
3	Motor current A	1 count= 0.1 A
4	Motor torque %	1 count= 0.1% of motor nominal torque
5	Motor power %	1 count= 0.1% of motor nominal power
6	Motor voltage V	1 count= 0.1 V
7	DC link voltage V	1 count= 1 V
8	Unit temperature °C	1 count= 1°C
9	Motor temperature	1 count= 0.1% of motor nominal temperature (estimated)
13	Analogue input 1	1 count= 0.01% of signal range
14	Analogue input 2	1 count= 0.01% of signal range
15	DIN1, DIN2, DIN3	binary coded b0= DIN3, b1= DIN2, b2= DIN1, b3-15 not used
16	DIN4, DIN5, DIN6	binary coded b0= DIN6, b1= DIN5, b2= DIN4, b3-15 not used
17	DO1,RO1,RO2	binary coded b0= RO2, b1= RO1, b2= DO, b3-15 not used
26	Analogue output	1 count= 0.01 mA
1553	Control mode	0= Basic 1= DI Logic 2= AutoReverse 3= Manual
1555	Inertia compensation %	1 count= 0.1% of motor nominal torque
1585	Reference for speed window Hz	1 count= 0.01Hz
1592	Status Word	See description at page 46
37	Active fault code	See fault codes list on Vacon User Manual
1603	Encoder frequency	1 count= 0.01Hz
1623	Drum diameter	1 count= 1mm

5. Keypad control (Menu M3)

Code	Parameter	Min	Max	Unit	Default	Note
P3.1	Control Place	1	3		1	**1** = I/O terminals **2** = Keypad **3** = Fieldbus
R3.2	Panel Reference	0,00	Par. 2.1.1	Hz	0,00	
P3.3	Direction	0	1		0	**0** = Forward **1** = Reverse
R3.4	Stop Button	0	1		1	**0**= Limited function for Stop button **1**= Stop button always active
P3.5	String	0	3		1	Power on string on display: **0**= None **1**= Cavotec Group Specimas **2**= VAHLE **3**= Konecranes
P3.6	Automatic Multimonitor	0	3		1	**0** = not active **1** = active

Table 1- 19. Keypad Control parameters, M3

3.1 Control Place

Active control place can be changed by this parameter. See also Vacon NX User's Manual chapter 7.3.3.1 for more information.
When keypad is the active control place, motor regulation is the same as in manual control mode, i.e. speed control with torque limit.

3.2 Keypad reference

Frequency reference can be set on panel with this parameter.

3.3 Keypad direction

0 Forward: motor runs forward when keypad is the active control place
1 Reverse: motor runs in reverse direction when keypad is the active control place.

3.4 Stop Button

If it is wished that Stop Button always stops the drive, regardless active control place, value 1 must be set on this parameter.

3.5 String

Selection of the company name that is shown on display at power on, for about 15 seconds. This parameter is normally locked. To unlock it, it is needed to push right arrow button for at least 10 seconds. Parameter is anyway unlocked when the drive is in serial communication with NCDrive tool.

3.6 Automatic Multimonitor

When active, Multimonitor page will be shown after 1 minute of drive keypad inactivity.

6. Expander Boards (Menu M7)

The Expander Boards menu makes it possible for the user :
1) to see what expander boards are connected to the control board
2) to reach and edit the parameters associated with the expander boards.

It is possibile to browse through slots from A to E to see what expander boards are connected.
On the lowermost line of the display you see the number of parameters associated with the board.
Some boards also display a Monitor menu, useful for diagnostic.

Code	Parameter	Min	Max	Default	Selection
P7.1.1.1	AI1 Mode	1	5	3	**1**=0…20 mA **2**=4…20 mA **3**=0…10 V **4**=2…10 V **5**=–10…+10 V
P7.1.1.2	AI2 Mode	1	5	1	See P7.1.1.1
P7.1.1.3	AO1 Mode	1	4	1	**1**=0…20 mA **2**=4…20 mA **3**=0…10 V **4**=2…10 V

Table 1-20. Parameters for basic I/O board NXOPTA1.

Code	Parameter	Min	Max	Default	Selection
P7.3.1.1	Pulses/revolution	1	65535	1024	
P7.3.1.2	Invert Direction	0	1	0	0: No 1: Yes
P7.3.1.3	Reading Rate	0	4	1	0: No 1: 1ms 2. 5ms 3: 10ms 4: 50ms

Table 1-21. Parameters for encoder board NXOPTA4.

7. Faults

Application specific fault codes

Code	Text	Action	Cause
61	Upper limit	Alarm	Motor speed is above higher window limit (when speed is not automatically limited)
62	Lower limit	Alarm	Motor speed is below lower window limit (when speed is not automatically limited)
63	Overspeed	From P2.7.28	Motor speed is over the threshold in P2.7.29
64	Speed error	Fault	Motor speed is out from window limits
70	License key	Fault	The right license key must be set in P2.10.1

Generic fault codes

Fault code	Fault	Possible cause	Correcting measures
1	Overcurrent	Frequency converter has detected too high a current (>4*I_H) in the motor cable: – sudden heavy load increase – short circuit in motor cables – unsuitable motor Subcode in T.14: S1 = Hardware trip S2 = Reserved S3 = Current controller supervision	Check loading. Check motor. Check cables. Make identification run.
2	Overvoltage	The DC-link voltage has exceeded the limits defined. – too short a deceleration time – high overvoltage spikes in supply Subcode in T.14: S1 = Hardware trip S2 = Overvoltage control supervision	Make deceleration time longer. Use brake chopper or brake resistor (available as options) Activate overvoltage controller. Check input voltage.
3	Earth fault	Current measurement has detected that the sum of motor phase current is not zero. – insulation failure in cables or motor	Check motor cables and motor.
5	Charging switch	The charging switch is open, when the START command has been given. – faulty operation – component failure	Reset the fault and restart. Should the fault re-occur, contact the distributor near to you.
6	Emergency stop	Stop signal has been given from the option board.	Check emergency stop circuit.
7	Saturation trip	Various causes: – defective component – brake resistor short-circuit or overload	Cannot be reset from the keypad. Switch off power. DO NOT RE-CONNECT POWER! Contact factory. If this fault appears simultaneously with Fault 1, check motor cables and motor

Fault code	Fault	Possible cause	Correcting measures
8	System fault	- component failure - faulty operation Note exceptional fault data record. S1 – Reserved S2 – Reserved S3 – Reserved S4 – Reserved S5 – Reserved S6 – Reserved S7 – Charging switch S8 – No power to driver card S9 – Power unit communication (TX) S10 – Power unit communication (Trip) S11 – Power unit comm. (Measurement)	Reset the fault and restart. Should the fault re-occur, contact the distributor near to you.
9	Undervoltage	DC-link voltage is under the voltage limits defined. – most probable cause: too low a supply voltage – frequency converter internal fault – defect input fuse – external charge switch not closed Subcode in T.14: S1 – DC-link too low during run S2 – No data from power unit S3 – Undervoltage control supervision	In case of temporary supply voltage break reset the fault and restart the frequency converter. Check the supply voltage. If it is adequate, an internal failure has occurred. Contact the distributor near to you.
10	Input line supervision	Input line phase is missing.	Check supply voltage, fuses and cable.
11	Output phase supervision	Current measurement has detected that there is no current in one motor phase.	Check motor cable and motor.
12	Brake chopper supervision	– no brake resistor installed – brake resistor is broken – brake chopper failure	Check brake resistor and cabling. If the these are ok, the chopper is faulty. Contact the distributor near to you.
13	Frequency converter under-temperature	Heatsink temperature is under –10°C	
14	Frequency converter over-temperature	Heatsink temperature is over 90°C (or 77°C, all FR6 690V drives). Overtemperature warning is issued when the heatsink temperature exceeds 85°C (72°C).	Check the correct amount and flow of cooling air. Check the heatsink for dust. Check the ambient temperature. Make sure that the switching frequency is not too high in relation to ambient temperature and motor load.
15	Motor stalled	Motor stall protection has tripped.	Check motor and load.
16	Motor over-temperature	Motor overheating has been detected by frequency converter motor temperature model. Motor is overloaded.	Decrease motor load. If no motor overload exists, check the temperature model parameters.
17	Motor underload	Motor underload protection has tripped.	Check load.
18	Unbalance	Unbalance between power modules in parallelled units. Subcode in T.14: S1 – Current unbalance S2 – DC voltage unbalance	Should the fault re-occur, contact the distributor near to you.

Fault code	Fault	Possible cause	Correcting measures
22	EEPROM checksum fault	Parameter save fault – faulty operation – component failure	Should the fault re-occur, contact the distributor near to you.
24	Counter fault	Values displayed on counters are incorrect	
25	Microprocessor watchdog fault	– faulty operation – component failure	Reset the fault and restart. Should the fault re-occur, contact the distributor near to you.
26	Start-up prevented	Start-up of the drive has been prevented. Run request in ON when new application is loaded to the drive	Cancel prevention of start-up if this can be done safely. Remove Run request
29	**Thermistor fault**	**The thermistor input of option board has detected increase of the motor temperature**	**Check motor cooling and loading Check thermistor connection (If thermistor input of the option board is not in use it has to be short circuited)**
30	Safe disable	The input on OPTAF board has opened	Cancel Safe Disable if this can be done safely.
31	IGBT temperature (hardware)	IGBT Inverter Bridge overtemperature protection has detected too high a short term overload current	Check loading. Check motor size. Make identification run.
32	Fan cooling	Cooling fan of the frequency converter does not start, when ON command is given	Contact the distributor near to you.
34	CAN bus communication	Sent message not acknowledged.	Ensure that there is another device on the bus with the same configuration.
35	Application	Problem in application software	Contact your distributor. If you are application programmer check the application program.
36	Control unit	NXS Control Unit can not control NXP Power Unit and vice versa	Change control unit
37	Device changed (same type)	Option board or control unit changed. Same type of board or same power rating of drive.	Reset. Device is ready for use. Old parameter settings will be used.
38	Device added (same type)	Option board or drive added.	Reset. Device is ready for use. Old board settings will be used.
39	Device removed	Option board removed.	Reset. Device no longer available.
40	Device unknown	Unknown option board or drive. Subcode in T.14: S1 – Unknown device S2 – Power1not same type as Power2	Contact the distributor near to you.
41	IGBT temperature	IGBT Inverter Bridge overtemperature protection has detected too high a short term overload current	Check loading. Check motor size. Make identification run.
42	**Brake resistor**	**Problems with brake resistor.**	**Set the deceleration time longer. Use external brake resistor.**
43	Encoder fault	Problem detected in encoder signals. Subcode in T.14: 1 – Encoder 1 channel A is missing 2 – Encoder 1 channel B is missing 3 – Both encoder 1 channels are missing 4 – Encoder reversed 5 – Encoder board missing	Check encoder channel connections. Check the encoder board. Check encoder frequency in open loop.

Fault code	Fault	Possible cause	Correcting measures
44	Device changed (different type)	Option board or power unit changed. New device of different type or different power rating.	Reset Set the option board parameters again if option board was changed. Set converter parameters again if power unit was changed.
45	Device added (different type)	Option board of different type added.	Reset Set the option board parameters again.
49	Division by zero in application	Division by zero has occurred in application program.	Contact your distributor if the fault re-occurs while the converter is in run state. If you are application programmer check the application program.
50	**Analogue input I_{in} < 4mA (sel. signal range 4 to 20 mA)**	**Current at the analogue input is < 4mA.** **– control cable is broken or loose** **– signal source has failed**	**Check the current loop circuitry.**
51	External fault	Digital input fault.	Remove fault situation on external device.
52	Keypad communication fault	The connection between the control keypad (or NCDrive) and the frequency converter is broken.	Check keypad connection and possible keypad cable.
53	Fieldbus fault	The data connection between the fieldbus Master and the fieldbus board is broken	Check installation. If installation is correct contact the nearest Vacon distributor.
54	Slot fault	Defective option board or slot	Check board and slot. Contact the nearest Vacon distributor.

APITQ224_v320



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